

(12) **United States Patent**
Simi et al.

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(54) **SURGICAL TOOL FOR ROBOTIC SURGERY
AND ROBOTIC SURGICAL ASSEMBLY**

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 1012 days.

This patent is subject to a terminal dis-
claimer.

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(65) **Prior Publication Data**
US 2021/0386496 A1 Dec. 16, 2021

Related U.S. Application Data

(63) Continuation of application No. 15/768,525, filed as
application No. PCT/EP2016/074808 on Oct. 14,
2016, now Pat. No. 11,141,233.

(30) **Foreign Application Priority Data**

Oct. 16, 2015 (IT) 102015000062500

(51) **Int. Cl.**
A61B 34/37 (2016.01)
A61B 1/313 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **A61B 34/37** (2016.02); **A61B 1/3132**
(2013.01); **A61B 10/04** (2013.01);
(Continued)

(58) **Field of Classification Search**

CPC A61B 34/37; A61B 1/3132; A61B 10/04;
A61B 17/00234; A61B 17/062;
(Continued)

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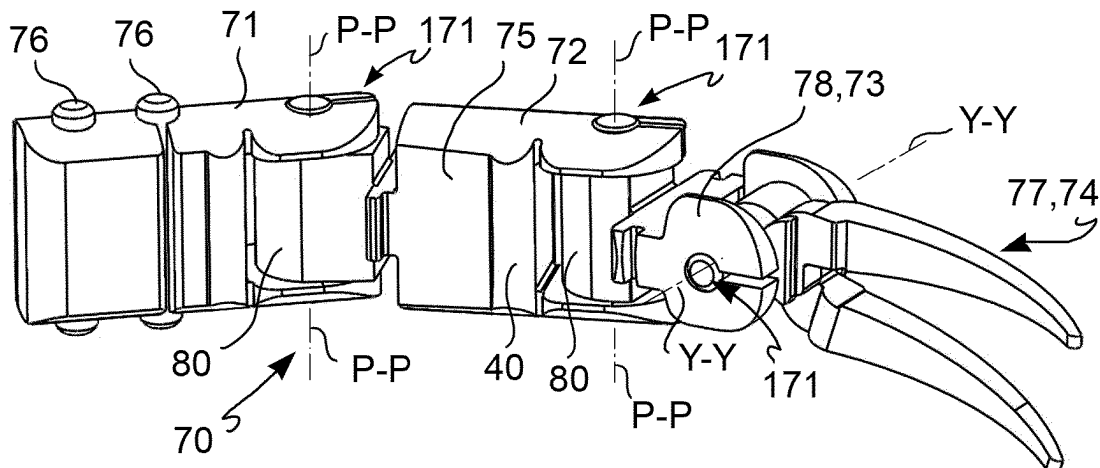
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Assistant Examiner — Rachael L Geiger

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(57) **ABSTRACT**

A medical instrument for surgery includes at least one frame
and at least one jointed device. The jointed device includes
at least one first joint member, or first link, adapted to
connect to at least one portion of the frame and at least one
second joint member, or second link. The first joint member
is connected by a rotational joint to the second joint member.
The medical instrument includes at least a pair of tendons,
adapted to move the second joint member with respect to the
first joint member. Each of the first joint member and the
second joint member includes a main structural body made
in a single piece with one or more convex contact surfaces.
Each of the convex contact surfaces is a ruled surface
(Continued)



formed by straight line portions all parallel to each other and substantially parallel to a joint movement axis.

G05B 2219/45117; B23H 7/02; B23H 11/003

See application file for complete search history.

21 Claims, 24 Drawing Sheets

(56)

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A61B 17/00 (2006.01)
A61B 17/062 (2006.01)
A61B 17/29 (2006.01)
A61B 17/32 (2006.01)
A61B 17/3211 (2006.01)
A61B 34/00 (2016.01)
A61B 34/20 (2016.01)
A61B 34/30 (2016.01)
A61B 34/35 (2016.01)
A61B 90/00 (2016.01)
A61B 90/20 (2016.01)
A61B 90/25 (2016.01)
B25J 3/04 (2006.01)
B25J 9/10 (2006.01)
B25J 9/16 (2006.01)
B25J 15/00 (2006.01)
G05B 19/402 (2006.01)
A61B 17/34 (2006.01)

(52) **U.S. Cl.**

CPC *A61B 17/00234* (2013.01); *A61B 17/062* (2013.01); *A61B 34/30* (2016.02); *A61B 34/35* (2016.02); *A61B 34/71* (2016.02); *A61B 34/72* (2016.02); *A61B 90/20* (2016.02); *A61B 90/25* (2016.02); *A61B 90/361* (2016.02); *B25J 3/04* (2013.01); *B25J 9/1015* (2013.01); *B25J 9/1669* (2013.01); *B25J 15/0052* (2013.01); *G05B 19/402* (2013.01); *A61B 2017/00345* (2013.01); *A61B 17/29* (2013.01); *A61B 17/320016* (2013.01); *A61B 17/3211* (2013.01); *A61B 17/34* (2013.01); *A61B 2034/2048* (2016.02); *A61B 2034/2051* (2016.02); *A61B 2034/301* (2016.02); *A61B 2034/715* (2016.02); *A61B 2090/061* (2016.02); *A61B 2090/064* (2016.02); *G05B 2219/45117* (2013.01)

(58) **Field of Classification Search**

CPC *A61B 34/30*; *A61B 34/35*; *A61B 34/71*; *A61B 34/72*; *A61B 90/20*; *A61B 90/25*; *A61B 90/361*; *A61B 17/29*; *A61B 17/320016*; *A61B 17/3211*; *A61B 17/34*; *A61B 2017/00345*; *A61B 2034/2048*; *A61B 2034/2051*; *A61B 2034/301*; *A61B 2034/715*; *A61B 2090/061*; *A61B 2090/064*; *A61B 90/40*; *A61B 2034/302*; *A61B 2034/305*; *A61B 34/75*; *B25J 3/04*; *B25J 9/1015*; *B25J 9/1669*; *B25J 15/0052*; *B25J 9/1689*; *G05B 19/402*;

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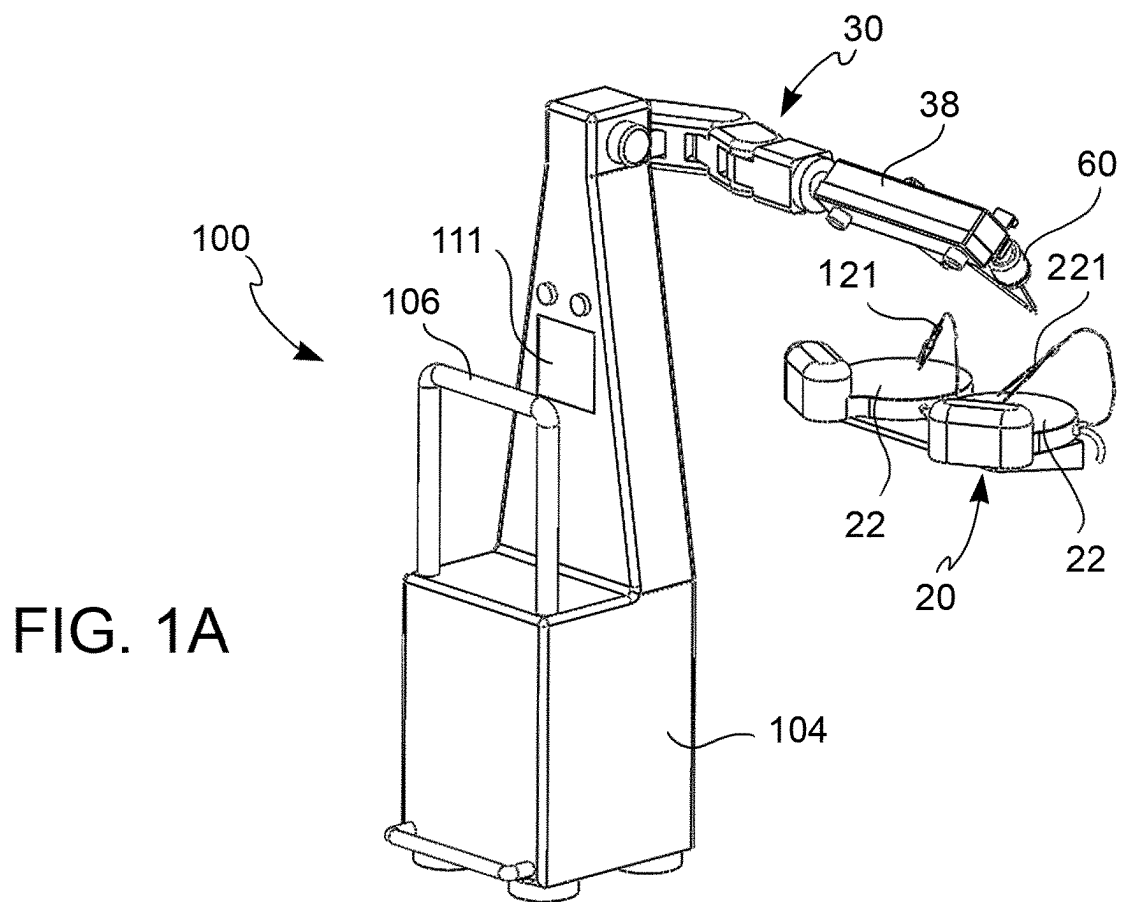
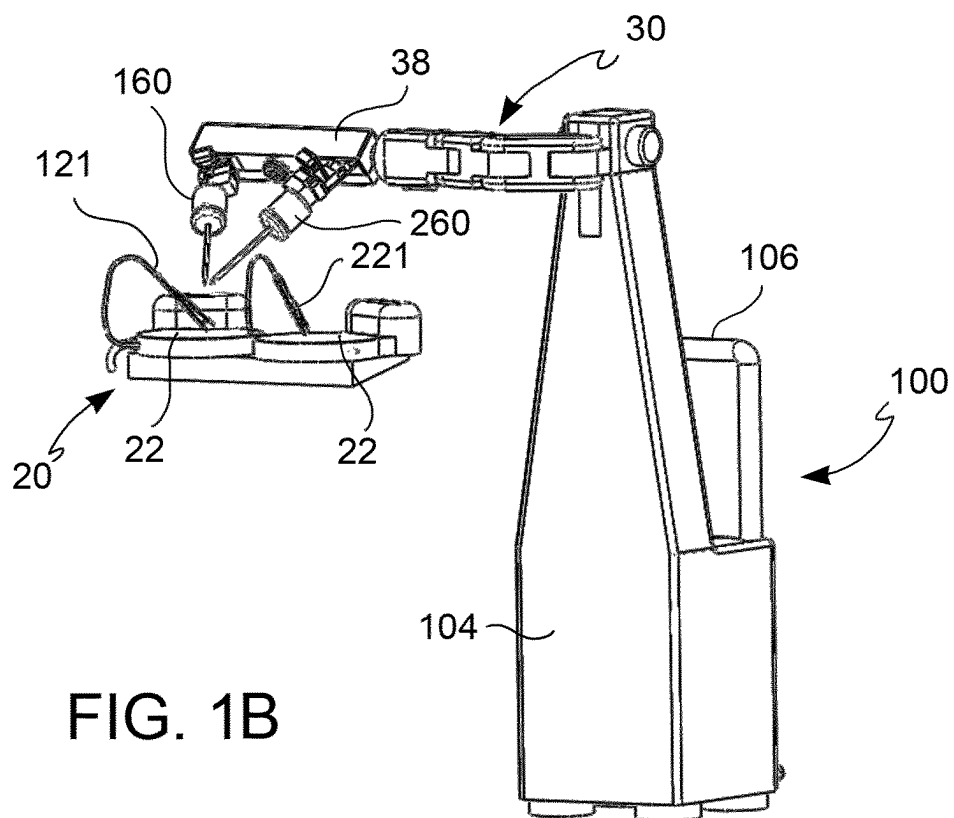
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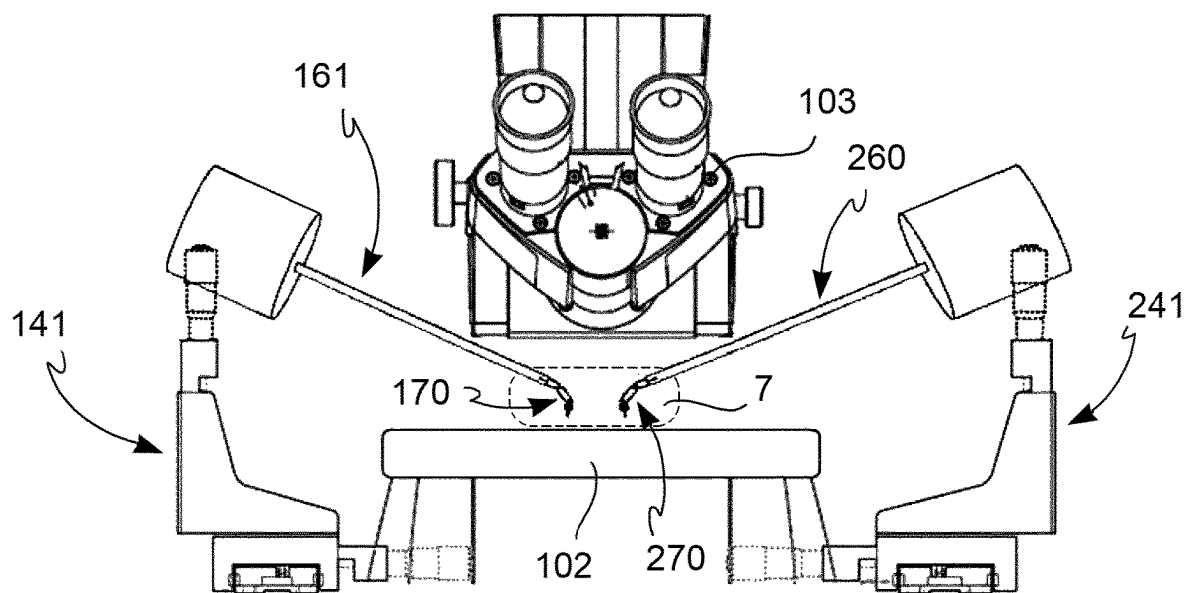
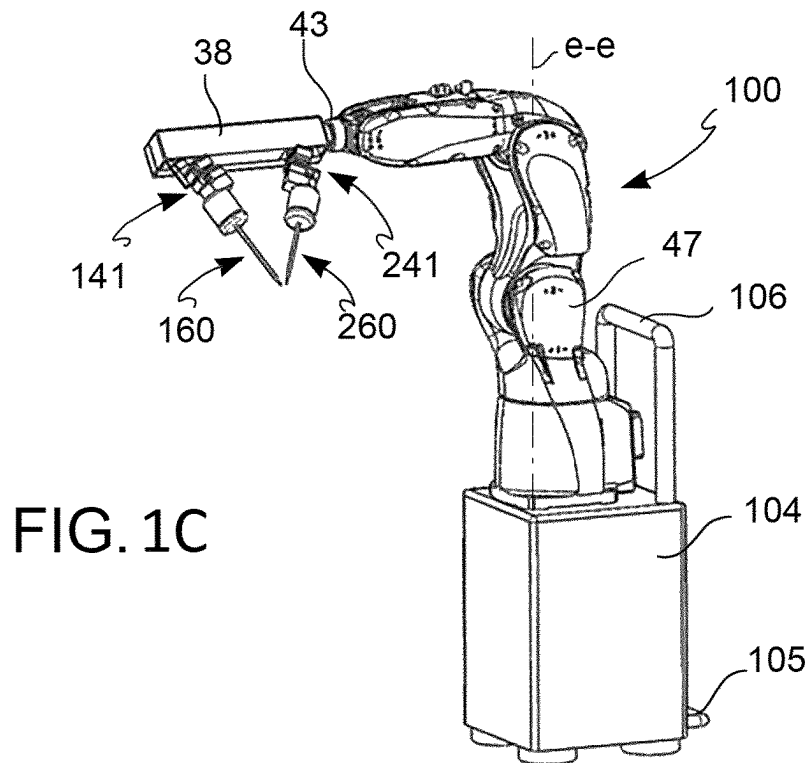
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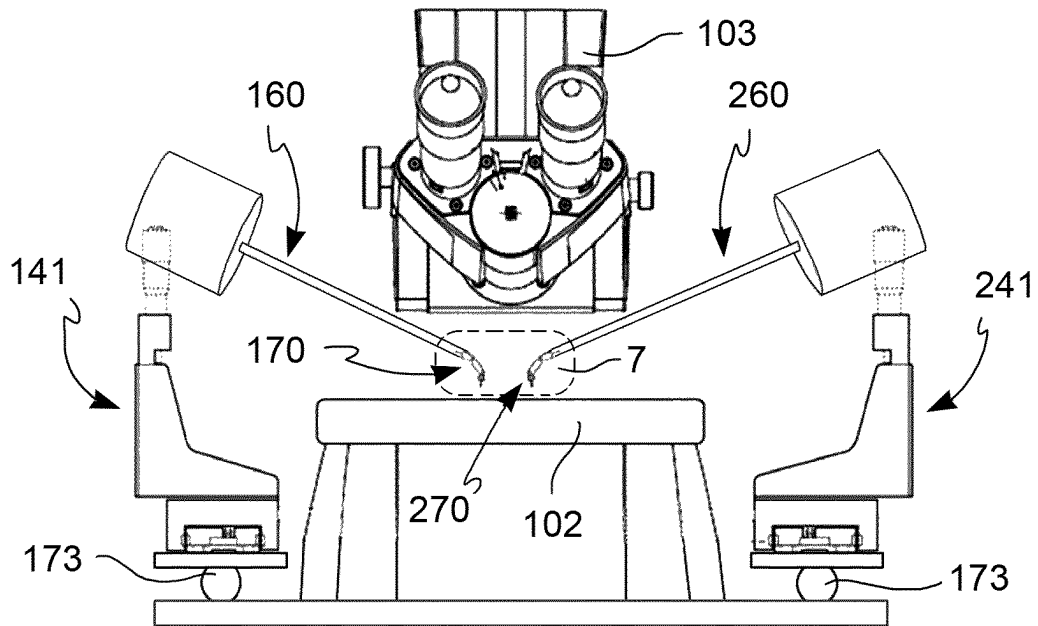


FIG. 2B

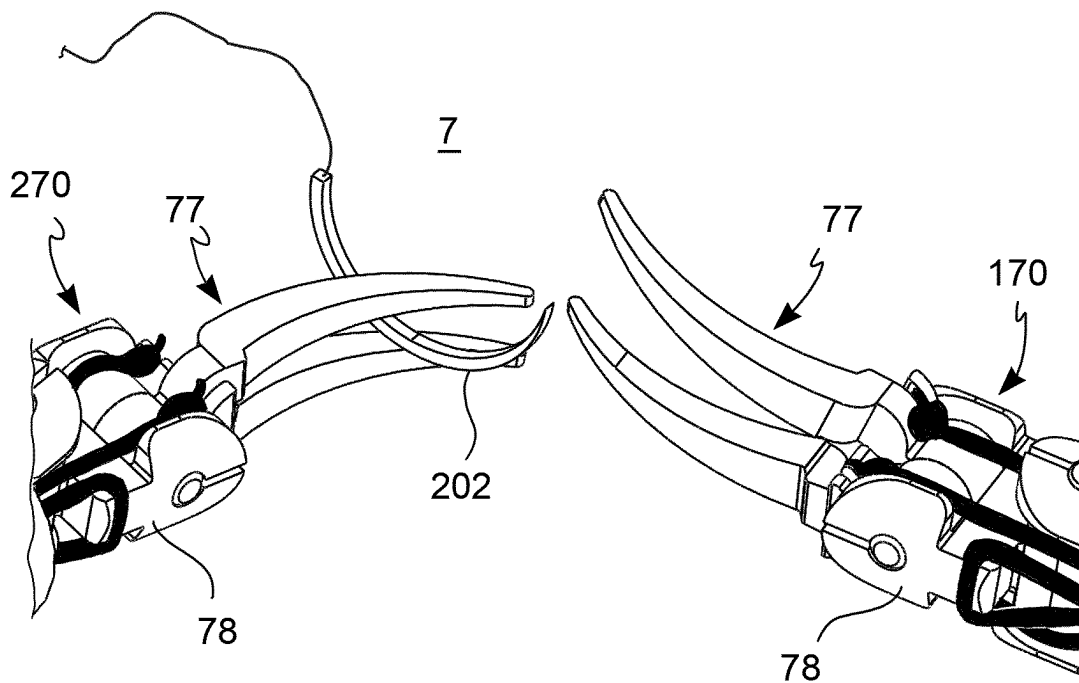


FIG. 3

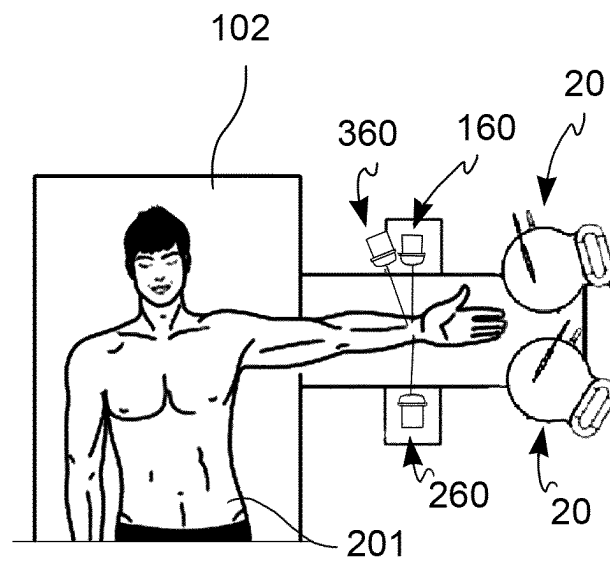


FIG. 4B

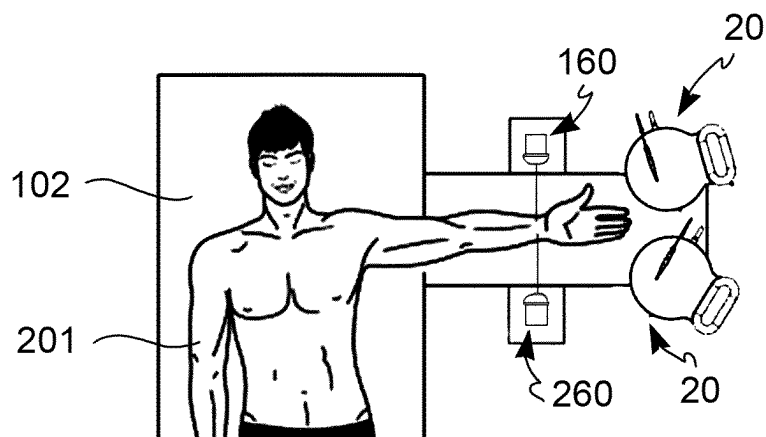


FIG. 4A

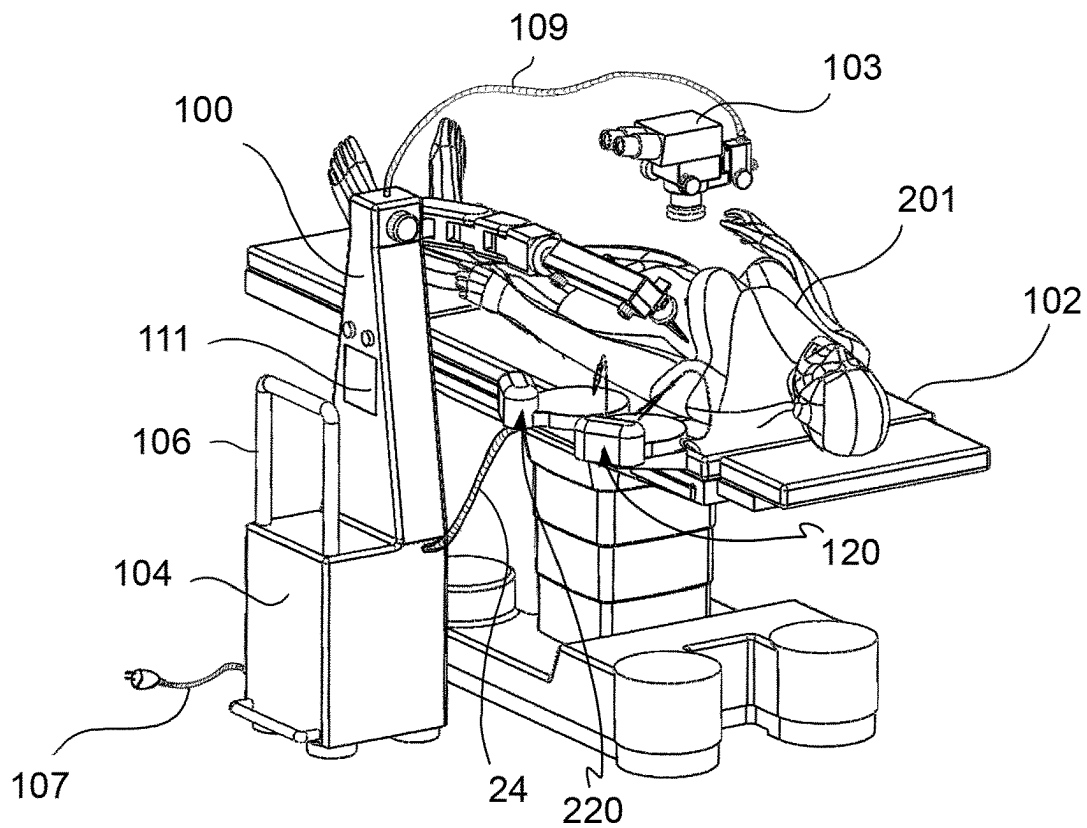


FIG. 5

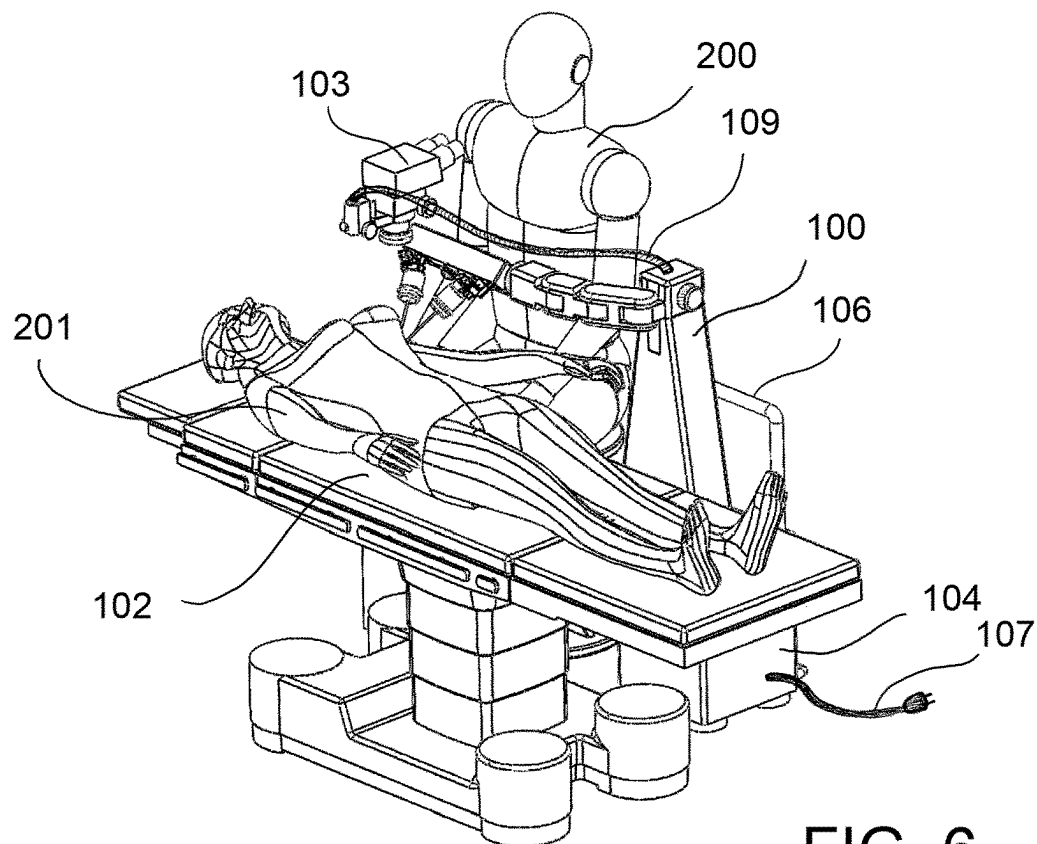


FIG. 6

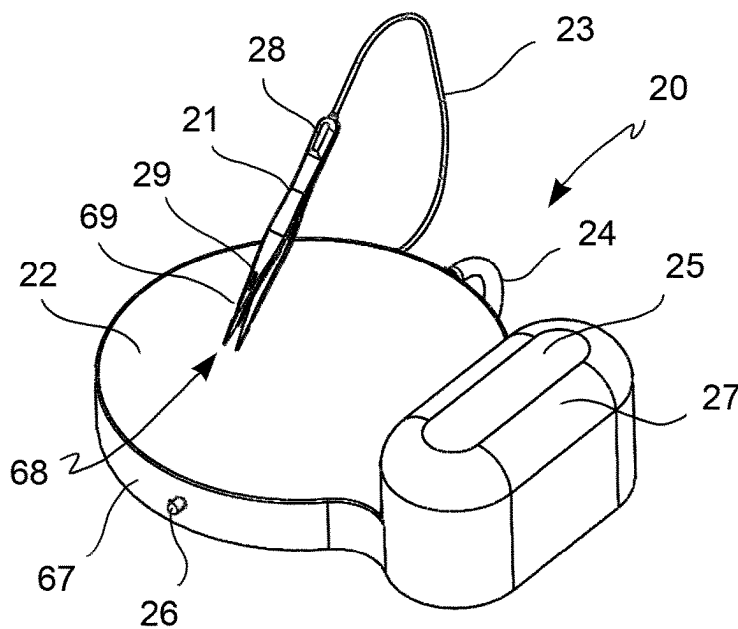


FIG. 7

FIG. 8

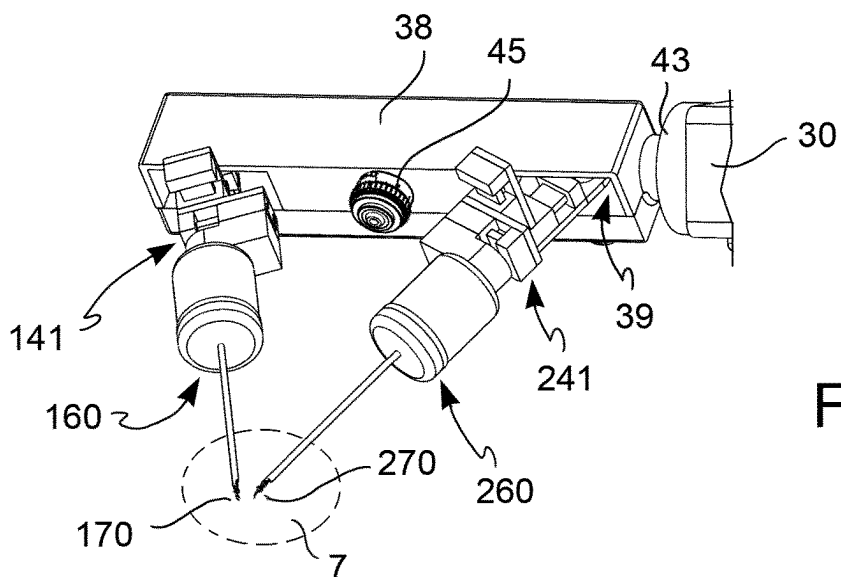
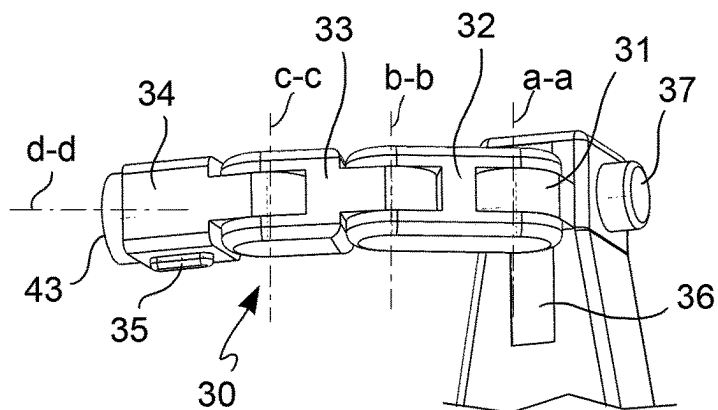


FIG. 9A

FIG. 9B

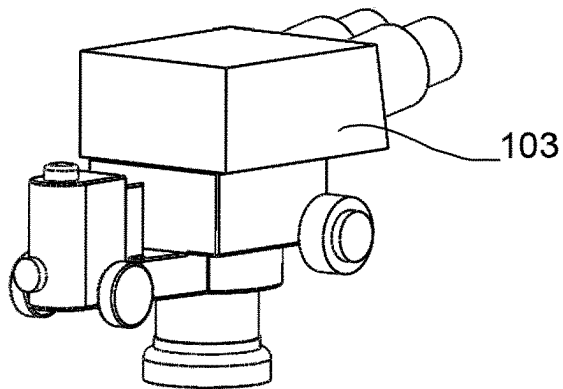
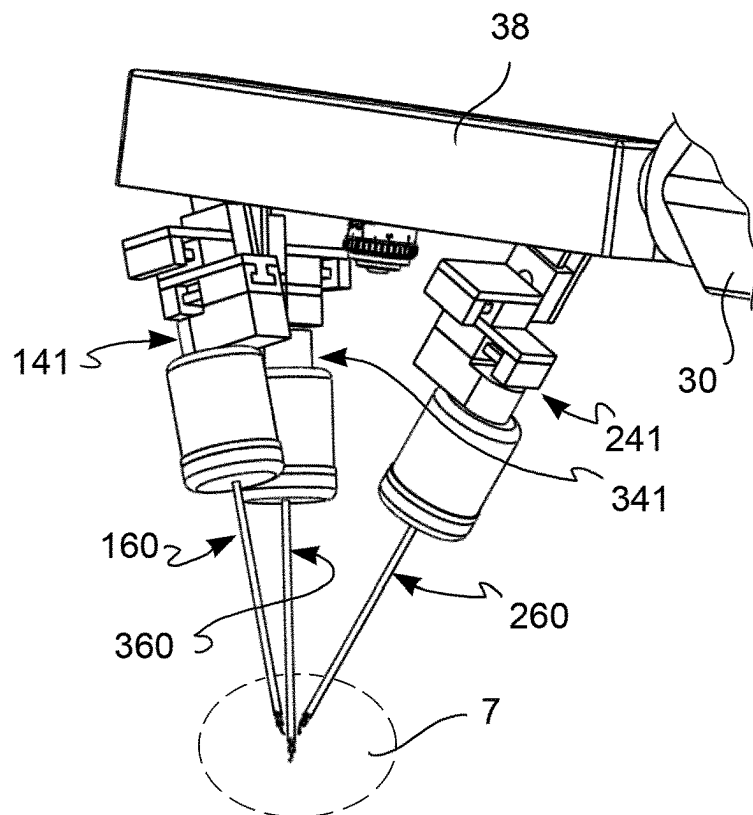
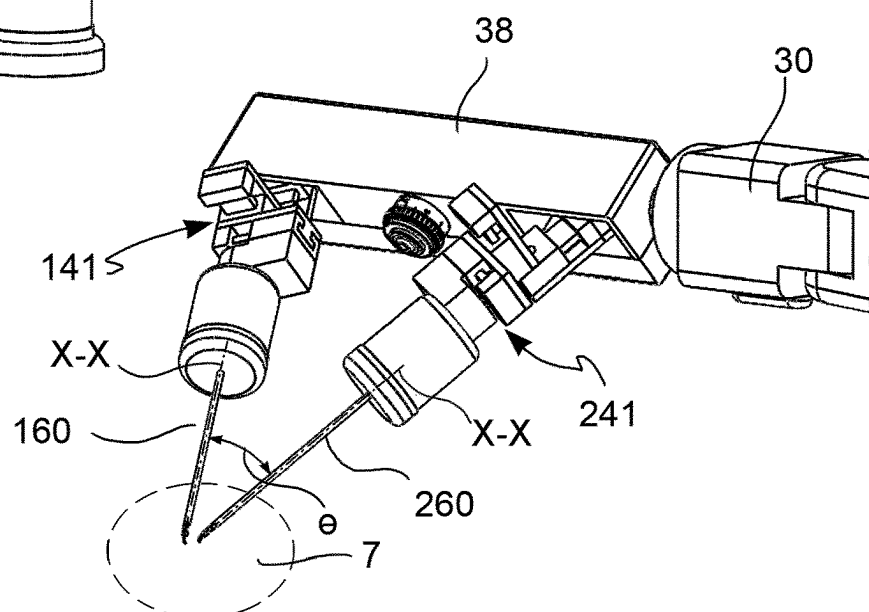
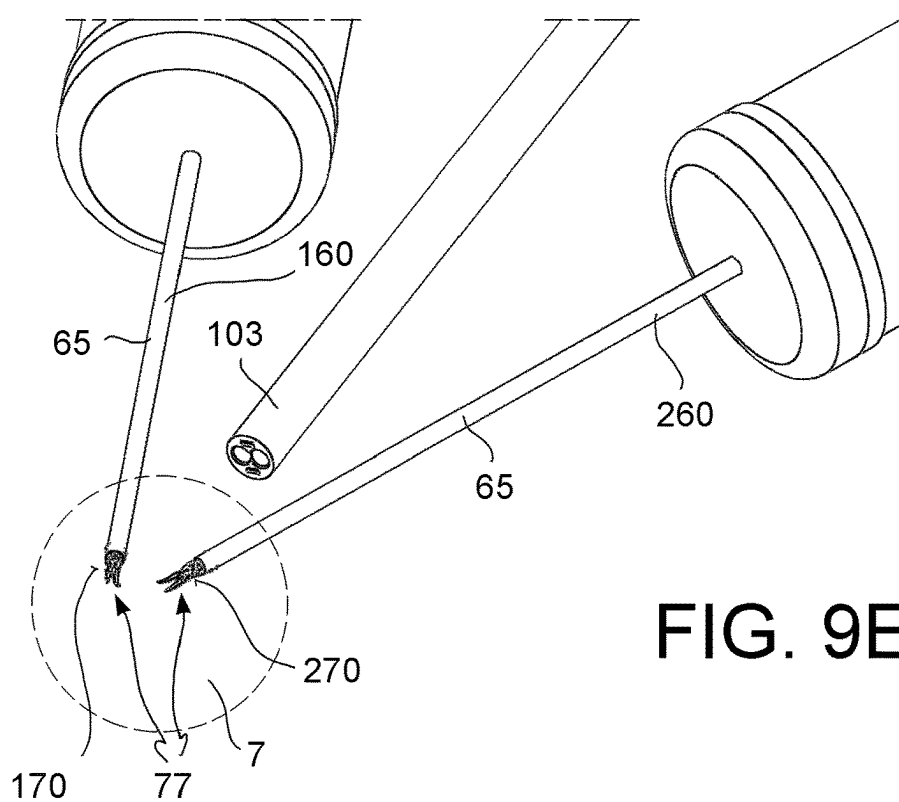
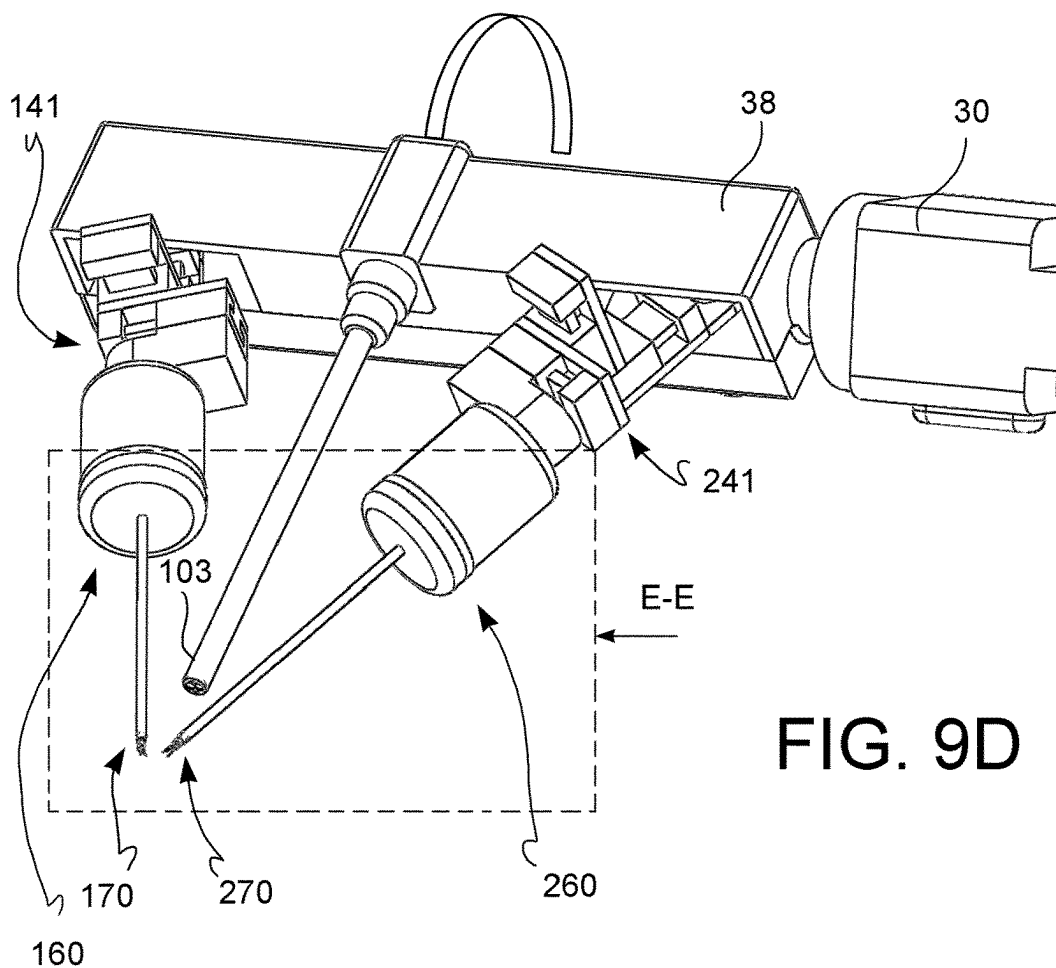


FIG. 9C





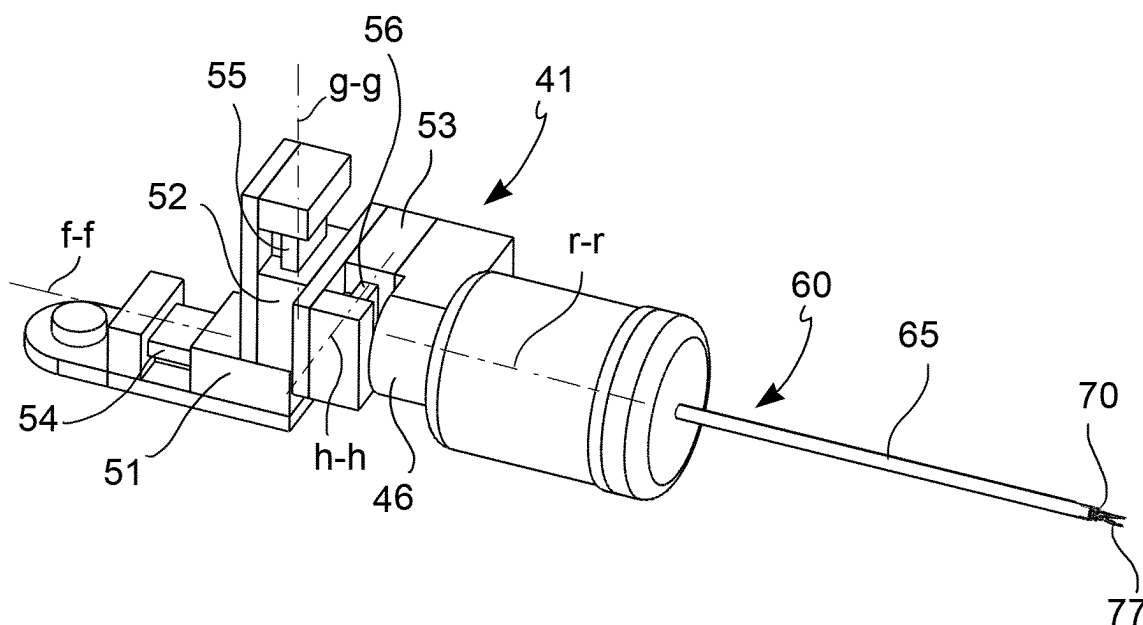


FIG. 10

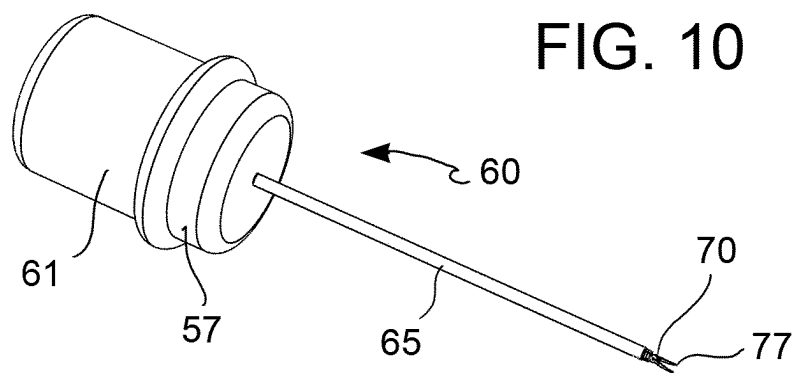


FIG. 11

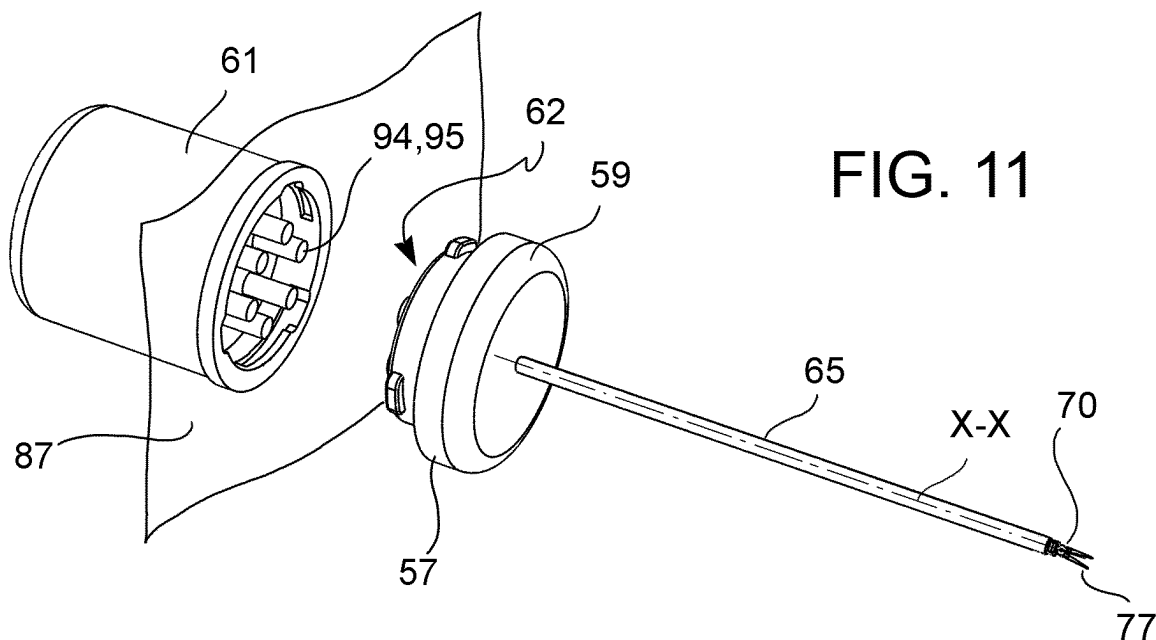


FIG. 12

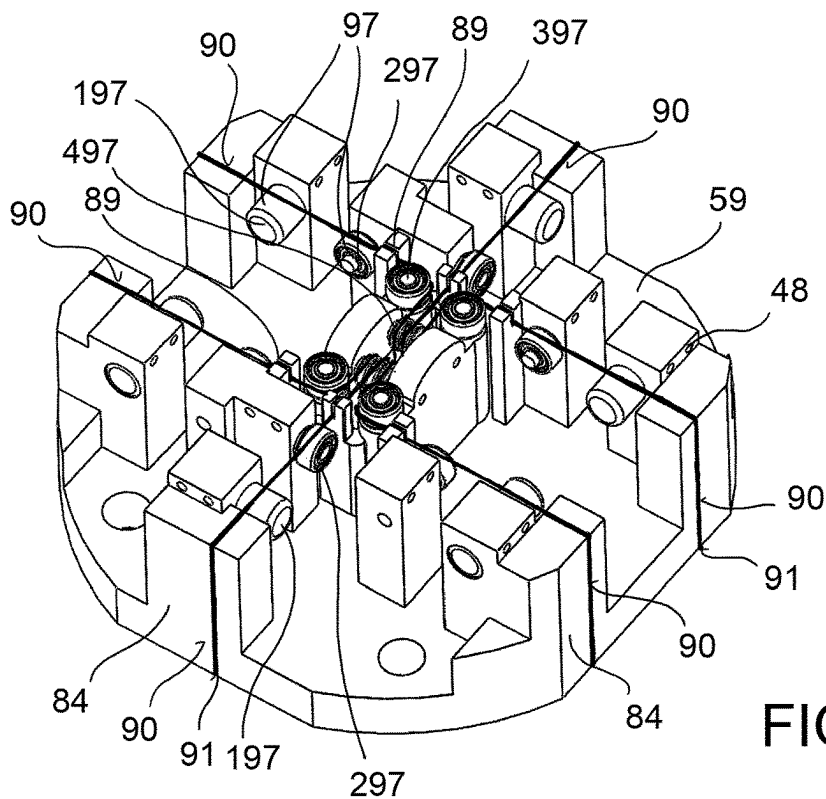


FIG. 13A

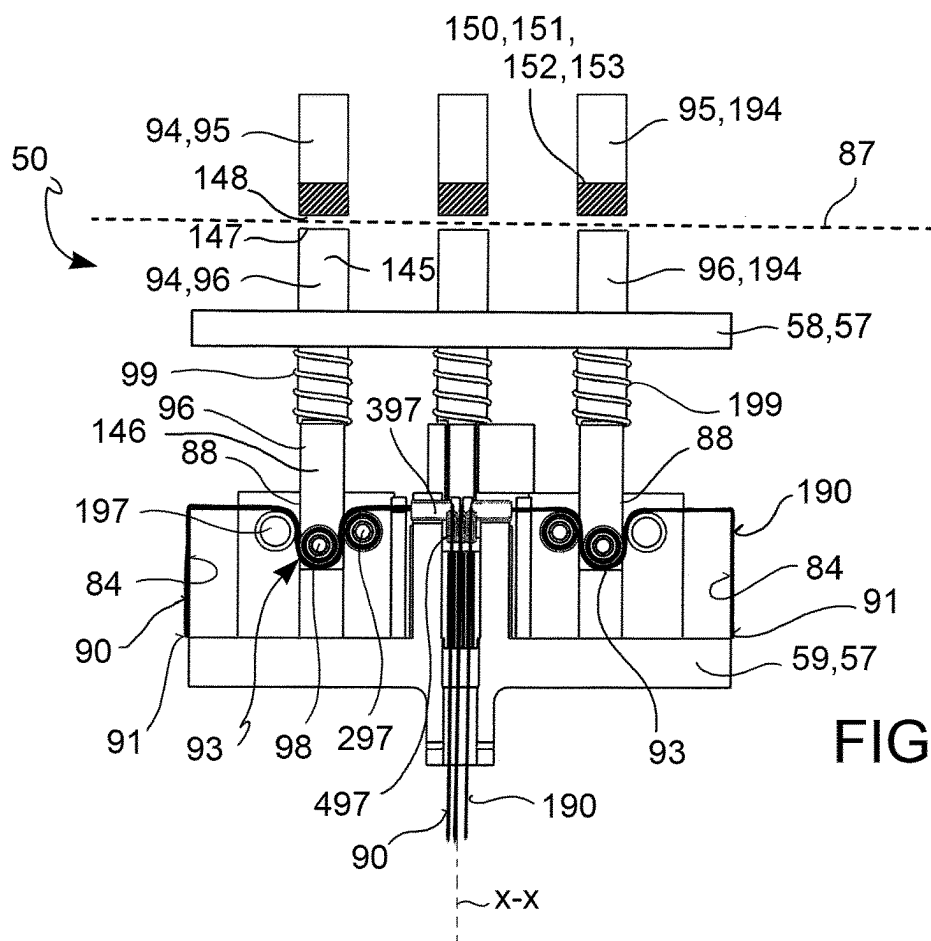


FIG. 14A

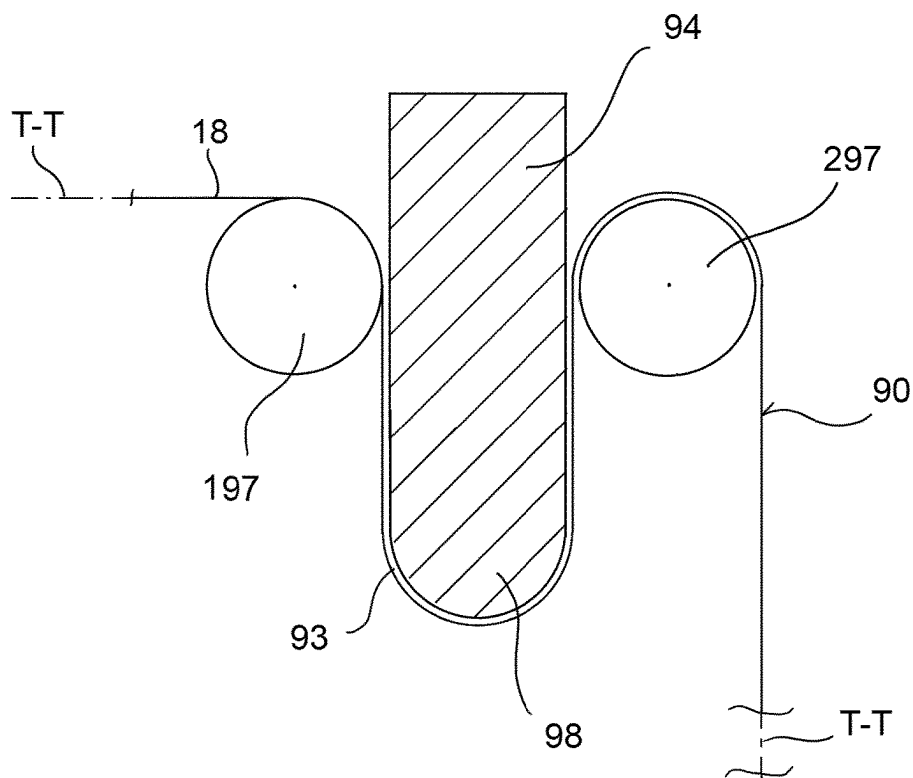


FIG. 14B

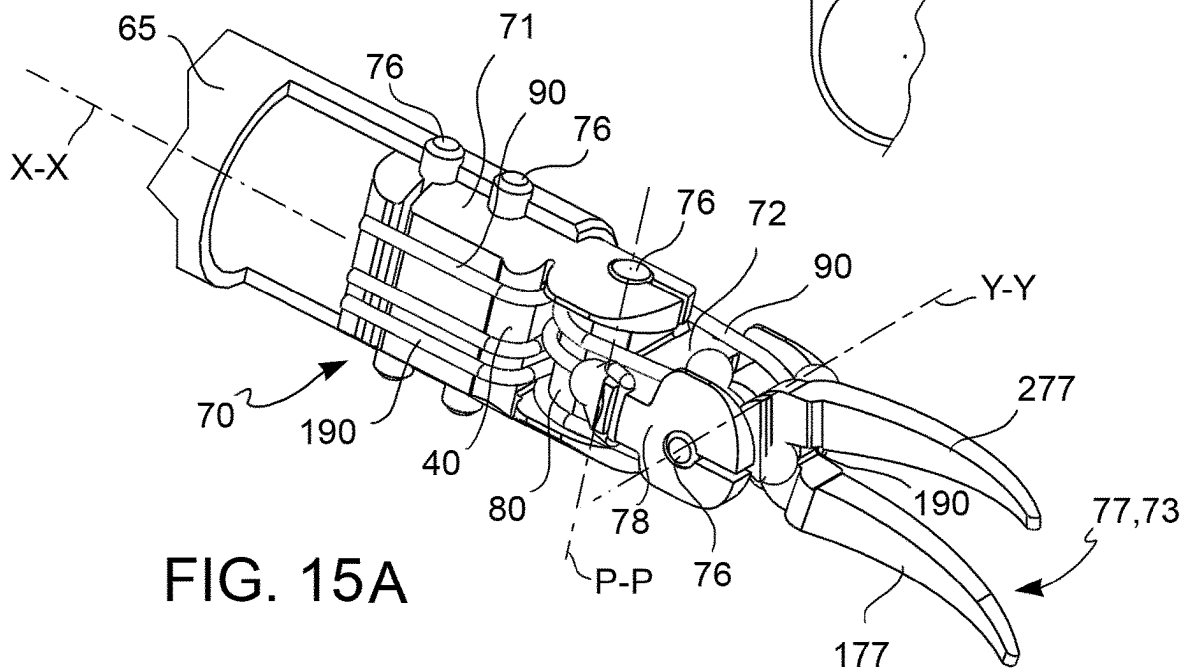


FIG. 15A

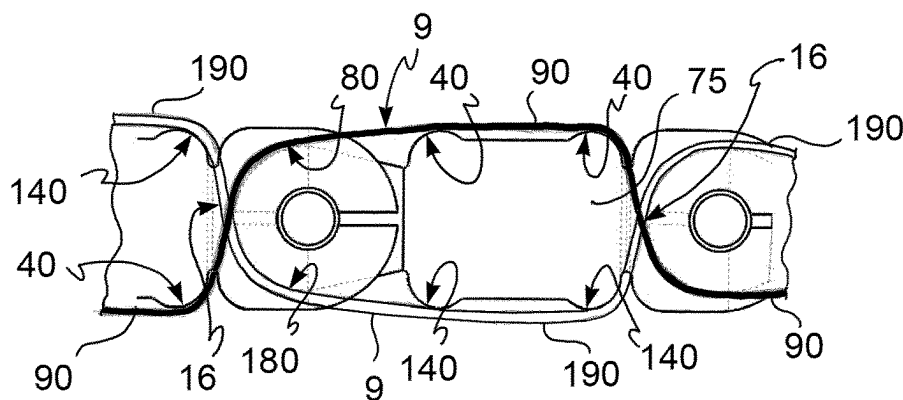


FIG. 16

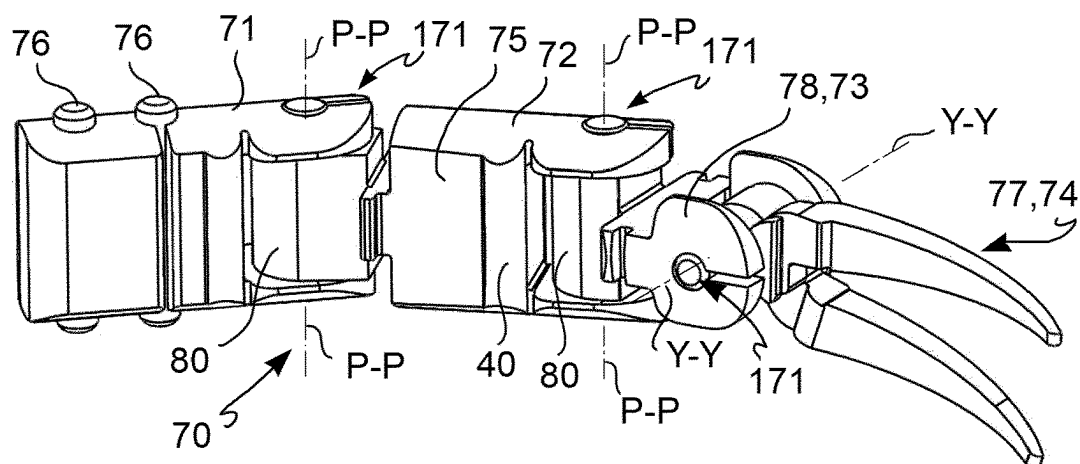


FIG. 17

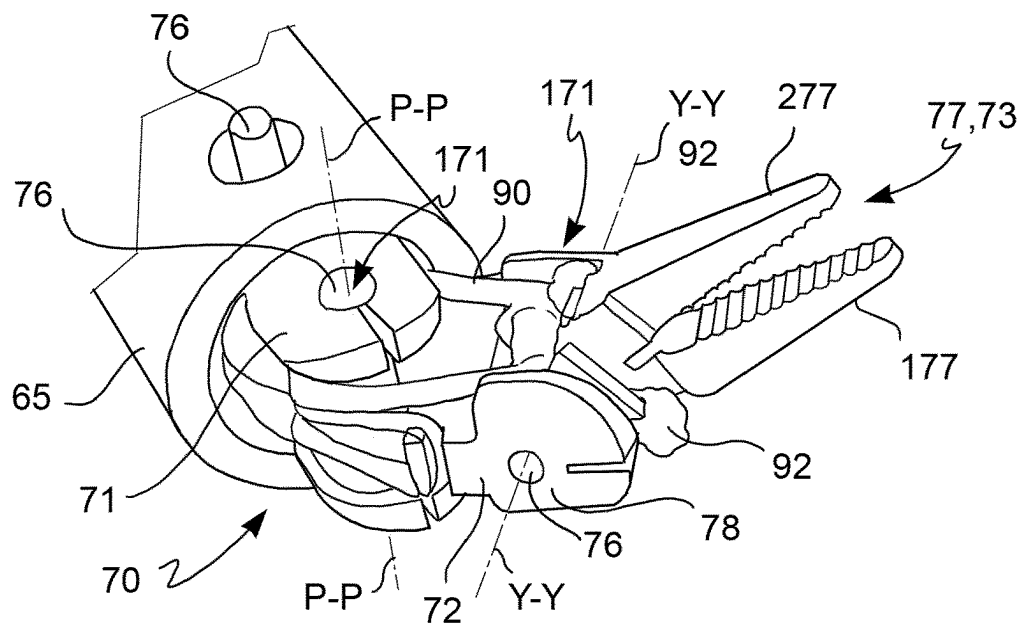


FIG. 15B

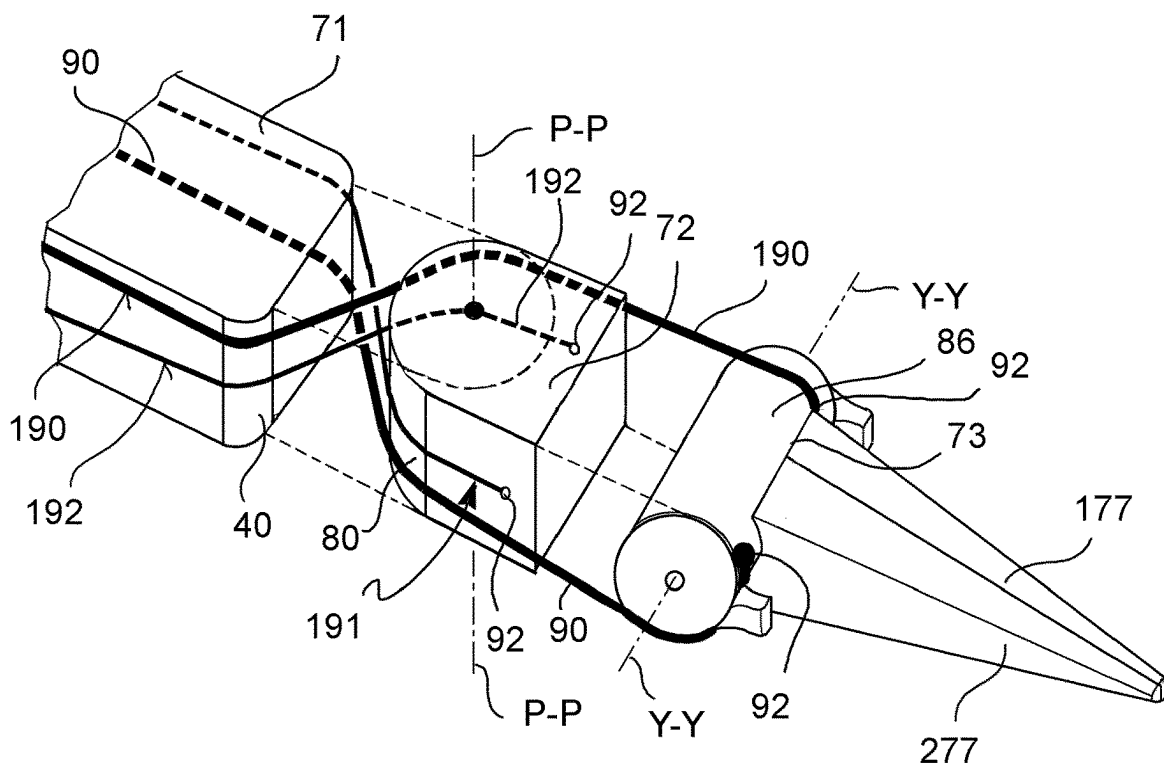


FIG. 15C

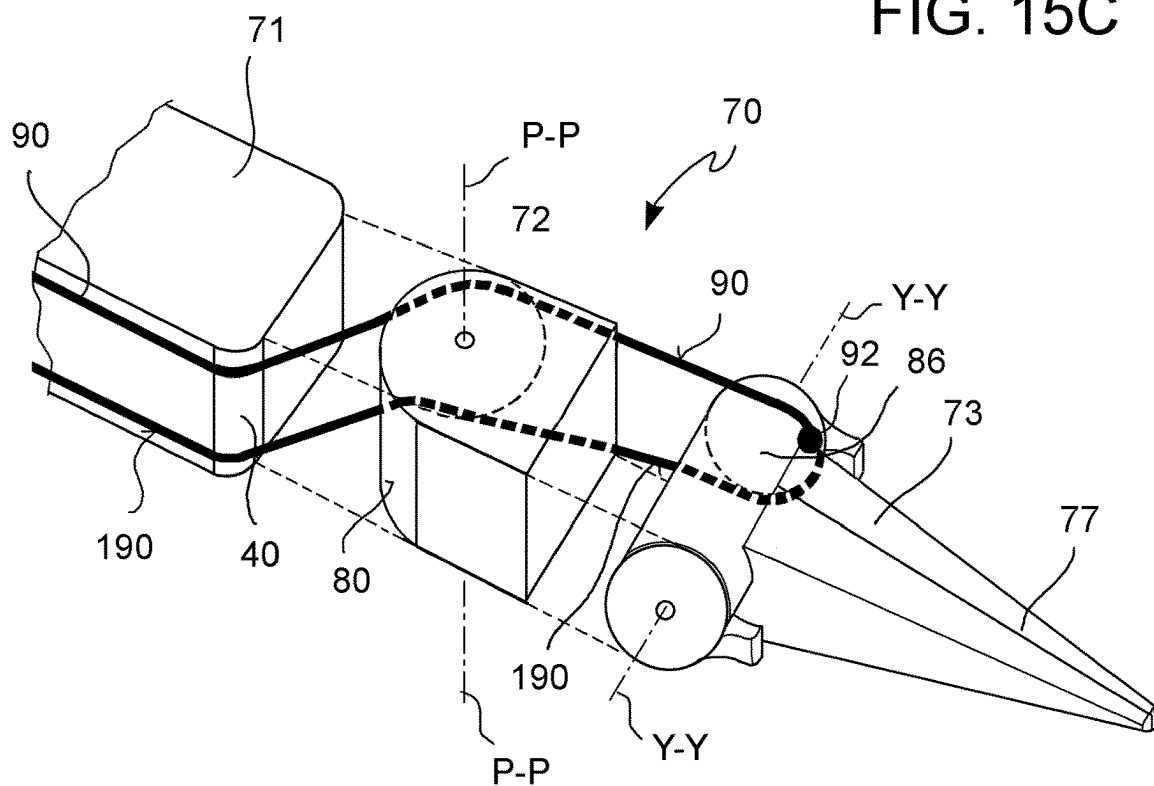


FIG. 15D

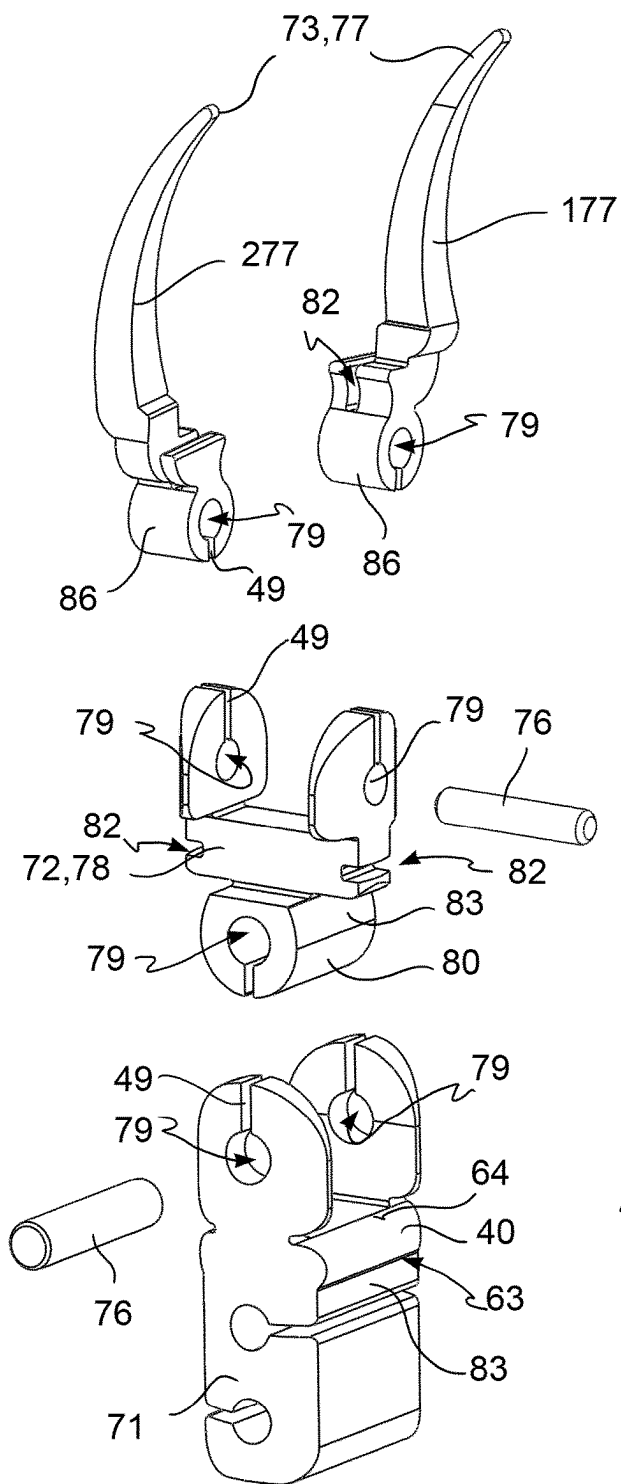


FIG. 18

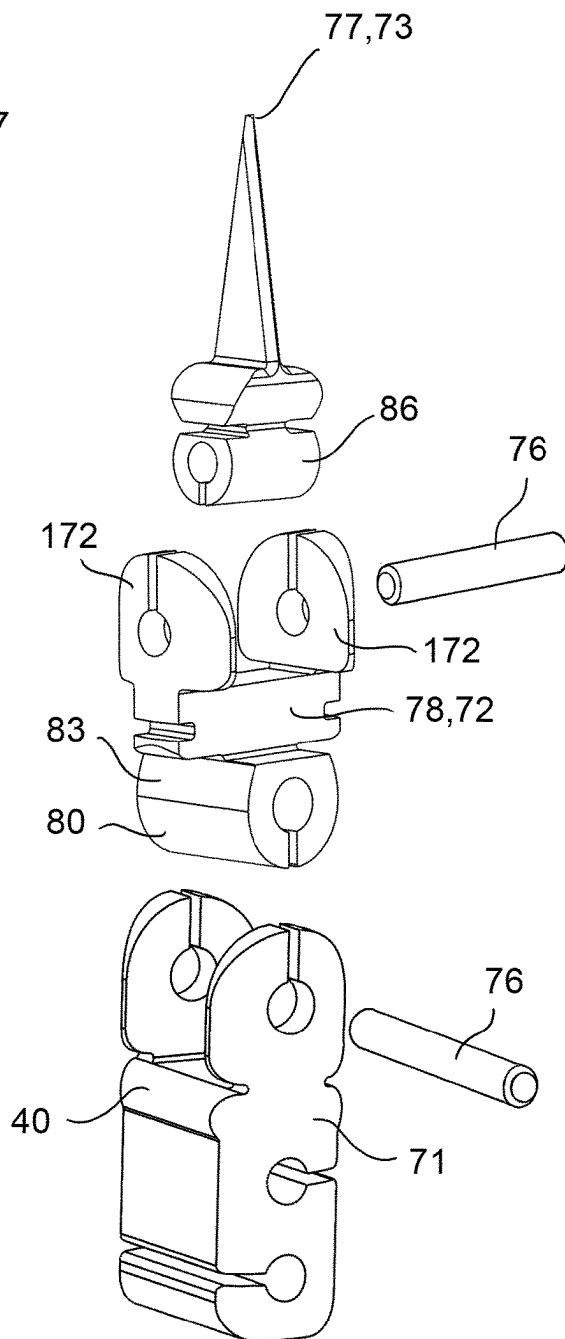


FIG. 19

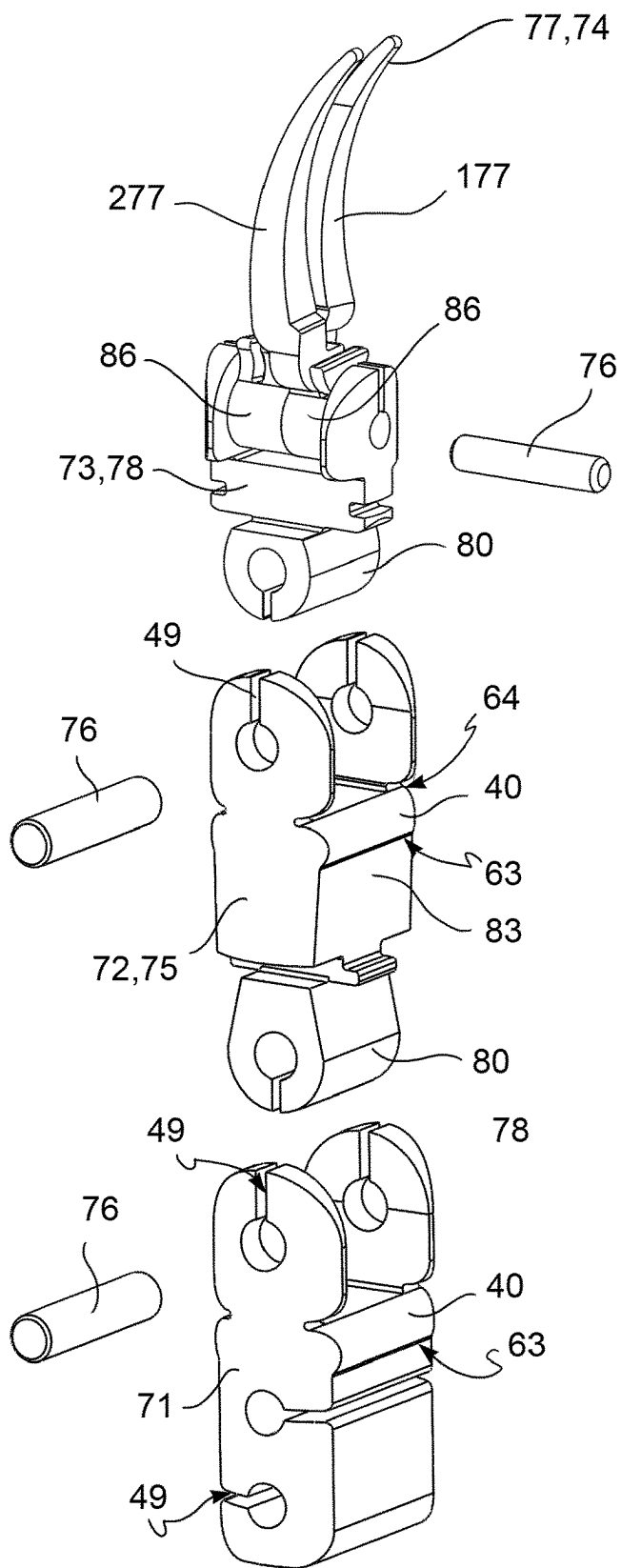


FIG. 20

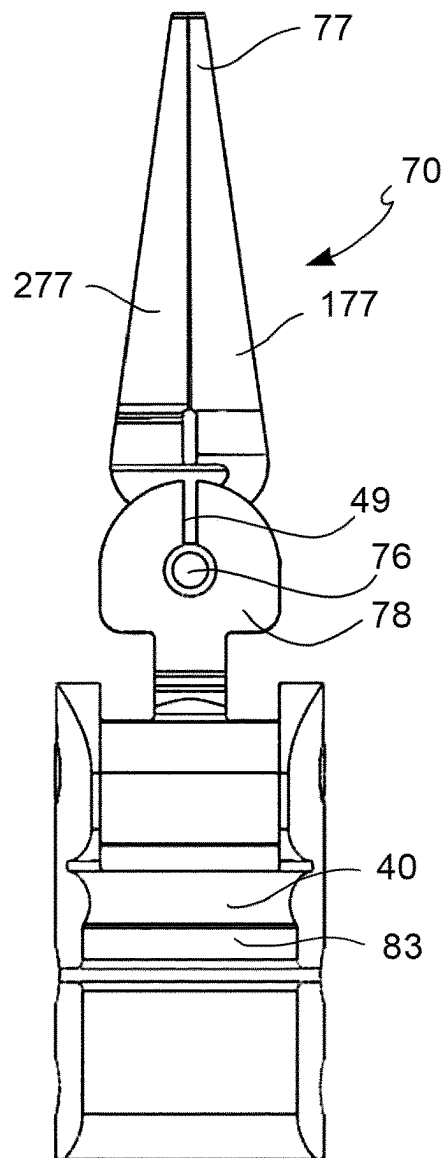


FIG. 21

FIG. 22

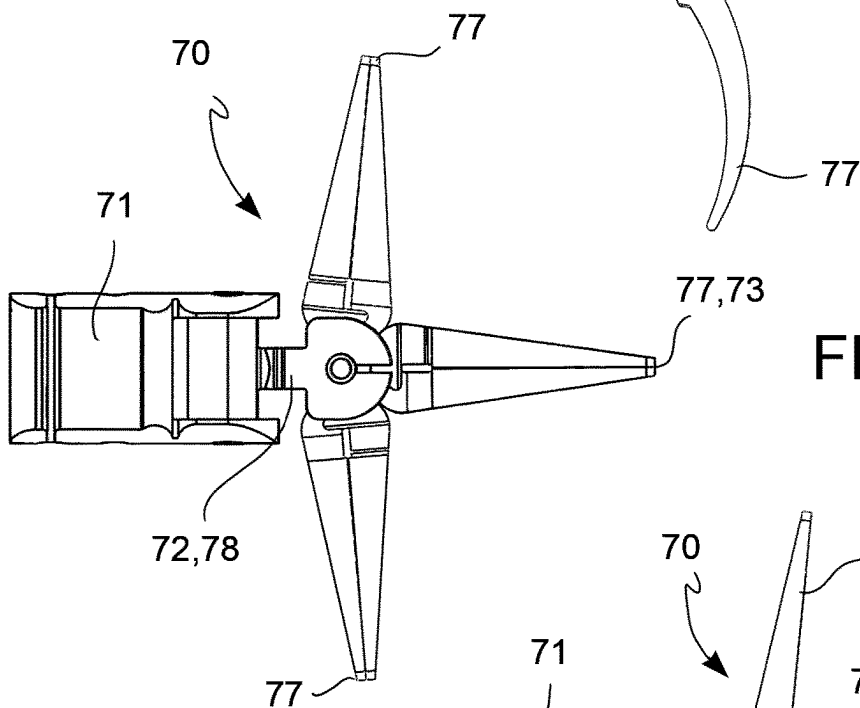
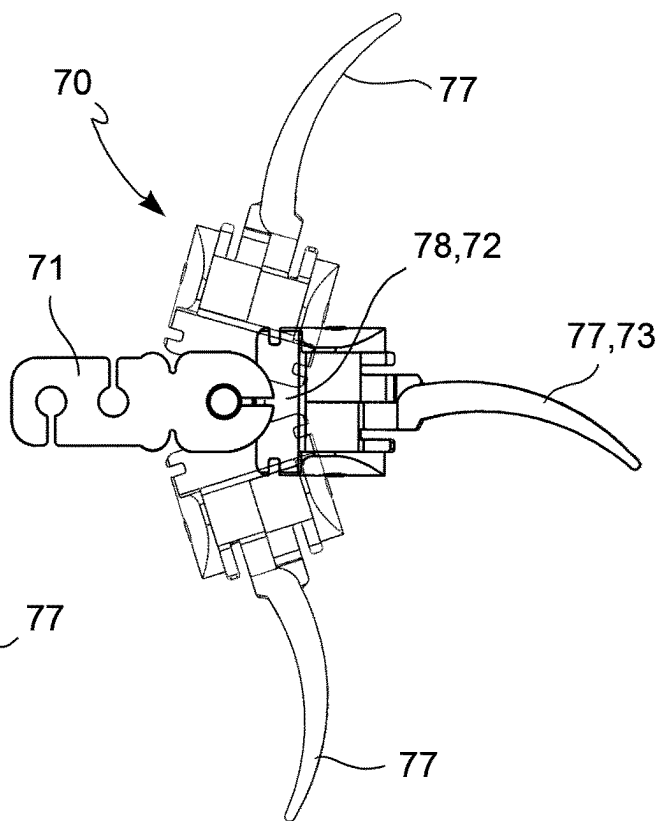
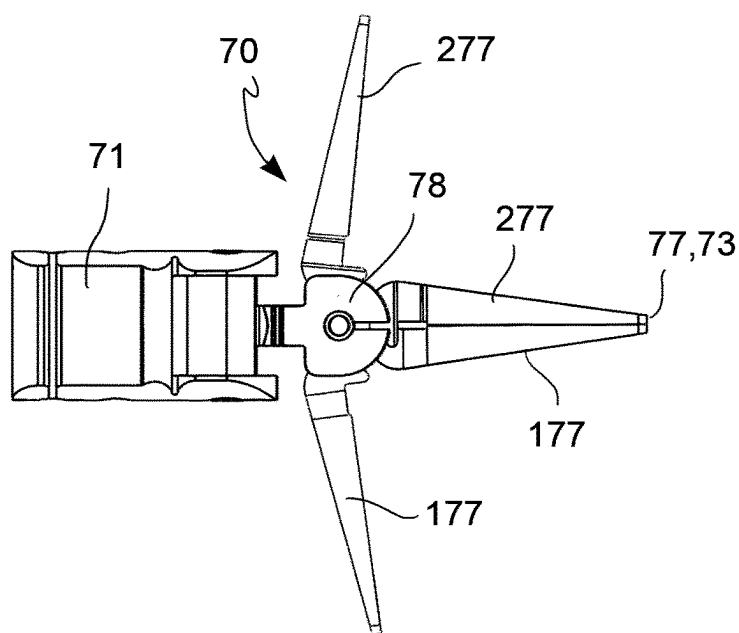
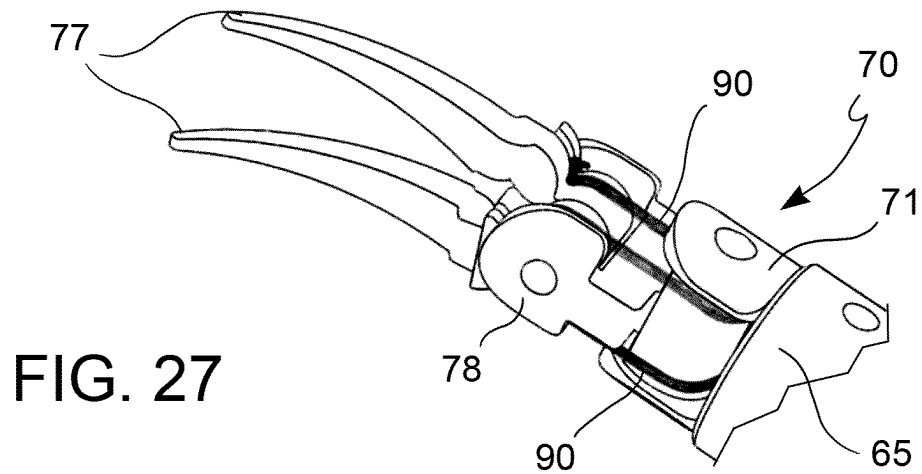
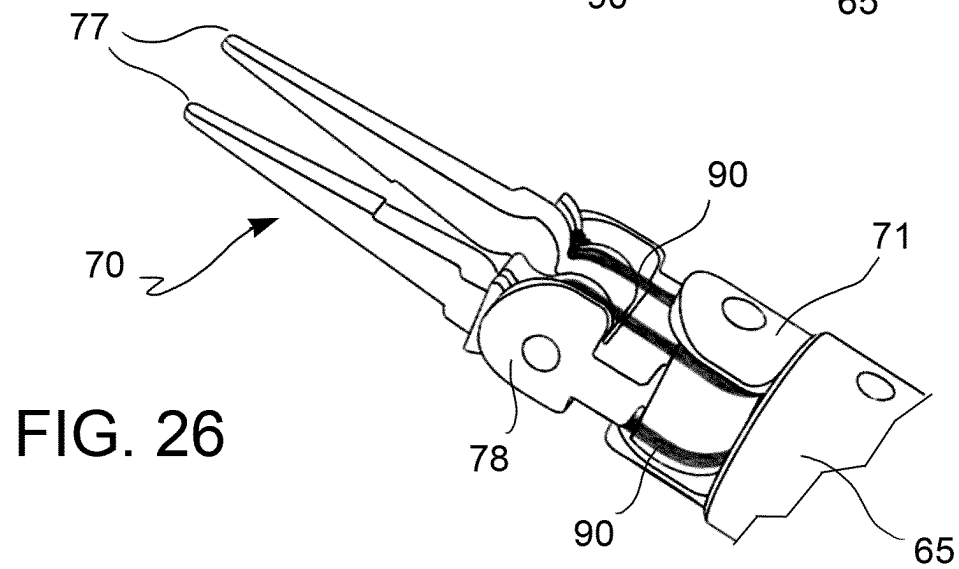
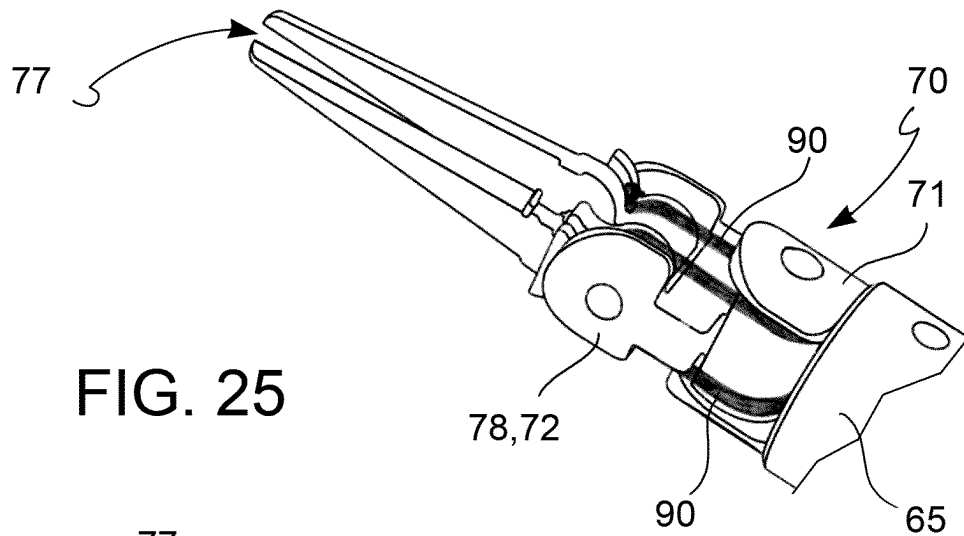


FIG. 23

FIG. 24





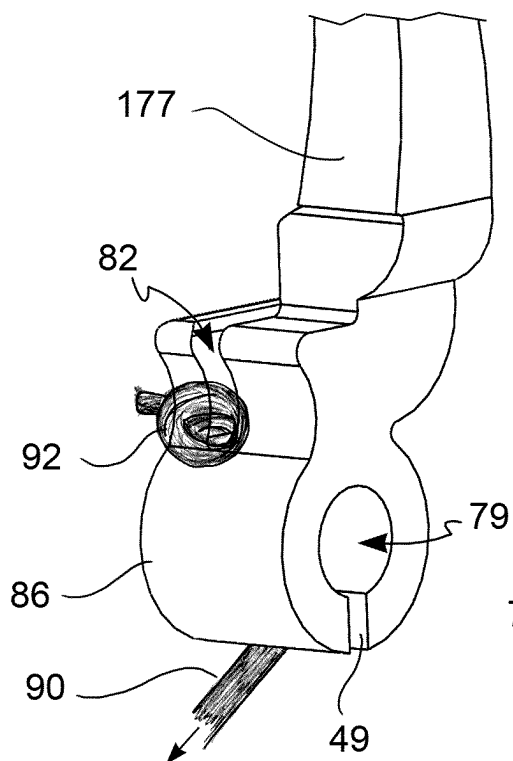


FIG. 28

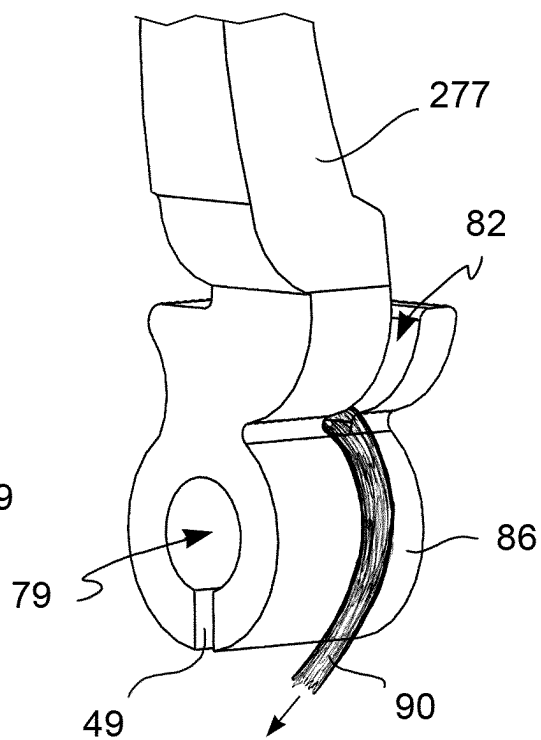


FIG. 29

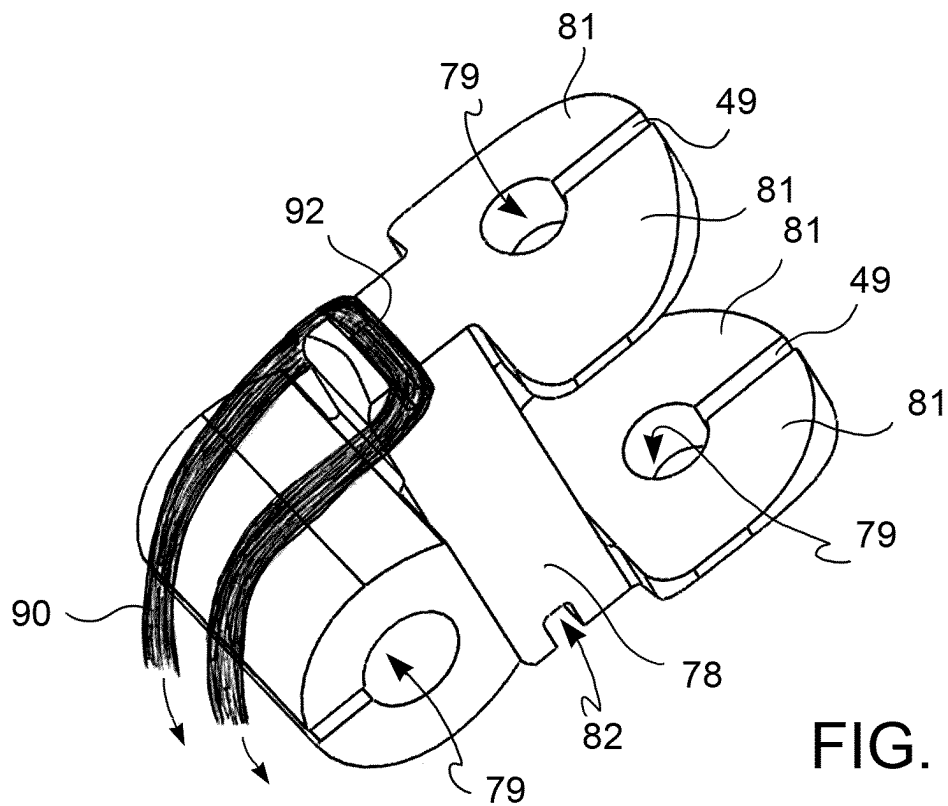
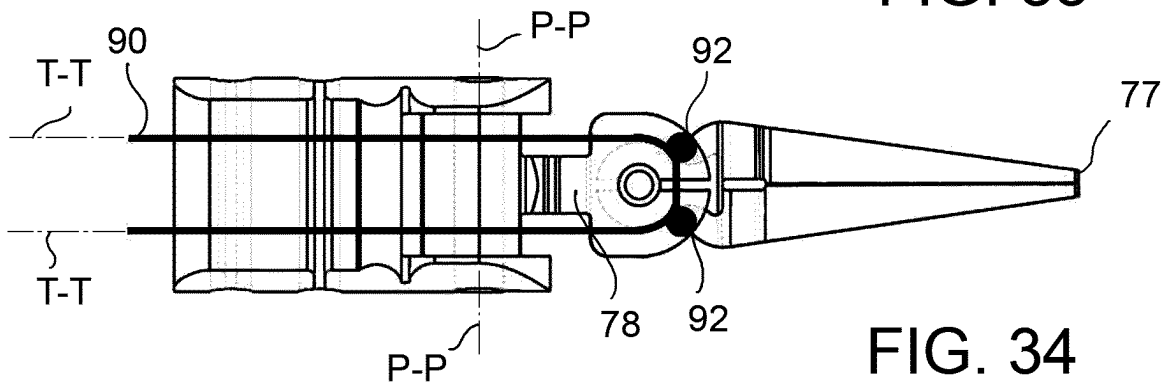
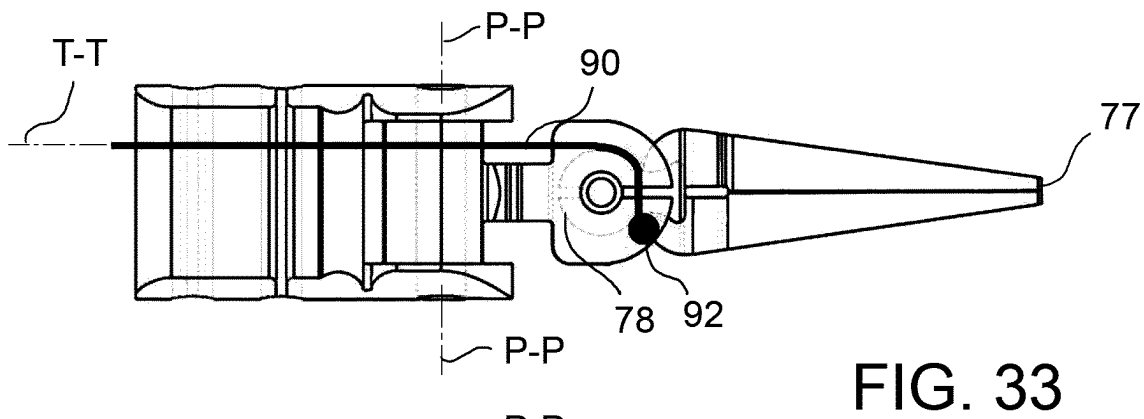
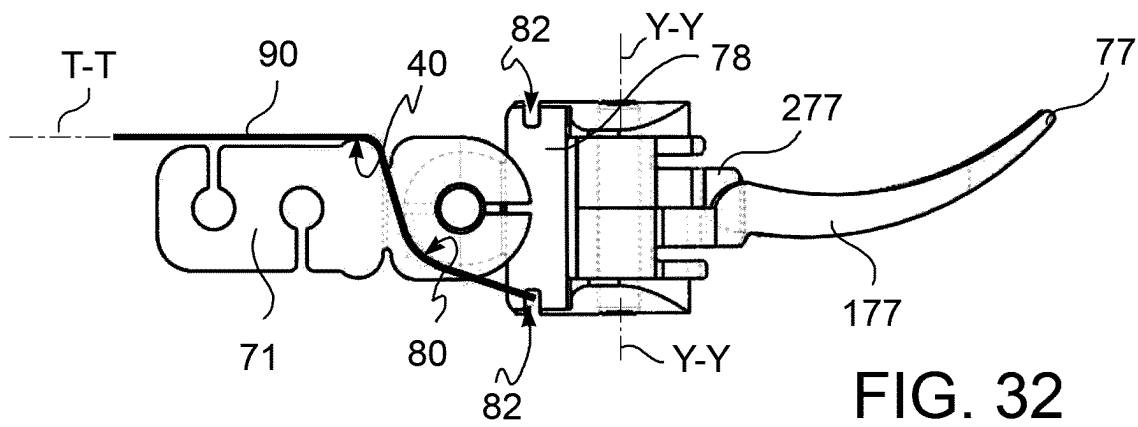
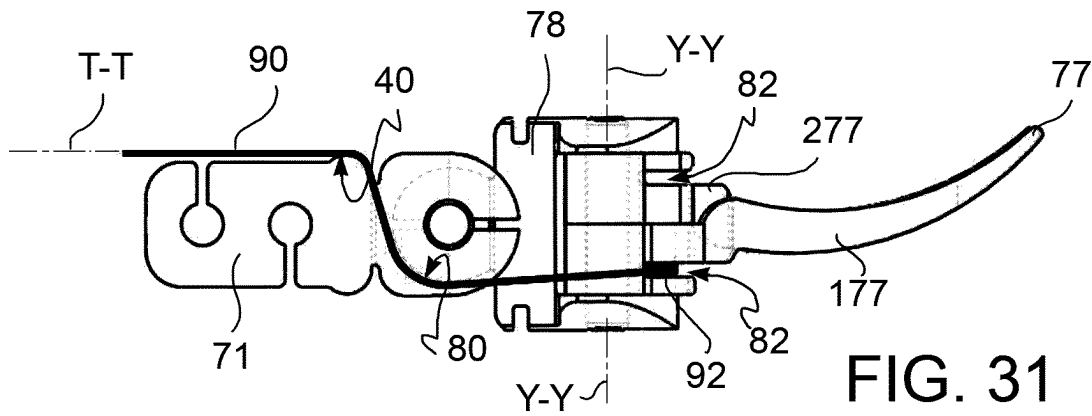


FIG. 30



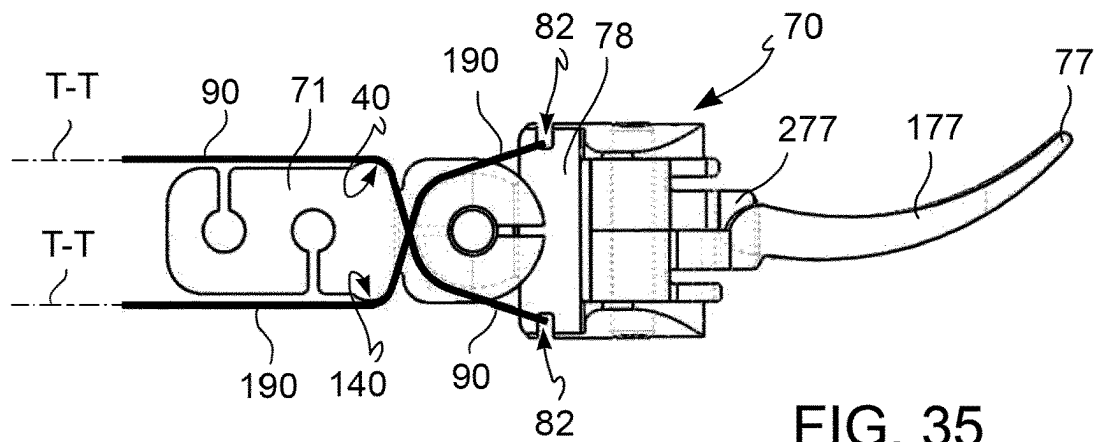


FIG. 35

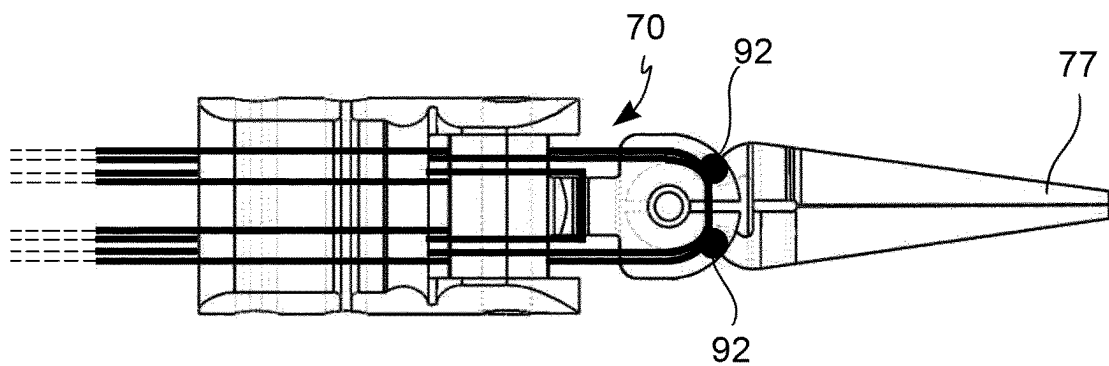


FIG. 36

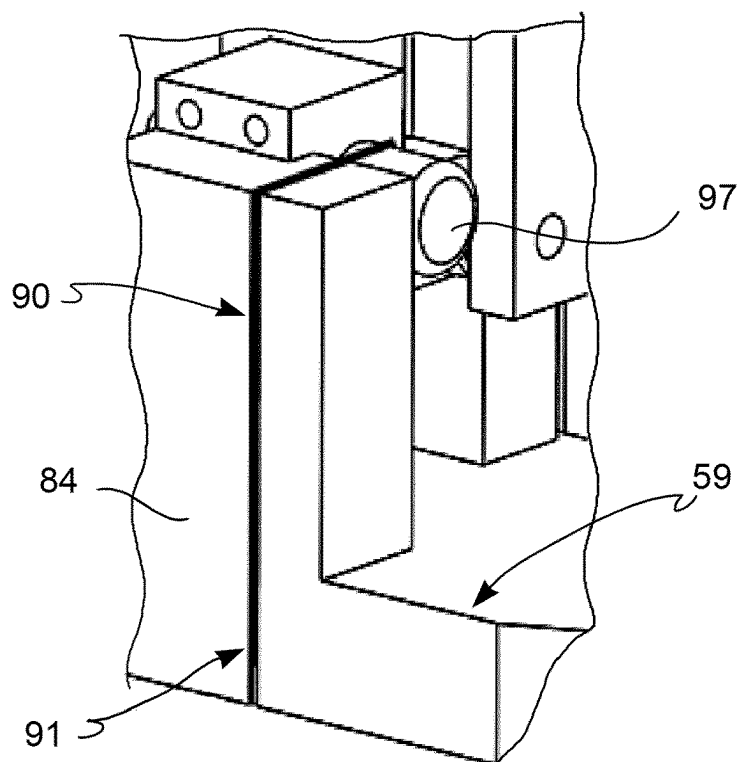


FIG. 13B

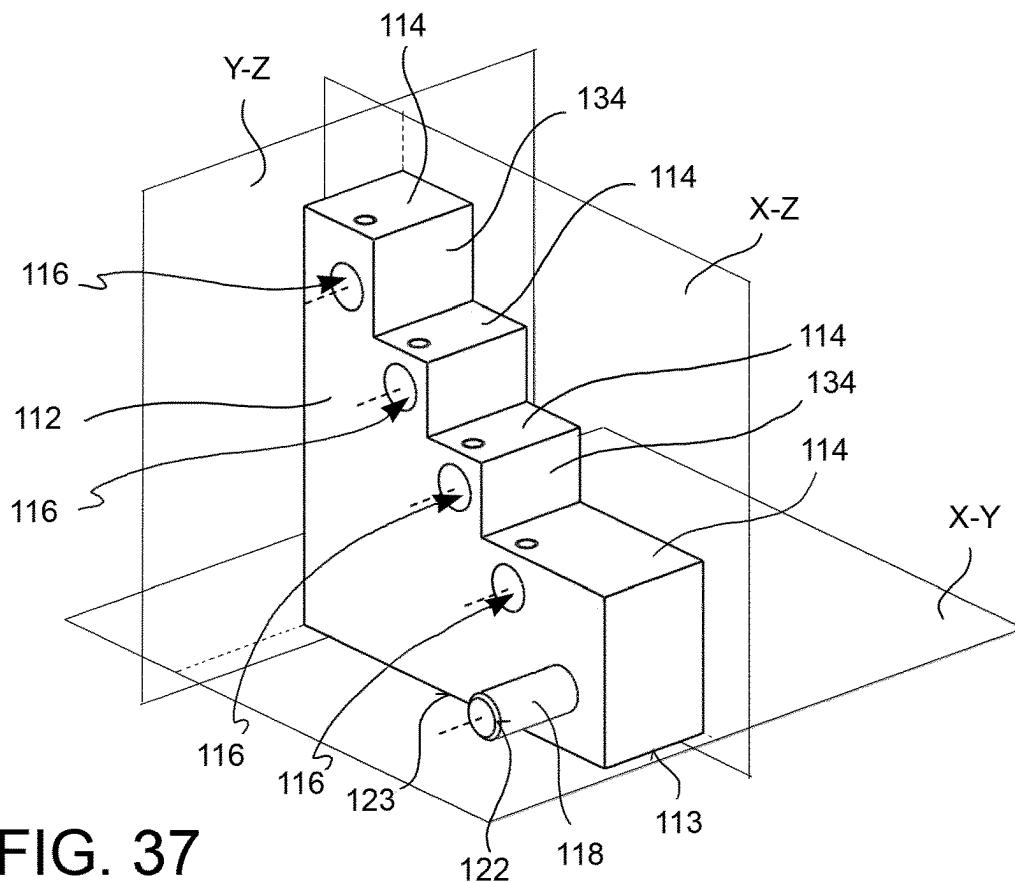


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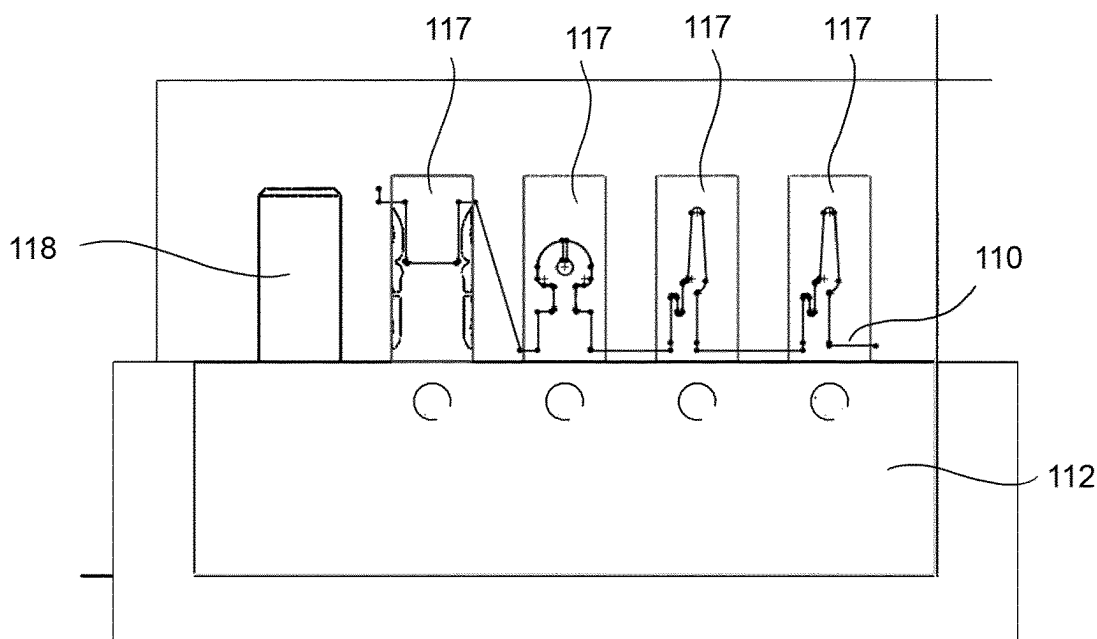


FIG. 38

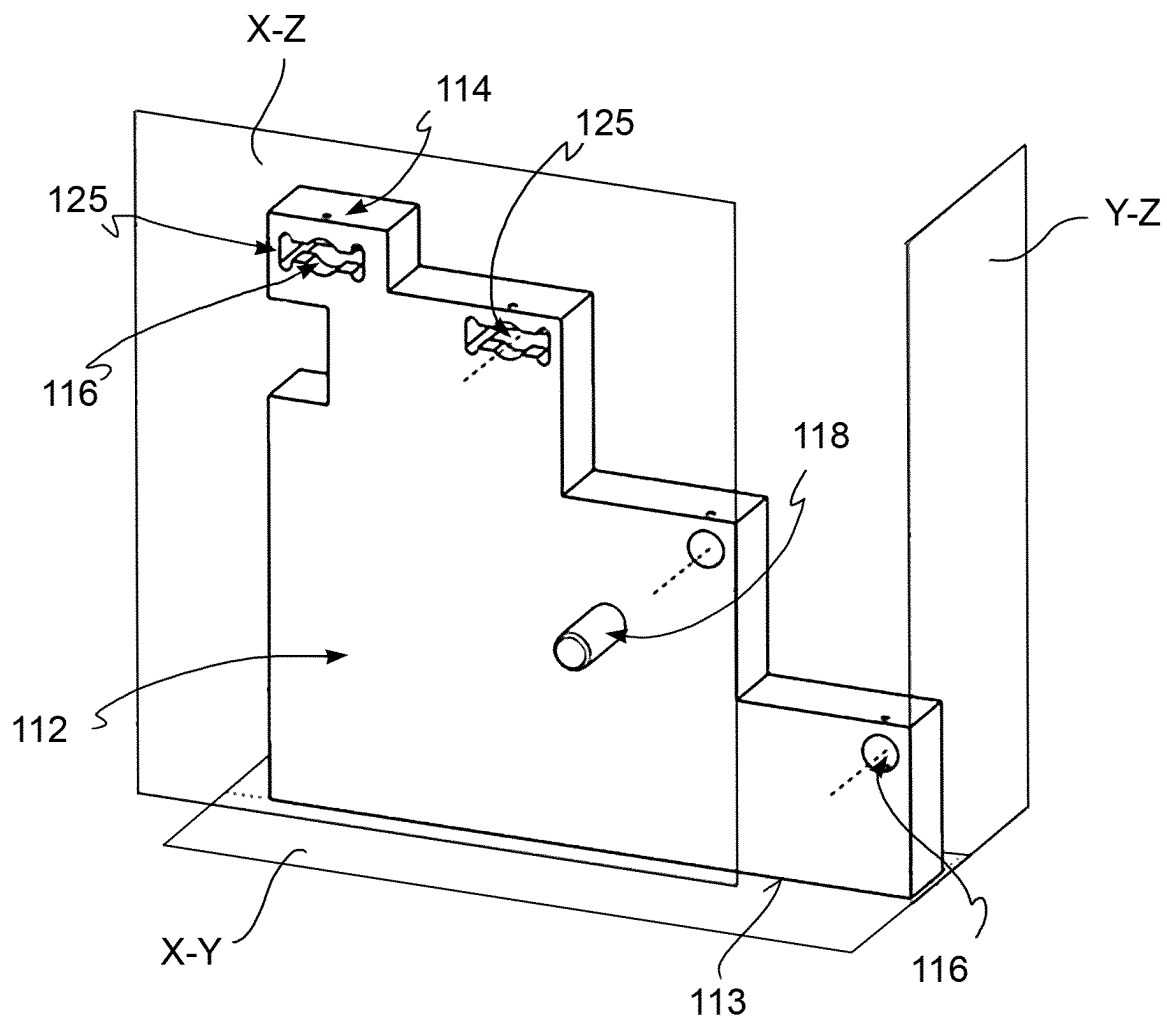


FIG. 40B

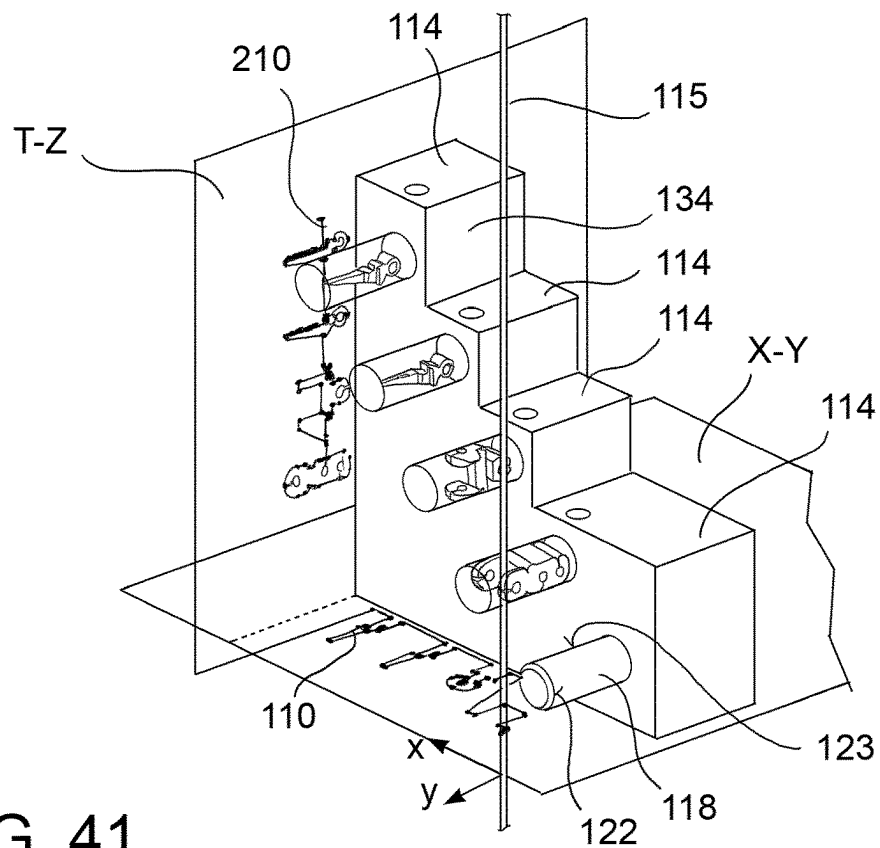


FIG. 41

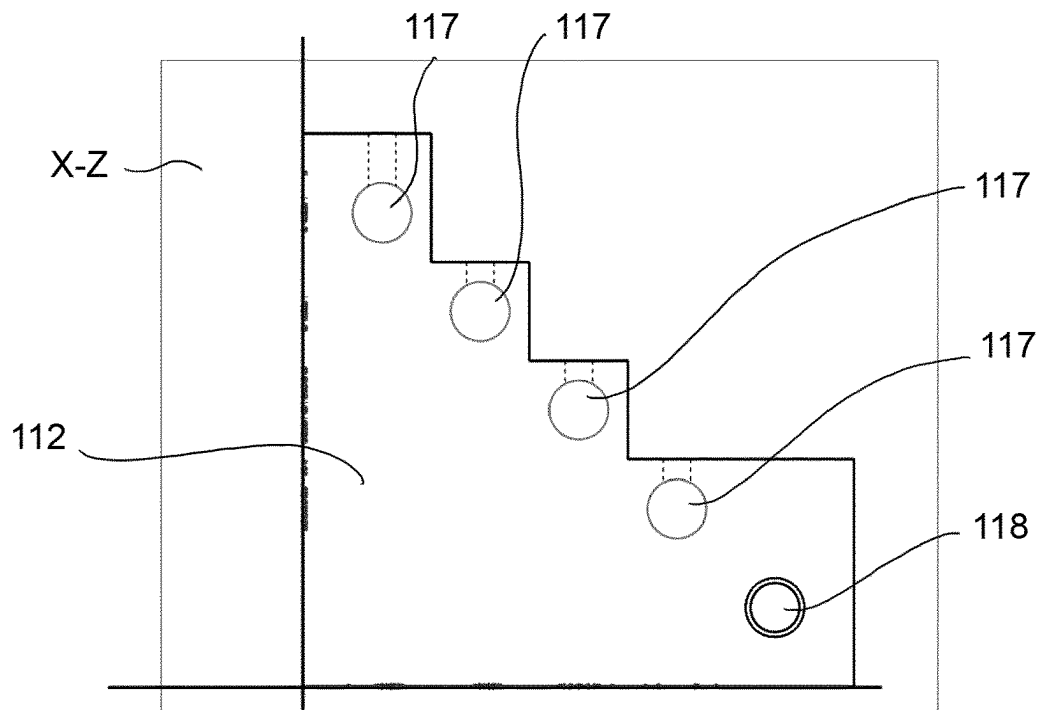


FIG. 42

SURGICAL TOOL FOR ROBOTIC SURGERY AND ROBOTIC SURGICAL ASSEMBLY

This application is a Continuation of U.S. patent application Ser. No. 15/768,525, filed 13 Apr. 2018, which is a National Stage Application of PCT/EP2016/074808, filed 14 Oct. 2016, which claims benefit of Serial No. 102015000062500, filed 16 Oct. 2015 in Italy and which applications are incorporated herein by reference. To the extent appropriate, a claim of priority is made to each of the above-disclosed applications.

FIELD OF INVENTION

The present invention relates to a medical instrument.

In particular, the present invention relates to a medical instrument specifically suitable for acting as a robotic end effector for robotic microsurgery.

In addition, the present invention relates to a method of manufacturing of said medical instrument.

Moreover, the present invention relates to a robotic surgical assembly comprising said medical instrument.

Further, the present invention relates to a tendon driving system for a medical instrument and to a medical instrument comprising said tendon driving system.

STATE-OF-THE-ART

Robotic assemblies for surgery or microsurgery comprising multi-joint robotic arms terminating with surgical instruments are known in the field. For instance, document U.S. Pat. No. 7,155,316-B2 discloses a robotic assembly for performing brain microsurgery under MRI (Magnetic Resonance Imaging) guidance comprising an MRI-based image acquisition system and two multi-joint arms, each with three rotary joints with vertical axes to avoid direct gravity loads (as shown for instance in FIG. 7 of said document U.S. Pat. No. 7,155,316-B2), each connected to its respective end-effector endowed with an internal degree of freedom of motion for gripping.

Solutions available in the state-of-the-art, although offering partial advantages, require a motion strategy that simultaneously involves a plurality of independent movements even for small motions of the surgical instrument in the operating work-field, which results both in a difficult control of the kinematic accuracy and in a large encumbrance in the operating work-field, that in practice becomes inaccessible to the Surgeon. As a matter of fact, the application field of the majority of robotic assemblies for surgery that are based on the master-slave paradigm are dedicated to use in minimally invasive surgery (or MIS), such as laparoscopic or endoscopic surgery. In both such applications, the kinematics of the robotic assembly is aimed to optimize the access of the surgical instruments to the operating field through the surgical ports or orifices, a feat that requires the coordination of a plurality of degrees of freedom of movement. In contrast, surgical, and microsurgical, applications in open surgery require an accurate kinematic control of translational movements, over a workspace limited by the field of view of the operating microscope, without the limiting kinematic constraints represented by the surgical ports or natural orifices, and thus benefit hugely from the surgeon's ability to directly access the operating field.

It is also notable that the execution of the principal surgical primitives, such as tissue tensioning and anastomotic suturing, requires the ability to orient the surgical instrument tip in a large spatial cone of directions and to

rotate the instrument around its longitudinal axis (roll), for example to guide the needle through the tissue with the tip of the needle holder instrument, in a similar manner as the human hand is jointed at the wrist and the elbow.

Robotic assemblies for surgery or microsurgery comprising a teleoperated master-slave system are generally known, as described, for example, in document U.S. Pat. No. 6,963,792-A and, more specifically for the microsurgical application by U.S. Pat. No. 6,385,509-B2, and US-2014-0135794-A1, that describe kinematic solutions for the movement of the surgical instrument tip that require coordination of a plurality of joints in a serial kinematic chain that clutter the operating field. Such encumbrance effect is increasingly pronounced as the joints articulating the tip of the instrument are further away from the tip itself. Moreover said micro-surgical systems do not allow adequate movement, and more specifically adequate reorientation, of the instrument tip when in an operating site inside a lesion as little as 10 centimeters from the surface of the skin.

Generally, even a specialized operator requires long training to acquire mastery of the master command devices adopted in known master-slave systems. In fact, known master devices have a long learning curve, primarily because they are mechanically linked to motion recording stations, which necessarily limit their movement in an unfamiliar way and often are of large dimensions. Hence, known master devices are intrinsically unfit to replicate the function of traditional open surgery instruments and lack the ability to carry out a large spectrum of linear as well as angular movements in three dimensional space.

For example, document U.S. Pat. No. 8,521,331-B2 discloses a robotic device for laparoscopic surgery, where the master command device has a shape that allows the surgeon to wear it as a glove on his-her fingers. According to another embodiment shown in FIG. 2B of said patent, the master command device has a joystick shape, attached on one part to the surgeon's wrist and extends so that it is held with just one hand, comprising a cylindrical stem having a pair of lateral wings that can register the grip movement. A surgeon makes use of a laparoscopic display device integral to said command device.

The above solution, although partly advantageous for laparoscopic surgery, does not entirely solve the issue, making long training still necessary for the surgeon before becoming proficient at handling said command devices instead of the familiar open surgery instruments.

As is well known, the practice of microsurgery requires the use of either an optical microscope or magnifying loupes, demands an high level of dexterity and experience of the surgeon, who works at the limits of physiological tremor and the accuracy that human hand motions can reach at such dimensional scale.

The adoption of robotic technologies can bring about great benefits, allowing both a high degree of miniaturization of the instruments and scaling the size of the movements in the operating field, hence eliminating the effect of physiological tremor and easing the manual task. For example, microsurgical procedures are carried out in several phases of the reconstruction of biological tissues, such as for example in the execution of blood vessel anastomosis, comprising small diameter vessels, and nerves. Such procedures are carried out to reconstruct anatomy after the occurrence of traumatic lesions or of lesions produced by surgical removal of tissue, to reattach limbs and to revascularize tissues, all performed in an open surgery set-up given the pre-existence of a superficial lesion.

Other examples of application of microsurgical techniques are found in transplant surgery, neurosurgery or in vascular surgery, as well as in surgery around and inside the eye, and in the inner ear, as in the case of cochlear implants. Also the prominent surgical procedure of cardiac by-pass comprises the critical step of anastomosis of the coronary arteries. The need for instrument miniaturization is also felt in other surgical techniques, for example in minimal invasive surgery, such as laparoscopy and endoscopy, that are aimed at limiting the invasiveness of surgical instruments on biological tissue. With reference to laparoscopy, the technical solutions known in the art do not allow a satisfactory miniaturization of the diameter of the laparoscopic instruments employed in Single Incision Laparoscopic Surgery or Single Port Surgery. Moreover, it is worth noticing that the endoscopes typically employed in MIS have an instrument channel with a diameter between 1 mm and 3.2 mm. Such dimensions limit the functionality of current surgical instrumentation available through the endoscope instrument channel, which at present is typically just capable of gripping action.

Medical instruments comprising a jointed device suitable to work on the patient, are generally known in the art. For example, document WO-2010-009221-A2 shows a robotic surgical instrument comprising a distally jointed device, capable of providing three degrees of freedom of motion, respectively pitch, yaw and grip, employing just four actuation cables. Such cables slide inside guiding channels, or sheaths, present inside the body of the articulating device.

Said technical solution limits the miniaturization of the robotic articulating device, because friction between the guiding channels surfaces and the cables that slide inside them limits the positioning precision achievable by the articulating device. As it is known in the art, as the physical dimensions of a medical instruments are reduced, difficulties arise which are related to the increase of relevance of superficial forces, such as friction, that become dominant over volume forces. Such a phenomenon requires to resort to solutions that minimize friction forces, and at the same time reduce lost motions of mechanics to a minimum. The loss of positioning precision of an articulating device is a fundamental technological obstacle to further miniaturization of articulating instrument, since, with miniaturization, also the stiffness of the driving members (tendons) goes down with the second power of their diameter, making it even more difficult to overcome friction for the precise positioning of the instrument tip. Moreover, such a solution requires a tendon guiding system comprising channels and guiding surfaces that surround the cables that make the pitch and yaw links, as well as the instrument shaft, very difficult to miniaturize using known fabrication methods, such as for example injection molding and machining, and would be prone to have several locations of mechanical weakness.

In order to simplify the miniaturization of a surgical instrument, the said document WO-2010-009221-A2 indicates the advantageous opportunity of reducing the number of actuation tendon terminations, associated to three degrees of freedom, from six to four, exploiting for actuation the torque that cables terminated on the yaw link apply on the pitch link (see FIG. 4-A of cited document) and requires to such purpose to pull and release selectively such cables, thanks to a kinematic mechanism comprising a number of gears. Moreover, the driving system described requires that each end of an actuation tendon is attached to a winch, that selectively winds the tendon inducing the pull. The presence of mechanical aspects such as said winch and said teeth, which are notoriously subject to lost motion, creates a

difficult to drive a miniature articulation, because lost motion in the drive system is translated into an angular play at the joint, that increase as the articulating device gets smaller. Said driving system is also unsuited to keep a low preload on the actuation cables to further limit friction and wear.

Moreover, the solutions described for tendon termination comprise tortuous paths meant to trap the tendon in some sections. Such solutions require the use of cables that are sufficiently resistant to survive such trapping, such as steel cables or cables with larger diameter than otherwise required.

Further examples of actuation cables for surgical instruments suited to slide, when pulled or pushed, inside sheaths or guiding channels, for example obtained on the lateral surfaces of pulleys, are disclosed in documents U.S. Pat. No. 6,371,952-B1, U.S. Pat. No. 6,394,998-B1 and WO-2010-005657-A2. Specifically, the latter document discloses a solution where actuation cables follow trajectories that cross as they go around pulleys that comprise guiding channels to avoid that such cables interfere with one another, a condition that limits their efficacy in transmitting motion to the articulating device, such as for instance in case of bundling up or sliding of one tendon onto another one. The provision of idle pulleys, necessarily with a diameter close to half of the instrument diameter (as shown in FIG. 4 of cited document WO-2010-005657-A2) and attached to the links, for example to links integral with the instrument shaft, or to the pitch link, to guide the tendon to cross, is a considerable obstacle to miniaturization. Moreover the provision of grooves and walls to realize the channels for the actuation cables is a further obstacle to the miniaturization of the shaft or cannula diameter of a medical, or surgical, instrument.

The document US-2003-0034748-A discloses a solution suitable for reducing the diameter of the surgical instrument to 5.1 mm. This instrument foresees the use of a series of disks that function as vertebra, providing some flexibility.

Nevertheless, this solution is not appropriate for achieving a compact joint that can extend for approximately one instrument diameter, or in other words, has a radius of curvature similar to its diameter. This is instead achievable by those articulations described in the documents cited above which are based on a pivot-type joint, comprised of a pure axis of rotation.

A further obstacle to the miniaturization of jointed or articulated devices is the challenge of fabricating and assembling three dimensional micromechanical parts with sufficient precision at a reasonable process cost. The need to develop relatively high forces at the tip in devices with a sub-millimeter size suggests the use extremely rigid metals for such components, such as for example tool steel.

It is known that biomedical devices are generally fabricated using fabrication techniques derived from the micro-electronic industry. For example, laser or water-jet cutting is not appropriate for fast machining in three dimensions. Injection molding does not currently produce sufficiently high tolerance parts. In contrast, electrical discharge machining (EDM) are capable of producing satisfactory performance both in terms of surface finishing and with respect to the geometric tolerance required by the mechanical designs. EDM generally entails a slow and expensive process. For example, the document U.S. Pat. No. 6,768, 076-B2 discloses a fixture for EDM able to support pieces to be cut in a single plane.

Nevertheless, the fixture is not suitable for repeated placement of the piece, for example it is not possible to rotate the fixture while it is being machined in a way that the

EDM can work in multiple cutting planes, resulting in a laborious fabrication process that necessitates complex operations of recalibration every time a cut is carried out in a different plane. This results in a loss of precision and hence less precise dimensional and geometric tolerances.

There is a felt need for a surgical robotic assembly able to carry out precise motions and simply control a wristed medical instrument within the surgical workspace, for example an anatomical district of a patient. At the same time, there is a need to develop a reliable robotic assembly characterized by a simple driving method without compromising its precision. Furthermore, there is a need for a robotic assembly that is more versatile than known assemblies and is able to carry out a wider variety of surgical procedures.

Hence there is a felt need to provide a driver device for microsurgery, suitable to form a master interface in a robotic assembly for microsurgery that comprises a master-slave type teleoperation system which is simpler and more intuitive to manipulate for the microsurgeon than known solutions, without limiting its functionality. Equally, there is a felt need to provide a master interface, which can be mastered more quickly and easily by the surgeon. Furthermore, there is a felt need to provide a command device that is more versatile than the known solutions and can be applied to different types of microsurgical procedures.

Hence there is a felt need to provide a jointed or articulated medical instrument, or an assembly comprising an jointed or articulated device, which is structurally and functionally suitable for extreme miniaturization without compromising its reliability and safety. There is also a felt need to provide a jointed or articulated medical instrument, or an assembly comprising a jointed device, suitable for carrying out a wide variety of medical-surgical therapies. Finally, there is a felt need to provide a jointed or articulated medical instrument, or an assembly comprising an jointed or articulated device, that is durable and able to undergo periodic maintenance without compromising its sterility or reliability.

There is a felt need to provide a jointed or articulated medical instrument, or an assembly comprising a jointed device, that requires simplified manufacturing compared to known solutions.

There is a felt need to provide a fabrication method of said medical instrument that is more efficient with respect to known solutions and that guarantees the required level of precision for the assembly.

There is a felt need to provide a manufacturing method for a medical instrument that guarantees a faster machining process, without compromising precision in production.

Furthermore, there is a felt need to provide a method of fabrication suitable for producing parts subject to extreme miniaturization without reducing neither the precision of detailed manufacturing nor the ease of assembly of the parts produced.

Hence, there is a felt need to provide a driver device based on tendons, or actuation cables, for a medical instrument suitable to be subject to extreme miniaturization, without compromising its precision or reliability in use.

Furthermore, there is a felt need to provide a driver device based on tendons, or actuation cables, for a medical instrument that guarantees said tendons, a predetermined preload, even if light, and in which the force of the preload can be independently defined for each of said tendons.

Furthermore, there is a felt need to provide a drive system based on tendons, or actuation cables, for a medical instrument that can guarantee an adequate level of sterility to the

medical instrument itself, particularly to the portion of the medical instrument that is meant to come into contact with the patient anatomy.

Furthermore, there is a felt need to provide a drive system based on tendons without backlash.

Furthermore, there is a felt need to provide a drive system based on tendons that can replicate the drive precision achieved by precision micrometric slitters or piezoelectric drive systems, for example.

Hence, there is a felt need to provide a tendon, or actuation cable, for a medical instrument with characteristics that render it suitable for extreme miniaturization without compromising its resistance or reliability in use. Furthermore, there is a felt need to provide a tendon for a medical instrument that is suitable for gliding over at least one portion of said instrument with improved performance in terms of friction with respect to known solutions. Furthermore, there is a felt need to provide a tendon for a medical instrument exclusively meant to work under tensile load applied at its endpoints, without comprising solutions that might result in deflecting the path of the tendon, that would diminish its resistance. Furthermore, there is a felt need to provide a tendon for a medical instrument, as well as a method for replacing the tendon, which is suitable for increasing the lifespan of the medical instrument, with respect to known solutions, without compromising its performance in terms of sterility and reliability.

The need is felt to miniaturize medical instruments.

The need is felt to reduce the known dimensions of medical instruments.

For example, document WO-2014-151952-A1 shows a medical instrument comprising a plurality of links forming a joined device, said medical instrument having actuation cables wrapping around a plurality of pulleys rotatably supported on shaft provided cantilevered on the links of the medical instrument. This solution is characterized by a high number of parts, and the layout of said pulleys provided on said shafts forces to machining said shaft to resist to the stresses arising from the use of the medical instrument, therefore this solution results unsuitable for miniaturization, in fact this device could not measure less than 10 millimeters in diameter. Similar solutions are shown, for example, in documents U.S. Pat. No. 6,676,684-A, US-2009-0112230-A and US-2003-135204-A1.

It is therefore felt the need of reducing the number of parts which forms the medical instrument.

For example, document WO-03-001986-A1 shows a medical instrument comprising a plurality of disc-shaped links forming a joined device, wherein each of said links comprises a plurality of holes for guiding the actuation cables. Therefore, this solution is unsuitable for miniaturization as it is highly unsatisfactory performing micrometric holes in such links, and at the same time, it is unsatisfactory providing actuation cables which slide inside such holes without damaging.

It is therefore felt the need of obtaining an accurate guiding of the actuation cables without providing micrometric holes in the links and at the same time to reduce the number of parts which forms the medical instrument.

For example, document US-2008-177285-A1 discloses a medical instrument comprising a plurality of links, wherein some links comprise two protruding pins suitable to guide the deflection of the actuation cables. Although satisfactory under some points of views, such solution is also unsuitable for miniaturizing, as the protruding pins dimension cannot be reduced without compromise the integrity of the links composing the medical instrument. Therefore, the need is

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felt to provide a miniaturized medical instrument, having a plurality of links actuated by means of actuation cables, without compromise the structural resistance, and thus the safety when in use, of the medical instrument.

Solution

One of the goals of the invention described here is to overcome the limitations of known solutions described above and to provide a solution to the needs mentioned with reference to the state of the art.

FIGURES

Further characteristics and advantages of the invention will appear from the description reported below of preferred embodiments, which are given as examples and are not meant to be limiting, which makes reference to the attached Figures, in which:

the FIG. 1A is a perspective view, which shows the surgical robotic assembly according to one aspect of the invention;

the FIG. 1B is a perspective view, which shows the surgical robotic assembly according to one aspect of the invention;

the FIG. 1C is a perspective view, which shows a surgical robotic assembly according to one aspect of the invention;

the FIG. 2A is a perspective view, which shows a surgical robotic assembly according to one aspect of the invention associated with other elements of the operating room;

the FIG. 2B is a frontal view, which shows a surgical robotic assembly according to one aspect of the invention associated with other elements of the operating room;

the FIG. 3 is a perspective view, which shows a portion of a couple of jointed or articulated devices according to one aspect of the invention.

the FIG. 4A is a view from above, which shows a portion of a surgical robotic assembly according to one aspect of the invention associated with other elements of the operating room and a patient.

the FIG. 4B is a top view, which shows a portion of a surgical robotic assembly according to one aspect of the invention associated with other elements of the operating room and a patient.

the FIG. 5 is a perspective view, which shows a portion of a surgical robotic assembly according to one aspect of the invention associated with other elements of the operating room and a patient.

the FIG. 6 is a perspective view, which shows a portion of a surgical robotic assembly according to one aspect of the invention associated with other elements of the operating room, the surgeon and a patient.

the FIG. 7 is a perspective view, which shows a control device according to one aspect of the invention.

the FIG. 8 is a perspective view, which shows a macro-positioning arm according to one aspect of the invention.

the FIG. 9A is a perspective view, which shows a portion of a robotic assembly according to one aspect of the invention.

the FIG. 9B is a perspective view, which shows a portion of a robotic assembly according to one aspect of the invention.

the FIG. 9C is a perspective view, which shows a portion of a robotic assembly according to one aspect of the invention, associated with a microscope.

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the FIG. 9D is a perspective view, which shows a portion of a robotic assembly according to one aspect of the invention, associated with an endoscope.

the FIG. 9E is an enlarged view of the detail indicated with the arrow E-E of FIG. 9C.

the FIG. 10 is a perspective view, which shows a portion of a robotic assembly according to one aspect of the invention.

the FIG. 11 is a perspective view, which shows a medical instrument according to one aspect of the invention.

the FIG. 12 is a perspective view, and a depiction of separate parts, which shows a medical instrument according to one aspect of the invention.

the FIGS. 13A and 13B show, in a perspective view, portions of a driving system according to one aspect of the invention.

the FIG. 14A is a schematic section view of a portion of a driving system according to one aspect of the invention.

the FIG. 14B is a schematic section view of a portion of a driving system according to one aspect of the invention.

The FIG. 15A is a perspective view that shows a medical instrument according to one aspect of the invention.

the FIG. 15B is a perspective view that shows a medical instrument according to one aspect of the invention.

the FIG. 15C is a sketch in perspective view that shows a medical instrument according to one aspect of the invention.

the FIG. 15D is a sketch in perspective view that shows a medical instrument according to one aspect of the invention.

the FIG. 16 is a schematic drawing, viewed from top and with partially transparent parts, which shows a tendon path of two tendons according to one aspect of the invention.

the FIG. 17 is a perspective view of an articulated device according to an aspect of the invention.

the FIGS. 18 to 20 are perspective views with isolated parts of some embodiments of an articulated device, according to several aspects of the invention.

the FIG. 21 shows a profile of an articulated device according to one aspect of the invention.

the FIGS. 22 to 24 show several poses of some embodiments of an articulated device according to some aspects of the invention.

the FIGS. 25 to 27 shows several embodiments of a terminal tool according to some aspects of the invention.

the FIG. 28 shows a perspective view of a detail of a tendon according to one aspect of the invention.

the FIG. 29 shows a perspective view of a detail of a tendon according to one aspect of the invention.

the FIG. 30 shows a perspective view of a detail of a tendon according to one aspect of the invention.

the FIGS. 31 to 36 are schematics, which show a path of the tendon according to some aspects of the invention.

the FIG. 37 is a schematic in perspective view, which shows a machining fixture according to one aspect of the invention.

the FIG. 38 is a schematic, which shows the profile of a machining cut according to one aspect of the invention.

the FIG. 39 is a schematic in perspective, which shows a phase of a fabrication method according to one aspect of the invention.

the FIG. 40A is a planar view, which shows a detail of a machining fixture according to one aspect of the invention.

the FIG. 40B is a perspective view, which shows a detail of a machining fixture according to one aspect of the invention.

the FIG. 41 is a schematic in perspective view, which shows a phase of a fabrication method according to one aspect of the invention.

the FIG. 42 is a frontal view of a tool according to one aspect of the invention.

DESCRIPTION OF PREFERRED EMBODIMENTS

According to an embodiment, the term “tendon”, or “actuation cable”, refers to an element which presents a prevalently longitudinal extension and is suitable to work under tensile loads applied at its endpoints. According to an embodiment, the term “opposite tendon” or “opposite actuation cable” refers to a further tendon suitable to work in an antagonistic way with respect to said tendon. According to an embodiment, in the attached figures, said tendon will generally be indicated by the numeric reference “90” and said opposite tendon will be indicated by the numeric reference increased by one hundred, that is “190”. Nonetheless, in figures in which distinguishing between said tendon and said opposite tendon is irrelevant, said tendon and said opposite tendon will both be indicated by the numeric reference 90. According to an embodiment, the concept of “opposite” extends itself to multiple elements and/or parts of elements, such as referred to for said “tendon” above. According to an embodiment, the tendons comprised in a first pair of tendons will be indicated with references “90, 190”, and the tendons belonging to a second pair of tendons will be indicated with the references “191, 192”.

According to an embodiment, the terms “master-slave”, “master” and “slave” refer to the known system of teleoperation.

According to an embodiment, the term “terminal tool” refers to a portion suitable to perform an assigned task, such as for example form the interface with at least on portion of the patient. For example, in a teleoperation system of the master-slave type, said terminal tool, or terminal portion, or terminal member, is at least one portion of an “end-effector”.

According to an embodiment, the term “jointed or articulated device” refers to a wrist joint, an elbow joint or a shoulder joint of a robotic or mechatronic structure, in other words, an interconnected assembly of members and articulations suitable to support and/or orient and/or position and/or influence the position of said terminal tool.

According to an embodiment, the members of a jointed or articulated device will be indicated by the progressive annotation “first member”, “second member”, and so on, to indicate their position within the kinematic chain, in which the “first member” indicates the most proximal member; in other words “first member” indicates the member furthest from the terminal organ. According to an embodiment, the members of the jointed device will be indicated with the terms “wrist member”, “elbow member” or “terminal member” to indicate the function exercised by said members. For example, the same member could be simultaneously a “second member” and a “wrist member”.

According to an embodiment, the term “work volume”, or “work space”, or “work field”, or “workspace volume” refers to the set of Cartesian poses accessible to the terminal portion of a jointed or articulated device. According to an embodiment, said volume is of a substantially parallelepiped form. According to an embodiment, said work volume is of a substantially cylindrical form.

According to an embodiment, the term “macro-positioning” refers to an initial operation of positioning of at least

one portion of the medical instrument from any position to a work position within or adjacent to the operating field; in other words, “macro-positioning” refers to the operation of making the work volume coincide with the operating field.

According to an embodiment, the term “micro-positioning” refers to an operation of positioning at least one portion of a medical instrument in a finer manner than the “macro-positioning”. According to an embodiment the micro-positioning takes place in a more limited space, in real time and under the direct control of the control device (master).

According to an embodiment, the prefix “micro-” before a certain object indicates that said object is primarily, but not exclusively, meant to operate on a sub-millimeter scale.

According to an embodiment, the term “rotational joint” refers to a junction between two elements suitable to permit a relative moment of rotation between said two elements around an axis of joint movement.

According to an embodiment, the term “medical instrument” refers to an instrument suitable to be used during at least one phase of a medical surgical and/or cosmetic therapy. According to an embodiment, the term “surgical instrument” refers to a medical instrument specifically suited to be generally used in at least one phase of a surgical therapy. According to an embodiment, the term “microsurgical instrument” or “surgical micro-instrument” refers to a medical instrument specifically suited to be used in at least one phase of a microsurgical therapy.

According to an embodiment, the term “frame” refers to a portion of a medical instrument primarily suited to have a structural holding function. According to an embodiment, the “frame” can comprise at least one shaft, that is a long rigid or flexible element that presents a primarily longitudinal extension. According to an embodiment, said shaft, for example can be of a hollow and/or tubular form.

According to an embodiment, the term “ruled surface” refers to a surface achieved by the union of multiple straight lines. According to an embodiment, if not otherwise explicitly stated, the term “ruled surface” refers to a surface achieved by the union of multiple straight lines substantially parallel to each other, or in other words, a ruled surface of substantially parallel generatrices.

Below, when reference is made to a device, or an assembly, or a method, for microsurgery, it is meant a device, assembly or method, suitable to be applied in microsurgery, i.e. with the simultaneous use of means of optical enlargement such as loupes or microscopes, but also suitable for applications in other surgical therapies, such as general surgery, laparoscopic surgery or endoscopic surgery.

According to an embodiment, to not burden the text or figures, when reference is made to a “first” or “second” element (for example a “first micro-positioning device” and a “second micro-positioning device”), they will be indicated with the same numeric reference, as long as they are functionally indistinguishable (for example “41” above); sometimes, due to a need for clarity, the numerical reference will be specified incremented by one hundred (for example “141” above and “241”); hence, for example, the numerical reference “41” will indicate both said “first micro-positioning device” and said “second micro-positioning device”, as well as a “third” micro-positioning device. While when the specific reference, for example “141”, is used, it will refer to the specific element, in this case the “first micro-positioning device”. Analogously, to not burden the text excessively, the numeric reference relating to an “opposite” element will be omitted, if an element is functionally indistinguishable from its opposite.

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According to a general embodiment, a medical instrument 60, 160, 260, 360 for surgery comprises at least one frame 57 and at least one jointed device 70, 170, 270.

Said jointed device 70, 170, 270 comprising:

- at least one first joint member 71, or first link 71, adapted to connect to at least one portion of said frame 57;
- at least one second joint member 72, or second link 72.

Said first joint member 71 is connected by means of a rotational joint 171 to said second joint member 72.

Said medical instrument 60, 160, 260, 360 further comprising at least a pair of tendons 90, 190, adapted to move said second joint member 72 with respect to said first joint member 71, pulling it. Said tendons acts as actuation cables suitable for working only in traction.

Each of said first joint member 71 and said second joint member 72 comprises a main structural body comprising in a single piece one or more convex contact surfaces 40, 80, 86, 140, 180,

Each of said convex contact surfaces 40, 80, 86, 140, 180 is a ruled surface formed by a plurality of straight line portions all parallel to each other and substantially parallel to a joint movement axis P-P, Y-Y.

According to an embodiment, all said convex contact surfaces 40, 80, 86, 140, 180 defining with their prolongations thereof at least partially a single convex volume. In other words, the prolongations of said convex contact surfaces 40, 80, 86, 140, 180 of a single main structural body define, together with said contact surfaces 40, 80, 86, 140, 180, a convex volume. According to an embodiment, the wording "convex volume" means that given a pair of points chosen inside said convex volume, the shorter straight conjunction between them is inside the convex volume in its entirety. This avoids providing grooves or channels on pulleys for guiding the tendons, allowing to miniaturize the dimensions of the main structural bodies and of the jointed device. According to an embodiment, all said convex contact surfaces 40, 80, 86, 140, 180 of said main structural body define with their prolongations thereof the convex hull of said main structural body.

According to an embodiment, wherein two tendons of said pair of tendons 90, 190 are parallel to each other and in contact with the same convex contact surface 40, 80, 86, 140, 180.

According to an embodiment, each tendon of said pair of tendons 90, 190 comprises: a first tendon termination 91, associated to said frame 57, a second tendon termination 92, secured to said second member 72, and a main portion, extending between said first tendon termination 91 and said second termination 92.

and wherein the main portion of each tendon of pair of tendons 90, 190 is in contact with said jointed device 70, 170, 270 only on said convex contact surfaces 40, 80, 86, 140, 180.

According to an embodiment, said contact surface 40, 80, 86, 140, 180 is a ruled surface formed by a plurality of straight line portions all parallel to each other and substantially parallel to a joint movement axis P-P, Y-Y of the rotational joint 171 closer to said contact surface 40, 80, 86, 140, 180.

According to an embodiment, said contact surface 40, 80, 86, 140, 180 is a sliding surface 40, 80, 140, 180 or a winding surface 86.

According to an embodiment, said sliding surface 40, 80, 140, 180 is a side sliding surface 40, 140, adapted to extend from said jointed device 70, 170, 270 so as to make at least one tendon portion slide in the air or slide off the contact with said jointed device 70, or said sliding surface 40, 80,

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140, 180 is a joint sliding surface 80, 180, which surrounds at least partially a joint movement axis.

According to an embodiment, on said joint sliding surface 80, 180 said two or more tendons 90, 190 overlap at least partially on a plane orthogonal to the direction of said joint movement axis of the closer rotational joint 171.

According to an embodiment, at least two tendons of said at least two pairs of tendons 90, 190, 191, 192 are in contact with a same convex contact surface 40, 80, 86, 140, 180 and are parallel to each other. According to an embodiment, the two tendons of a said pair of tendons are parallel to each other over a length of their tendon path in which they both are in contact with a said same convex contact surface 40, 80, 140, 180 of said first joint member 71 and said second joint member 72.

According to an embodiment, said pair of tendons connected with a same joint member is in contact with a same convex contact surface 40, 80, 86, 140, 180.

According to an embodiment, said main structural body is a rigid body.

According to an embodiment, said medical instrument 60, 160, 260, 360 comprises a further pair of tendons so as to comprise at least a third pair of tendons.

According to an embodiment, said convex contact surface 40, 80, 86, 140, 180 is a sliding surface 40, 80, 140, 180 on which a said tendon substantially slides. Alternatively, said convex contact surface 40, 80, 86, 140, 180 is winding surface 86 on which a said tendon substantially winds itself without sliding.

According to an embodiment, the projection of the tendon path T-T of a first tendon of a said two pairs of tendons 90, 190, 191, 192 and the projection of the tendon path T-T of a second tendon of same said pairs of tendons 90, 190 on a plane orthogonal to the direction of a said joint movement axis 171 crosses each other.

According to an embodiment, said medical instrument 60, 160, 260, 360 comprising a pair of tendons 90, 190 for each joint member 71 or 72 or 77.

According to an embodiment, said contact surface 40, 80, 86, 140, 180 defines at least partially the convex hull of the joint member 71 or 72 or 77 or 78;

According to an embodiment, each tendon 90, 190 defines a tendon path T-T which remains stationary with respect to the joint member 71 or 72 or 77 or 78 closer thereto.

According to a general embodiment, a medical instrument 60, 160, 260, 360 includes:

- at least a joint member 71, 72, 73, 74 of a jointed device 70,

- a frame 57 including a shaft 65

- a tendon 90, 190 suitable to move said joint member 71, 72, 73, 74 with respect to said frame 57

- a plunger 96 mobile along a degree of freedom with respect to said frame 57, in contact with said tendon 90, 190 and suitable to actuate said tendon 90, 190

- a pushing element 95 mobile along a linear trajectory and including an actuator

- a sterile barrier 87 suitable to substantially impede mutual bacteria contamination of the two environments it separates, placed between said pushing element 95 and said plunger 96,

wherein said plunger 96 is free to move away from said sterile barrier 87 and/or pushing element 95 and said pushing element 95 pushes on said sterile barrier 87 bringing it in contact with said plunger 96 and thus moves said plunger 96.

According to an embodiment, said pushing element 95 pushes on a plunger 96 in a pushing direction directed

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towards the inside of said frame 57, to move said plunger 96 along its degree of freedom with respect to said frame 57.

According to an embodiment, said pushing element 95 exchanges with said plunger 96 a force that is always directed in said pushing direction. In other words, said pushing element 95 is not suitable to exchange with said plunger 96 a pulling force, in other words said pushing element 95 cannot pull said plunger 96.

According to an embodiment, said pushing element 95 includes a lead screw and nut type actuator.

According to an embodiment, said actuator includes a ball screw.

According to an embodiment, said pushing element 95 includes a piston.

According to an embodiment, said plunger 96 has two portions one first portion of plunger 145 suitable to be in contact with said pushing element 95 and one second portion of plunger 146 to suitable to be in contact with said tendon 90, 190.

According to an embodiment, said first portion of plunger 145 is exposed from the frame 57 to be pushed by said pushing element 95.

According to an embodiment, said first portion of plunger 145 extends outside of frame 57 to be accessible by said pushing element 95.

According to an embodiment, said first portion of plunger 145 is flush with said frame 57 to be accessible by said pushing element 95.

According to an embodiment, said first portion of plunger 145 includes a pushing surface 147 suitable to be engaged with said pushing element 95.

According to an embodiment, said pushing elements 95 has a reciprocal pushing surface 148

According to an embodiment, said pushing element 95 pushes said plunger 96 transmitting a linear force through said a reciprocal pushing surface 148.

According to an embodiment, said pushing element 95 includes at least one pushing element idle pulley not represented, suitable to push on said pushing surface 147.

According to an embodiment, said pushing element 95 pushes said plunger 96 transmitting a linear force through said one pushing element idle pulley.

According to an embodiment, said pushing surface 147 and reciprocal pushing surface 148 are flat.

According to an embodiment, said pushing surface 147 and reciprocal pushing surface 148 are curved surface that mate with each other.

According to an embodiment, said pushing surface 147 and reciprocal pushing surface 148 are sliding surfaces that slide with respect to each other as said pushing element 95 moves along a linear trajectory.

According to an embodiment, said second portion of plunger 96 in contact with said tendon 90, 190.

According to an embodiment, said medical instrument 60, 160, 260, 360 includes at least one tensioning element 99, suitable for impose a preload on said tendon 90.

According to an embodiment, said tensioning element 99 is a spring.

According to an embodiment, said tensioning element 99 is suitable to apply a force between the frame 57 and the plunger 96, in the direction of moving said plunger 96 so as to impose a preload on said tendon 90.

According to an embodiment, said tensioning element 99 is suitable to apply a force between the frame 57 and the plunger 96, in the direction of moving said plunger 96 away from said pushing element 95.

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According to an embodiment, said tensioning element 99 is suitable to apply a force between the frame 57 and the plunger 96, in the direction of moving said plunger 96 towards the inside of said frame 57.

According to an embodiment, said preload is substantially proportional to the compression movement of said spring 99.

According to an embodiment, said second portion of plunger 146 pushes on at least one tendon deflectable portion 93 of said tendon 90.

According to an embodiment, said tendon deflectable portion 93 of said tendon 90 extends from a first guiding pulley 197 and second guiding pulley 297.

According to an embodiment, said second portion of plunger 146 moves in a space provided between said a first guiding pulley 197 and said second guiding pulley 297.

According to an embodiment, said plunger 96 changes the length of tendon 90 path between said a first guiding pulley 197 and said second guiding pulley 297 of an amount linearly proportional to the plunger 96 motion along said degree of freedom of plunger 96 with respect to said frame 57.

According to an embodiment, said second portion of plunger 146 includes at least one plunger idle pulley 98, suitable to push on said tendon deflectable portion 93,

According to an embodiment, said tendon 90 has a first tendon endpoint 91 fastened to said joint member 71, 72, 73, 74.

According to an embodiment, said tendon 90 has a second tendon endpoint 91 fastened to said frame 57.

According to an embodiment, said first tendon endpoint 91, is fastened to said second portion of plunger 146, instead than to said frame 57.

According to an embodiment, said frame 57 includes a upper frame portion 58 and a lower frame portion 59 the latter including a shaft 65.

According to an embodiment, said plunger 96 is mobile along a degree of freedom with respect to said frame 57.

According to an embodiment, said plunger 96 is jointed to said upper frame portion 58 with a linear joint.

According to an embodiment, said plunger 96 is jointed to said lower frame portion 58 with a rotational joint not represented.

According to an embodiment, said plunger 96 moves linearly along a degree of freedom with respect to said frame 57.

According to an embodiment, said plunger 96 is maintained in a proper alignment by means of linear bushings not represented inserted in the first frame section 58.

According to an embodiment, said plunger 96 is maintained in a proper alignment with upper frame 58 by means of respective shoulder surfaces 88.

According to an embodiment, said plungers 96 is a rocker that rotates around a pivot of said frame 57.

According to an embodiment, said sterile barrier 87 is of a form and material suitable to transmit the push of said pushing element 95 to said plunger 96.

According to an embodiment, said sterile barrier 87 is a flexible continuous layer of material.

According to an embodiment, said sterile barrier 87 lays in between said pushing element 95.

According to an embodiment, said sterile barrier 87 is trapped between said pushing surface 147 and said reciprocal pushing surface 148.

According to an embodiment, said medical instrument 60, 160, 260, 360 includes a plurality of tendons 90 and of pairs of plungers 96 and associated pushing element 95.

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According to an embodiment, said sterile barrier **87** is a flexible continuous layer of material.

According to an embodiment, said sterile barrier **87** is trapped between each plunger **96** and associated pushing element **95**.

According to an embodiment, said sterile barrier **87** is made of a stretchable material that stretches as said plungers **96** move with respect to said frame **57** exerting forces that do not substantially impede the motion of said plungers **96**.

According to an embodiment, said sterile barrier **87** is a drape.

According to an embodiment, said sterile barrier **87** is a loose fitting drape that stretches as said plungers **96** move with respect to said frame **57** exerting forces that do not substantially impede the motion of said plungers **96**.

Due to the provision of a pushing element **95** of a medical instrument **60, 160, 260, 360** according to one aspect of the invention, suitable to move a jointed device across a sterile barrier allows the production of a medical instrument, which is highly reliable and sterile.

Due to the provision of a plunger **96** of a medical instrument **60, 160, 260, 360** according to one aspect of the invention, it is possible to employ a simple sterile barrier in a shape of a drape or continuous flexible sheet of material.

Due to the provision of a plunger **96** of a medical instrument **60, 160, 260, 360** according to one aspect of the invention, it is possible to increase the precision of the commanded motion using pushing elements with high precision linear actuators.

Due to the provision of a plunger **96** of a medical instrument **60, 160, 260, 360** according to one aspect of the invention, it is possible to protect the tendons **90** inside said frame **57** while allowing sterile barrier **87** to be external to said frame.

Due to the provision of a plunger **96** of a medical instrument **60, 160, 260, 360** according to one aspect of the invention, it is possible to provide tensioning to said tendons **90, 190** for any joint member **71** position of said jointed device **70**.

Due to the provision of a plunger **96** of a medical instrument **60, 160, 260, 360** according to one aspect of the invention, it is possible to avoid lost motion and backlash effects associated to changes of direction of motion of which are altogether avoided making use of a continued pushing action of said pushing element on said plunger.

Due to the provision of a sterile barrier **87** of a medical instrument **60, 160, 260, 360** according to one aspect of the invention, it is possible to provide a sterile barrier that is not attached to pushing element and so it is easier to deploy for the surgical staff.

According to an embodiment, said pushing element **95** includes a sensor **150**.

According to an embodiment, said pushing element **95** includes a sensor **150** suitable to detect contact between said pushing element **95** and said plunger **96** through said sterile barrier **87**.

According to an embodiment, said pushing element **95** includes a force sensor **151** suitable to measure the pushing force exchanged between said pushing element **95** and plunger **96** through said sterile barrier **87**.

According to an embodiment, said force sensor **151** is a mono-axial load sensor measuring a component of pushing force along the linear trajectory of motion of said pushing element **95**.

According to an embodiment, said pushing element **95** includes a pressure sensor **152** suitable to measure the

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pressure exchanged between said pushing element **95** and plunger **96** through said sterile barrier **87**.

According to an embodiment, said pressure sensor **152** is a thin film pressure sensor glued to said reciprocal pushing surface **148** of said pushing element **95**.

According to an embodiment, said pushing element **95** includes a non contact proximity sensor **153** suitable to measure the distance between said reciprocal pushing surface **148** and pushing surface **147** through said sterile barrier **87**.

Due to the provision of a pushing element **95** of a medical instrument **60, 160, 260, 360** according to one aspect of the invention, suitable to move a jointed device across a sterile barrier allows the production of a medical instrument, which is highly reliable and sterile.

Due to the provision of a sensor **150** of a medical instrument **60, 160, 260, 360** according to one aspect of the invention, it is possible to sense through a sterile barrier a sensed quantity related to the interaction between said jointed device **70** and patient **201** anatomy.

Due to the provision of a sensor **150** of a medical instrument **60, 160, 260, 360** according to one aspect of the invention, it is possible to detect contact between said pushing element **95** and plunger **96** through a sterile barrier.

Due to the provision of a sensor **150** of a medical instrument **60, 160, 260, 360** according to one aspect of the invention, it is possible to sense a pushing force through a sterile barrier related to tension of tendon **90, 190**.

According to a general embodiment, a robotic surgery assembly **100** comprises at least one medical instrument **60, 160, 260, 360** according to any one of the embodiment as previously described.

According to one aspect of the invention, a surgical robotic assembly **100** comprises:

at least one micro-positioning device **41, 141, 241, 341** having multiple degrees of freedom at least of translation.

at least one medical instrument **60**, comprising one jointed device **70**, or articulated device **70**, having multiple rotational degrees of freedom.

Said medical instrument **60** is connected in series, to said micro-positioning device **41** such that said articulated device **70** reaches a predefined position in a work volume **7** with its terminal portion **77**.

According to an embodiment, said robotic assembly **100** comprises a support **104** and at least one macro-positioning arm **30**, connected to said support **104**, with respect to which said macro-positioning arm **30** provides multiple degrees of freedom of macro positioning.

According to an embodiment, said micro-positioning device **41, 141, 241** and **341** is connected in cascade, that is in series, to said macro-positioning arm **30**.

The provision of a kinematic chain comprising a macro-positioning arm **30** connected in series to at least one micro-positioning device **41** comprising multiple degrees of freedom at least in translation, connected in series with a medical instrument **60**, allows to decouple the positioning movements in translation of the terminal portion **77** of said medical instrument **60** within said work volume **7**, and the positioning movements in orientation of the terminal portion **77** of said medical instrument **60** within said work volume **7**.

According to an embodiment, said micro-positioning device **41** comprises degrees of freedom exclusively of translation.

According to an embodiment, said micro-positioning device **41** is a cartesian kinematic mechanism, suitable to

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determine translational movements along at least two mutually orthogonal directions. According to an embodiment, said micro-positioning device **41** is a cartesian kinematic mechanism, suitable to determine translational movements along at least three mutually orthogonal directions.

According to an embodiment, said micro-positioning device **41** comprises a X-Y-Z cartesian kinematic mechanism and a further rotational degree of freedom, around a rotational axis which substantially coincides with the longitudinal direction in which the medical instrument develops.

According to an embodiment, said at least one medical instrument **60** comprising one jointed device **70**, has multiple degrees of freedom that are exclusively rotational.

According to an embodiment, a robotic surgical assembly **100** comprises a further micro-positioning device **41**, such that it comprises at least a first micro-positioning device **141** and a second micro-positioning device **241**.

According to an embodiment, said at least two micro-positioning devices **141**, **241** are placed parallel to each other. According to an embodiment, said at least two micro-positioning devices are placed side-by-side to move one medical instrument on the right and one medical instrument on the left.

According to an embodiment, a surgical robotic assembly **100** comprises a further medical instrument **60** such as to comprise at least a first medical instrument **160**, connected in cascade, or in series, to said first micro-positioning device **141** and at least a second medical instrument **260**, connected in cascade, or in series, to said second micro-positioning device **241**.

According to an embodiment, said first medical instrument **160** comprises one jointed device **170** and said second medical instrument comprises a second jointed device **270**.

According to an embodiment, said first micro-positioning device **141** and said second micro-positioning device **241** are placed in such a way that the respective terminal portions **77** of each jointed device **70** reach respective work volumes **7** which must at least partially overlap.

The provision of work volumes **7** that at least partially overlap permits an operation in context using at least two medical instruments on one single portion of the patient.

According to an embodiment, said at least two medical instruments **160**, **260** are placed parallel to each other.

According to an embodiment, said respective work volumes **7** substantially coincide.

According to an embodiment, said macro-positioning arm **30** comprises at least one support member **38**, comprising at least one attachment feature **39**, suited to hold at least one portion of at least one micro-positioning device **41**.

According to an embodiment, said support member **38** is suited to simultaneously carry/receive at least one portion of said first micro-positioning device **141** and at least one portion of said second micro-positioning device **241**.

According to an embodiment, said support member **38** comprises at least one other attachment feature **39**, such that it comprises at least three attachment features **39**, said further attachment feature **39** being suitable to hold at least one portion of a further micro-positioning device **41**.

According to an embodiment, said robotic assembly **100** comprises at least three micro-positioning devices **41**, **141**, **241**, **341**.

According to an embodiment, said robotic assembly **100** comprises at least three medical instruments **60**, **160**, **260**, **360**.

According to an embodiment, said three medical instruments **60**, **160**, **260**, **360** are positioned in cascade, or in

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series, with a co-respective micro-positioning device **41**, **141**, **241**, **341**, of said at least three micro-positioning devices **41**, **141**, **241**, **341**.

According to an embodiment, said first micro-positioning device **141**, said second micro-positioning device **241** and said third micro-positioning device **341** are located such that the terminal positions **77** of each jointed device **70** reach respective work volumes that are at least partially overlapping.

According to an embodiment, said support member **38** comprises at least three attachment features **39**, each suited to hold at least one portion of a micro-positioning device **41**.

According to an embodiment, said macro-positioning arm **30** has three degrees of freedom.

According to an embodiment, said macro-positioning arm **30** has five degrees of freedom, and in which said five degrees of freedom are both of rotation as of translation.

According to an embodiment, said five degrees of freedom of said macro-positioning arm **30** are a translational movement which is substantially vertical, three movements which are substantially rotational around said first, second and third axis of movement of the arm a-a, b-b, c-c and at least one rotational movement around said fourth axis of movement of the arm d-d.

According to an embodiment, said axes of movement of the arm can be fixed or mobile with respect to a common reference system.

According to an embodiment, said macro-positioning arm **30** is a passive mechanism. In other words, according to an embodiment, said macro-positioning arm **30** is meant to be manually moved by an operator.

According to an embodiment, said macro-positioning arm **30** has six degrees of freedom, of which at least one of rotation. The provision of this characteristic allows the formation of an active anthropomorphic robot, as shown in a non-limiting example in FIG. 1C. According to an embodiment, said macro-positioning arm **30** is an active anthropomorphic robot. In other words, according to an embodiment, said macro-positioning arm is moved by a motorized system comprising a stepper motor or a servo-motor.

According to an alternative embodiment, said macro-positioning arm **30** is a passive anthropomorphic robot.

According to an embodiment, said macro-positioning arm **30** has a radius of extension of movement of 650 mm.

According to an embodiment, said macro-positioning arm **30** comprises:

one first arm member **31**, connected to said support **104** and mobile with respect to said support **104** along a linear sliding guide **36**,

a second arm member **32**, connected to said first arm member **31** around a first axis of movement a-a.

The provision that said first member of the arm **31** is mobile with respect to said support **104** along a linear sliding guide **36**, allows for a up and down movement to get closer or further from the operating field.

According to an embodiment, said macro-positioning arm **30** further comprises a third arm member **33** connected to a second arm member **32** and mobile with respect to said second arm member **32** around a second axis of movement of the arm b-b.

According to an embodiment, said macro-positioning arm **30** further comprises a fourth arm member **34** connected to said third arm member **33** and mobile with respect to said third arm member **33** around a third axis of movement of the arm c-c.

According to an embodiment, said macro-positioning arm **30** further comprises at least one rotational dial nut **43**,

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which is mobile around a fourth axis of movement of the arm d-d, and is suitable to be manipulated to move said support member **38** around said fourth axis of movement of the arm d-d.

According to an embodiment, said five degrees of freedom of said macro-positioning arm **30** are a translational movement which is substantially vertical, three substantially rotational movements around said first, second and third axis of movement of arm a-a, b-b, c-c and at least one rotational movement around said fourth axis of movement of the arm d-d.

According to an embodiment, said rotational dial nut **43** comprises a click or non-continuous movement mechanism defining pre-established displacements.

According to an embodiment, there is a reduction in the transmission of rotational movement between said rotational dial nut **43** and said support member **38**. In other words, big angular movements of said rotational dial nut correspond to small angular movements of said support member **38**, in a similar manner to an objective of a camera.

Provisioning said support member **38** to be mobile by a rotational movement around said fourth axis of movement of the arm d-d allows the positioning of said terminal portion **77** of said at least one medical instrument **60**, associated to said macro-positioning arm **30**, in proximity of a predetermined portion of the patient **201** with a favourable angle between the instrument shaft and the anatomy plane, steeper or shallower to facilitate suturing on different anatomical planes.

According to an embodiment, said rotational dial nut **43** comprises at least one milled handle. This provides for finer control.

According to an embodiment, said first axis of movement of the arm a-a, said second axis of movement of the arm b-b and said third axis of movement of the arm c-c are substantially parallel to each other.

According to an embodiment, said fourth axis of movement of the arm d-d is substantially orthogonal to said third axis of movement of the arm c-c.

According to an embodiment, a manual knob **37** moving a rack and pinion mechanism controls the movement of said first member of the arm **31** in said linear sliding guide **36** by its rotational movement.

According to an embodiment, said macro-positioning arm **30** comprises at least one braking system, suitable for blocking the relative movement of at least two of said support **104**, said first member of the arm **31**, said second member of the arm **32**, said third member of the arm **33**, said fourth member of the arm **34**.

According to an embodiment, said braking system comprises at least one electromagnetic brake device.

According to an embodiment, said macro-positioning arm **30** comprises at least one release button **35**, or unlocking button, which can be switched between a brake (or lock) and a release (or unlock) position.

According to an embodiment, said braking system can be released by a release button **35**.

According to an embodiment, said release button **35** can be switched between a brake position and a release position.

According to an embodiment, said release button **35**, when in the release position, allows the operator to move, by carrying it around, at least one of the degrees of freedom of said macro-positioning arm **30**.

According to an embodiment, when it is in its release position, said release button **35** is able to release the braking system, allowing the simultaneous relative movement of at least two of said support **104** and said first member of the

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arm **31**, said second member of the arm **32**, said third member of the arm **33** and said fourth member of the arm **34**.

According to an embodiment, when it is in the release position, said release button **35** is suitable to inactivate said arrest system, allowing the simultaneous relative movement of said first member of the arm **31**, said second member of the arm **32**, said third member of the arm **33** and said fourth member of the arm **34**.

According to an embodiment, said release button **35** is suitable to work by pressure, when it is depressed it is in said release position, and when it is raised or undepressed it is in said arrest position.

According to an embodiment, said robotic assembly **100** comprises:

said macro-positioning arm **30**, passively mobile by releasing said release system,
said at least one micro-positioning device **41** and said at least one articulated device **70**, actively controlled by master slave teleoperation, from the movement of said control instrument **21** as performed by the surgeon **200**.

According to an embodiment, said micro-positioning device **41**, **141**, **241** has three degrees of freedom of translation.

According to an embodiment, said micro-positioning device **41**, **141**, **241** has four degrees of freedom, of which three are of translation.

According to an embodiment, each micro-positioning device **41** comprises a spherical joint **173**, said spherical joint **173** is positioned in cascade, or in series, upstream of each micro-positioning device **41**.

According to an embodiment, for example shown in FIG. **2B**, each micro-positioning device **41**, **141**, **241** comprises a spherical joint **173**, suitable to change the orientation of the medical instrument **60**, **160**, **260** by moving the micro-positioning device **41**, **141**, **241**, from its base, i.e. most proximal portion. According to an embodiment, said spherical joint **173** is a universal joint that can be blocked.

According to an embodiment, said micro-positioning device **41** comprises a first motorized slide **51**, mobile along a first sliding rail **54** along a first sliding direction f-f.

According to an embodiment, said micro-positioning device **41** comprises a second motorized slide **52**, mobile along a second sliding rail **55** along a second sliding direction g-g.

According to an embodiment, said micro-positioning device **41** comprises a third motorized slide **53**, mobile along a third sliding rail **56** along a third sliding direction h-h.

According to an embodiment, said first sliding direction f-f is substantially rectilinear.

According to an embodiment, said second sliding direction g-g is substantially rectilinear.

According to an embodiment, said second sliding direction g-g is substantially orthogonal with respect to said first sliding direction f-f.

According to an embodiment, said third sliding direction h-h is substantially rectilinear.

According to an embodiment, said third sliding direction h-h is substantially orthogonal with respect to both said first sliding direction f-f and said second sliding direction g-g. According to an embodiment, the third sliding direction h-h is aligned with the shaft **65**.

According to an embodiment, said micro-positioning device **41** is suitable for working with a stepper motor or a servo-motor. According to an embodiment, said micro-positioning device **41** is suitable to work with a piezoelectric motor or an ultrasonic motor.

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According to an embodiment, at least one motorized slide **51**, **52**, **53** of said first, second and third motorized slides, is connected to a motor via a transmission mechanism comprising a ball screw which rotates with respect to the respective slide rail **54**, **55**, **56** and is held by a nut. According to an embodiment, said nut is solid to at least one motorized slide **51**, **52**, **53** of said first, second and third motorized slides.

The provision of a transmission mechanism comprising a coupling of a preloaded ball or lead screw-nut type confers an improved control of movement to the motorized slide as well as decreased backlash.

According to an embodiment, at least one motorized slide **51**, **52**, **53** of said first, second, third motorized slides, is connected to a motor by a transmission mechanism comprising a cogged belt.

According to an embodiment, said motorized slides **51**, **52**, **53** are precision micro-slides having a stroke between 1 cm and 10 cm, and having precision in the 0.1 micron and 25 micron range.

According to an embodiment, said motor is a servo-motor. According to an embodiment, said motor is a stepper motor.

According to an embodiment, said medical instrument **60** comprises a motorized rotary joint **46**, suitable for moving said medical instrument **60** around a longitudinal axis of rotation r-r.

According to an embodiment, said micro-positioning device **41** also comprises a motorized rotary joint **46**, suitable for moving said medical instrument **60** around a longitudinal axis of rotation r-r.

According to an embodiment, said axis of longitudinal rotation r-r substantially coincides with its longitudinal axis of development, or axis of the instrument X-X, or longitudinal axis of the shaft X-X, of said medical instrument **60**. According to an embodiment, a shaft angle θ is defined as the angle between the shaft direction X-X of the shaft **65** of said first medical instrument **160** and the shaft direction X-X of the shaft **65** of said second medical instrument **260**.

According to an embodiment, said medical instrument **60** comprises one articulated device **70** with two degrees of freedom of rotation. According to an embodiment, said medical instrument **60** comprises one articulated device **70** with two degrees of freedom of rotation orthogonal to each other to form a jointed wrist.

According to an embodiment, said medical instrument **60** comprises a jointed device **70** with at least three degrees of freedom. According to an embodiment, said jointed device **70** has three degrees of freedom of rotation, of which two degrees of freedom of rotation around axes parallel to each other and a third degree of freedom of rotation around said longitudinal axis of rotation r-r.

According to an embodiment, said jointed device **70** has three degrees of freedom of rotation, of which one first degree of freedom of rotation, around a first axis of rotation orthogonal to the axis of the instrument X-X, one second degree of freedom of rotation parallel to the first axis of rotation and a third degree of freedom of rotation orthogonal to the second axis of rotation, such that said second and third degrees of freedom of rotation are close to each other and form a sub-articulation of the wrist.

According to an embodiment, said medical instrument **60** comprises a jointed device **70**, which has a further degree of freedom in its terminal portion **77**, said further degree of freedom allows an opening and/or closing movement of said terminal portion **77**. According to an embodiment, said jointed device **70** comprises a terminal device **77** in said distal portion, in which said terminal device **77** comprises

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said further degree of freedom of opening and/or closing. For example, said further degree of freedom determines the opening and/or closing of forceps or of a cutting instrument, such as scissors.

According to an embodiment, said at least one medical instrument **60** is connected in a detachable fashion to said robotic assembly **100**.

According to an embodiment, said medical instrument **60** comprises at least a shaft **65**, suitable to connect said frame **57** with said jointed device **70**.

According to an embodiment, said medical instrument **60** comprises at least one shaft **65** such as to position its jointed device **70** at a predefined distance from said micro-positioning device **41**. According to an embodiment, said shaft **65** is suitable for distancing said jointed device **70** from said micro-positioning device **41** by a predefined distance.

According to an embodiment, said predefined distance is a multiple of the longitudinal extension of said jointed device **70**. According to an embodiment, said predefined distance is equal to at least five times the longitudinal extension of said jointed device **70**. According to an embodiment, said predefined distance is equal to at least twenty-five times the longitudinal extension of said jointed device **70**. According to an embodiment, said predefined distance is equal to substantially twenty times the longitudinal extension of said jointed device **70**. According to an embodiment, said predefined distance is measured along the longitudinal direction of the shaft X-X. According to an embodiment, said predefined distance is equal to substantially fifty times the longitudinal extension of said jointed device **70**.

The provision of said shaft **65** which distances said micro-positioning device **41** and said jointed device **70** allows for the fabrication of said micro-positioning device **41**, as well as said jointed device **70** to be of dimensions that are appropriate for them to fulfill their functions when in operating conditions. When said robotic assembly **100** comprises a plurality of medical instruments **60**, **160**, **260**, **360**, the provision of said shaft **65** in each medical instrument **60**, **160**, **260**, **360** which distances the respective micro-positioning devices **41**, **141**, **241**, **341** from the associated jointed devices, allows for the terminal portions **77** of each medical device to reach their own work volumes, while keeping their ability to move independently.

According to an embodiment, said shaft **65** is suitable to connect to said frame with said terminal device **77** at a predefined distance from said frame **57**.

According to an embodiment, said shaft **65** is rigid.

According to an embodiment, said shaft **65** has a longitudinal extension between 30 mm and 250 mm, and preferably between 60 mm and 150 mm.

According to an embodiment, said shaft **65** has a longitudinal internal hole. According to an embodiment, said shaft **65** has a hollow tubular form.

According to an embodiment, said medical instrument **60** comprises a motor box **61** suitable to house at least one driving system of at least said jointed device **70**, of said medical instrument **60**. In this way, the actuation of said jointed device **70** happens internally to said medical instrument **60**.

According to an embodiment, a robotic assembly **100** comprises at least one control device **20**, suitable to determine the movement of at least one portion of said medical instrument **60**, **160**, **260**, by a master-slave type communication system.

According to an embodiment, said assembly comprises a further control device **20**, such that it comprises at least two input devices **20**. According to an embodiment, said control

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device 20 is suitable to determine the motion of said jointed device 70 of said medical instrument 60. According to an embodiment, said control device 20 is suitable to determine the movement of said micro-positioning device 41. The provision of said characteristic allows a translational movement of said control instrument 21 as registered by said detection device 22 to be associated to a translational movement of said terminal device 77 within its workspace 7, 17.

According to an embodiment, said control device 20 is suitable to determine the motion of said micro-positioning device 41 and said medical instrument 60.

The provision of this characteristic allows to move at least a portion of said micro-positioning device 41 and at least a portion of said medical instrument 60 by means of said control instrument 21, such as to determine both rotational and translational movements of said terminal device 77 in said work volume 7.

According to one alternative embodiment, said micro-positioning device 41 comprises a plurality of passive degrees of freedom that can be braked or otherwise blocked. According to an embodiment, said plurality of degrees of freedom is placed immediately upstream and in series to said micro-positioning device 41.

According to an embodiment, said robotic assembly 100 is suitable to cooperate with a vision system 103 associable to said robotic assembly 100.

According to an embodiment, said vision system 103 is a microscope 103.

The provision of a microscope 103 associable to said robotic assembly allows for retro-fitting with pre-existing microscopes, making said robotic assembly 100 more versatile. For example, said robotic assembly 100 can be used in cooperation with microscopes that have a focusing distance between 100 mm and 500 mm, depending on the focal length of the objective lens used. Furthermore, it allows the swept volume of the robotic assembly 100 to be reduced, during the surgical operation given that it lacks as many parts as possible that require relatively large movements during the movement of the terminal portion of the instrument.

According to an embodiment, said microscope 103 is an optical microscope 103.

According to an embodiment, said microscope 103 is suitable to frame in its field of view said terminal portion 77 of said first medical instrument 160 and/or said terminal portion 77 of said second medical instrument 260 and/or said terminal portion of said third medical instrument 360.

According to an embodiment, said microscope 103 is suitable for framing the work volume 7.

According to an embodiment, at least one video-camera 45, is connected to said support member 38.

According to an embodiment, said video-camera 45 is suitable for framing said terminal portion 77 of said first medical instrument 160 and said terminal portion 77 of said second medical instrument 260.

According to an embodiment, said support 104 comprises at least one display 111, suitable to form a machine input interface.

According to an embodiment, said display 111 is suitable to visualize the images acquired by said video-camera 45.

According to an embodiment, said video-camera 45 is suitable to cooperate with said macro-positioning arm 30 to permit the correct positioning of said at least one medical instrument 60. The provision of this characteristic facilitates the positioning process of at least one portion of said at least one medical instrument 60 within the work volume 7.

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According to an embodiment, said first medical instrument 160, said second medical instrument 260 and said support member 38 are disposed in such a way that they substantially form a triangle. Such provision allows to reproduce of the same triangulation existing between the eyes and the arms of the surgeon by means of said robotic assembly 100.

According to an embodiment, said support 104 is at least one of: a mobile cart, a support structure of a microscope, an operating bed, an operating table.

According to one aspect of the invention, a control device 20 for microsurgery for a robotic assembly for microsurgery 100, in which said control device 20 is suitable to at least partially form the master interface of a master-slave pair for a robotic assembly for microsurgery 100, comprises:

at least one control instrument 21, mobile in space, of a shape and size which lends it to being held and handled like a traditional surgical instrument, that is to say a surgical instrument suitable to operate directly on at least one portion of the patient anatomy 201,

at least one detection device 22, suitable to detect the position of said control instrument 21 in at least one portion of space.

Said control instrument 21 comprises at least one position sensor 28, which cooperates with said detection device 22, to sense at least the position of said control instrument 21.

According to an embodiment, said detection device 22 generates an electromagnetic field such as to detect at least the position of said control instrument 21 by detecting the position of said at least one position sensor 28. According to an embodiment, said detection device 22 detects at least the position of said control instrument 21 by detecting the position of said position sensor 28 by measuring at least inertial accelerations components. According to an embodiment, said position sensor 28 comprises accelerometers.

According to an embodiment, said detection device 22 is positioned in a base structure 67 of said control device 20.

According to an embodiment, said control instrument 21 is connected to said detection device 22 by at least an electromagnetic communication system.

According to an embodiment, said control instrument 21 comprises at least one forceps articulation 69, effective in a tip portion 68 of said control instrument 21, such as to allow said tip portion 68 a grasping or cutting movement.

According to an embodiment, at least one tip sensor 29 measures an opening angle of said forceps articulation 69.

According to an embodiment, said control instrument 21 has a shape that substantially replicates the shape of a traditional surgical instrument.

According to an embodiment, said control instrument 21 has the shape of surgical forceps.

According to an embodiment, said control instrument 21 has the shape of a surgical scalpel.

According to an embodiment, said control instrument 21 has the shape of a surgical needle holder.

According to an embodiment, said control instrument 21 has the shape of surgical scissors.

According to an embodiment, said control instrument 21 has the shape of a surgical blade.

According to an embodiment, said control device 20 comprises at least one ergonomic support element for the operator 27, comprising at least one support surface for the operator 25, suitable to support at least one portion of the forearm of the micro-surgeon 200, at least when in operating conditions, such as to provide ergonomic support for the micro-surgeon 200. The provision of such a characteristic

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allows for improved comfort of the micro-surgeon, determining an improved operating efficiency.

According to an embodiment, said ergonomic support element **27** comprises at least one portion made of soft material or foam.

According to an embodiment, said control instrument **21** is connected to said detection device **22** by at least one system of electromagnetic communication. According to an embodiment, said position sensor is an electromagnetic position sensor with micro-bobbins and said sensor device comprises a generator of a magnetic field and an electric circuit that reads the circuit induced in said micro-bobbins by said magnetic field. The provision of this characteristic allows the control instrument **21** to preserve its functioning as a traditional surgical instrument, without affecting a response time for said detection device **22**.

According to an embodiment, said control instrument **21** is connected to said detection device **22** by a wired connection, or cable.

According to an embodiment, said control instrument **21** is connected to said detection device **22** by a wireless connection.

According to an embodiment, said detection device **22** is suitable to measure the position in space, this position measure being either by induced current, or it is an optic measure, or an ultrasound measure, or a measure by ionizing radiation.

According to an embodiment, said control device **20** comprises an on-off type switch, either implemented as a pedal or as a button, selectively suitable to activate or deactivate input from said control device **20**.

According to an embodiment, a robotic assembly **100**, comprises:

- at least one control device **20**, as described by one of the embodiments described above,

- at least one surgical micro-instrument **60, 160, 260, 360** comprising at least one terminal portion **77**.

According to an embodiment, said terminal portion **77** is suitable to operate on at least one portion of the patient **201**.

According to an embodiment, said terminal portion **77** is suitable to handle a surgical needle **202**, as shown for example in FIG. 3A-3B.

According to an embodiment, said control instrument **21** has the same dimensions and offers the same handling experience of a traditional surgical instrument, that is to say a surgical instrument that can be used to operate directly on at least one portion of a patient **201**, and said surgical micro-instrument **60** is suitable to replicate the same entire movement capability of said control instrument **21**.

According to an embodiment, said robotic assembly **100** is suitable to decouple the movements of said control instrument **21** and said surgical micro-instrument **60** in such a way that when the movements of said control instrument **21** are large and comprise vibrations, while the movements of said surgical micro-instrument **60** are filtered of vibrations and reduce the movement to a millimeter or to a micron scale. The provision of scaled movement introduced between the master interface and the slave interface allows for the reduction of tremor as well as an improvement of precision of said surgical micro-instrument without decreasing the ease of operation of the surgeon **200**.

According to an embodiment, said control instrument **21** is suitable to cooperate with said surgical micro-instrument **60** in such a way that, when in operating conditions, at a first 3D movement of said control instrument **21** with respect to said detection device, corresponds to a second 3D movement of said surgical micro-instrument **60**.

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According to an embodiment, said control instrument **21** is suitable to cooperate with said surgical micro-instrument **60, 160, 260**, in such a way that, when in operating conditions, a first translational movement of said control instrument **21** corresponds to a second translational movement of said surgical micro-instrument **60, 160, 260** equal to a fraction of the amplitude of said first movement of said control instrument **21**. In this way, it is possible to limit the transmission of tremor or vibration of the control instrument **21** to the surgical micro-instrument **60**.

According to an embodiment, said control instrument **21** is suitable to cooperate with said surgical micro-instrument **60, 160, 260** in such a way that, when in operating conditions, a first translational movement of said control instrument **21** corresponds to a second translational movement of said surgical micro-instrument **60, 160, 260** of an amplitude that is substantially equal to one tenth of the amplitude of said first movement of said control instrument **21**.

According to an embodiment, said control instrument **21** is suitable to cooperate with said surgical micro-instrument **60, 160, 260** such that, when in operating conditions, a first translational movement of said control instrument **21** corresponds to a second translational movement of said surgical microinstrument **60, 160, 260** of an amplitude substantially equal to one thirtieth of the amplitude of said first movement of said control instrument **21**.

According to an embodiment, said control instrument **21** is suitable to cooperate with said surgical micro-instrument **60, 160, 260** in such a way that, when in operating conditions, a first angular movement of said control instrument **21** corresponds to a second angular movement of said surgical micro-instrument **60, 160, 260**, said second angular movement of the micro-instrument being of an amplitude that is substantially equal to the amplitude of said first movement of the control instrument **21**. The provision of such a characteristic renders the use of said control instrument **21** familiar to a surgeon **200**.

According to an embodiment, said control instrument **21** is suitable to cooperate with said surgical micro-instrument **60**, such that, when in operating conditions, a first angular movement of said forceps articulation **69** of said control instrument **21** corresponds to a second angular movement of an articulation, situated on said terminal portion **77** of said surgical micro-instrument **60**, the amplitude of said second movement being substantially equal to said first angular movement of said forceps articulation of said control instrument **21**.

According to an embodiment, said a portion of control instrument **21** is of a shape that substantially reproduces the shape of said terminal portion **77** of said surgical microinstrument **60, 160, 260**.

According to an embodiment, said surgical micro-instrument **60, 160, 260** comprises at least one jointed device **70** and said control instrument **21** is suitable to cooperate with said jointed device **70, 170, 270** so that, when in operating conditions, a first movement of said control instrument **21** with respect to said detection device **22**, corresponds to a second movement of said jointed device **70, 170, 270**.

According to an embodiment, a robotic assembly **100** also comprises

- a support **104**,

- at least one macro-positioning arm **30**, connected to said support **104**, said macro-positioning arm having a plurality of degrees of freedom,

- at least one micro-positioning device **41, 141, 241** having a plurality of degrees of freedom of translation.

According to an embodiment, said at least one control device **20** is connected to at least one portion of said microsurgical robotic assembly **100**.

According to an embodiment, said at least one control device **20** is freely positionable with respect to said support **104**.

According to an embodiment, said surgical micro-instrument **60**, **160**, **260** comprises at least one micro-instrument sensor, suitable to cooperate with said detection device **22**, such that the position in space of at least one portion of the surgical micro-instrument **60**, **160**, **260** can be detected with respect to said detection device **22**.

According to an embodiment, said micro-positioning device **41**, **141**, **241** comprises at least one micro-manipulator sensor, suitable to cooperate with a detection device **22**, such as to detect the position in space of at least one portion of said micro-positioning device **41**, **141**, **241** with respect to said detection device **22**.

According to an embodiment, said macro-positioning arm **30** comprises at least one macro-positioning arm sensor, suitable to cooperate with said detection device **22**, such as to detect the position in space of at least one portion of said macro-positioning arm **30** with respect to said detection device **22**.

According to an embodiment, said microsurgical robotic assembly **100** is suitable to cooperate with a sensor, suitable to detect the position in space with respect to a single reference system of at least one of: said position sensor **28**, said tip sensor **29**, said macro-positioning arm sensor, said micro-positioning device sensor, said micro-instrument sensor. According to an embodiment, said microsurgical robotic assembly **100** is suitable to cooperate with a sensor, suitable to detect the position in space with respect to a single reference system of at least two of: said position sensor **28**, said tip sensor **29**, said macro-positioning arm sensor, said micro-positioning device sensor, said micro-instrument sensor. The provision of this characteristic allows for a teleoperation master-slave system to function adequately independently of the exact position of said detection device **22**, said support **104**, said macro-positioning arm **30** and said micro-positioning device **41**. In other words, said medical instrument **60** is able to follow the movement of control instrument **21** with respect to a same common reference system of coordinates.

According to an embodiment, said at least one surgical micro-instrument **60**, **160**, **260** is connected to said robotic assembly **100** in a detachable fashion.

According to an embodiment, a microsurgical robotic assembly **100** also comprises:

a further control instrument **21**, such as to comprise a first control instrument

and a second control instrument **221**;

a further surgical micro-instrument **60**, **160**, **260** such as to comprise a first surgical micro-instrument **160** and a second surgical micro-instrument **260**.

According to an embodiment, said first control instrument **121** is suitable to cooperate with said first surgical micro-instrument **160**, in such a way that, when in operating condition, a first movement of said first control instrument **121** with respect to said detection device **22**, corresponds to a second movement of said first surgical microinstrument **160**.

According to an embodiment, said second control instrument **221** is suitable to cooperate with said second surgical micro-instrument **260**, such that, when in operating conditions, a first movement of said second control instrument

221 with respect to said detection device **22**, corresponds to a second movement of said surgical micro-instrument **260**.

According to an embodiment, said first control instrument **121** is suitable to form the master interface of said robotic assembly **100** for one first hand of the surgeon **200**.

According to an embodiment, said second control instrument **221** is suitable to form the master interface of said robotic assembly **100** for one second hand of the surgeon **200**, different from said first hand.

According to an embodiment, said first and second control instruments **121**, **221** are of substantially mirrored in shapes and location, such as to form the master interface of said robotic assembly **100** for both hands of the surgeon. In this way, the interface has improved ergonomics and is more familiar to the surgeon.

According to an embodiment, said control device **20** comprises at least two control instruments **21**, **121**, **221**.

According to an embodiment, said microsurgical robotic assembly **100** comprises a further detection device **22** such as to comprise at least two detection devices.

According to an embodiment, said control device **20** comprises at least two detection devices **22**.

According to an embodiment, said microsurgical robotic assembly **100** comprises at least one further control device **20**, such as to comprise a first control device **120** and a second control device **220**.

According to an embodiment, said first control device **120** is suitable to form the master interface of said robotic assembly **100** for the first hand of the surgeon **200**.

According to an embodiment, said second control device **220** is suitable to form the master interface of said robotic assembly **100** for the second hand of the surgeon **200**, different from said first hand.

According to an embodiment, said first and second control devices **120**, **220** are of substantially mirrored shapes, such as to form the master interface of said robotic assembly **100** for both hands of the surgeon. In this way, the interface has improved ergonomics and is more familiar to the surgeon.

According to one aspect of the invention, a medical instrument **60**, **160**, **260**, **360** comprises at least one frame **57** and one jointed device **70**.

Said jointed device **70** comprises at least one first joint member **71**, or first link **71**, suitable to connect to at least one portion of said frame **57**, and at least a second joint member **72**, or second link **72**.

Said first link **71** is connected via a rotational joint **171** to said second link **72**.

Said medical instrument **60** also comprises at least one tendon **90**, **190**, suitable for moving at least said second link **72** with respect to said first link **71**, by pulling it.

At least one of said first link **71**, said second link **72** comprises at least a sliding surface **40**, **80**, **140**, **180**, suitable to allow the sliding of at least one portion of said tendon **90**, **190** over it.

Said sliding surface **40**, **80**, **140**, **180** is a ruled surface **40**, **80**, **140**, **180**, specifically a ruled surface formed by a plurality of portions of straight lines all parallel to each other and substantially parallel to a joint axis of movement P-P, Y-Y.

According to an embodiment, said sliding surface **40**, **80**, **140**, **180** is a ruled surface **40**, **80**, **140**, **180**, specifically a ruled surface formed by a plurality of portions of straight lines all parallel to each other and substantially parallel to a joint axis of movement P-P, Y-Y of the rotational joint **171** closest to said sliding surface **40**, **80**, **140**, **180**. According to

an embodiment, the closest rotational joint **171** is defined by measuring along the direction of the tendon path T-T.

According to an embodiment, said axes of joint movement can be fixed or mobile with respect to a base reference system.

According to an embodiment, said at least one second link **72** is a wrist member **78**, and said wrist member **78** comprises at least one sliding surface **40**, **80**, **140**, **180**, formed by a plurality of portions of straight lines parallel to each other and substantially parallel to a first joint axis of movement.

According to an embodiment, said wrist member **78** comprises at least one jointing portion **172**, suitable to form at least one portion of a second rotational joint **171** having a second joint axis of movement, not parallel to said first joint axis of movement.

According to an embodiment, said first joint axis of movement and said second joint axis of movement are substantially orthogonal to each other.

According to an embodiment, said first joint axis of movement is a pitch axis P-P.

According to an embodiment, said second joint axis of joint movement is a yaw axis Y-Y.

According to an embodiment, said medical instrument **60**, **160**, **260** has at least one terminal member **77**.

According to an embodiment, said terminal member **77** is suitable to contact with one portion with a patient **201**, when in operating conditions.

According to an embodiment, said terminal member **77** is suitable to handle a surgical needle **202**.

According to an embodiment, said terminal member **77** comprises a cutting surface or blade and can act as a scalpel.

According to an embodiment, said terminal member **77** comprises at least one winding surface **86**, made of a plurality of portions of straight lines all parallel to each other and substantially parallel to a joint axis of movement. According to an embodiment, said winding surface **86** is suitable to allow at least one portion of said tendon **90**, **190** to be wound up around it.

According to an embodiment, said second joint member **72** is a terminal member **77**.

According to an embodiment, said jointed device **70**, **170**, **270** comprises a third joint member **73**, suitable to connect to at least said second joint member **72** by a rotational joint **171**.

According to an embodiment, said third joint member **73** is a terminal member **77**.

According to an embodiment, said terminal member **77** is connected to said wrist member **78** by a rotational joint **171**.

According to an embodiment, said at least one joint member **72** is an elbow member **75**, and said elbow member **75** comprises a plurality of sliding surfaces **40**, **80**, **140**, **180** formed by a plurality of portions of straight lines all parallel to each other and substantially parallel to a single joint axis of movement.

According to an embodiment, said elbow member **75** comprises at least one jointing portion **172**, suitable to form at least one portion of a rotational joint **171**.

According to an embodiment, said jointed device **70** comprises a third joint member **73**, suitable to be connected to at least said second joint member **72** by a rotational joint **171**, in which said second joint member **72** is an elbow member **75** and said third joint member **73** is a wrist member **78**.

According to an embodiment, said elbow member **75** is connected by a rotational joint **171** to said first joint member

71, and in which said wrist member **78** is connected via a rotational joint **171** to said elbow joint member **75**.

According to an embodiment, said jointed device **70** comprises a fourth joint member **74**, suitable to connect to at least said third joint member **73** via a rotational joint **171**.

According to an embodiment, said fourth joint member **74** is a terminal member **77**.gmp

According to an embodiment, said terminal member **77** comprises at least one winding surface **86**, formed by a plurality of portions of straight lines all parallel to each other and substantially parallel to a joint axis of movement, wherein said winding surface **86** is suitable to allow the winding of at least one portion of said tendon **90**, **190** around it.

According to an embodiment, said jointed device **70** comprises said first member **71**, connected to said wrist member **78** via a rotational joint **171**, connected to said terminal member **77** via a rotational joint **171**.

According to an embodiment, said jointed device **70** comprises said first member **71**, connected to said elbow member **75** by a rotational joint **171**, connected to said wrist member **78** by a rotational joint **171**, itself connected to said terminal member **77** by a rotational joint **171**. It should be apparent to those skilled in the art that making use of joint members similar to **71**, **72**, **73**, a jointed device **70** can be assembled to include a serial sequence of members, of which from zero to a plurality of elbow joint members **75**, a plurality of, preferably orthogonal, pairs of wrist joint members **78** and at least one terminal joint member **77**.

According to an embodiment, said winding surface **86** is a ruled surface.

According to an embodiment, said winding surface is substantially unsuitable for said tendon **90**, **190** to slide over it. This is because said tendon **90**, **190** terminates close to said winding surface **86**, on the joint member which comprises said winding surface **86**.

According to an embodiment, said medical instrument **60** comprises at least one pair of tendons comprising one tendon **90** and one opposite tendon **190**, and said tendon **90** and said opposite tendon **190** are suitable to connect their second termination endpoints **92**, or second tendon termination **92**, to respective tendon fastening points **82**, or point of tendon termination **82**, of said second joint member **72**, such as to move said second joint member **72** around its joint axis in opposite directions.

According to an embodiment, said medical instrument **60** which comprises at least one pair of tendons comprising one tendon **90** and one opposite tendon **190**, and said tendon **90** and said opposite tendon **190** are suitable to connect in their second termination endpoints **92** to respective tendon fastening points **82**, or tendon termination features **82**, of said terminal member **77**, such as to move it around its joint axis in opposite directions.

The provision of such a feature makes sure that said tendon **90** and said opposite tendon **190** can work in an antagonistic fashion, for example both said tendon **90** and said opposite tendon **190** move said terminal member around the yaw axis Y-Y. Hence no passive or free joint movement can occur, and instead there are only positively guided and controlled movements.

According to an embodiment, said tendon **90** and opposite tendon **190** are suitable to connect by means of their second termination endpoints **92** to respective tendon fastening points **82**, or tendon termination features **82**, of at least one of said first, second, third and fourth joint members **71**, **72**, **73**, **74**.

According to an embodiment, said tendon **90** and opposite tendon **190** are suitable to connect by means of their second termination endpoints **92** in respective tendon fastening points **82**, or tendon termination feature **82**, of at least one of said elbow member **75**, wrist member **78** and terminal member **77**.

According to an embodiment, said medical instrument **60** comprises at least one shaft **65**, suitable to guide said at least one tendon **90**, **190**. Said shaft **65** is a shaft according to one of any of the previously described embodiments.

According to an embodiment, said shaft **65** has a substantially circular section and has a diameter smaller than 4 millimeters. This allows extreme miniaturization of the medical instrument.

According to an embodiment, said shaft **65** comprises a longitudinal hole such as to allow the passage of said at least one tendon **90**, **190** inside it.

According to an embodiment, said shaft **65** is integral to said frame **57**.

According to an embodiment, said jointed device **70** has a longitudinal extension smaller than 10 millimeters.

According to an embodiment, said jointed device **70** has a volume inferior to 10 cubic millimeters.

According to an embodiment, said terminal member **77** comprises at least one first portion of terminal member **177** and at least a second portion of terminal member **277**. According to an embodiment, said first portion of terminal member **177** and said second portion of terminal member **277**, are mobile with respect to each other around a joint axis of movement such as to determine a grasping or cutting movement. According to an embodiment, said joint axis of movement is said yaw axis Y-Y.

According to an embodiment, said medical instrument **60**, which comprises at least one pair of tendons, comprises a tendon **90** and an opposite tendon **190**, in which one of said tendon **90** and said opposite tendon **190** is suitable to connect by means of its second endpoint **92** to a respective tendon fastening point **82**, or tendon termination feature **82**, p on said first terminal member **177**, and in which the other one of said tendon **90** and said opposite tendon **190** is suitable to connect by means of its second endpoint **92** to a respective tendon fastening point **82**, or tendon termination feature **82**, on said second terminal member **277**, such as to move said first portion of terminal member **177** and said second portion of terminal member **277** with movements in opposite directions.

According to an embodiment, each of said first portion of terminal member **177** and said second portion of terminal member **277** comprise at least one winding surface **86**.

According to an embodiment, said medical instrument **60** comprises at least one pair of tendons comprising one tendon **90** and one opposite tendon **190**, in which said tendon **90** and said opposite tendon **190** are suitable to connect by means of their second endpoints **92** in respective tendon fastening points **82**, or tendon termination feature **82**, of said terminal member **77**, such as to move said third joint member **73** with respect to said fourth joint member **74** such as to determine a grasping or cutting movement.

According to an embodiment, said tendon **90** and said opposite tendon **190** wind their distal portions around at least one portion of said at least one winding surface **86** of terminal member **77**.

According to an embodiment, said sliding surface **40**, **80**, **140**, **180** is a lateral sliding surface **40**, **140** suitable to extend away from the center volume of said jointed device **70**, **170**,

270 such as to determine that at least one portion of tendon is deflected away and runs not in contact with said jointed device **70**.

According to an embodiment, said lateral sliding surface **40**, **140** joins a surface of the member on which it is built, with at least a continuity surface **64**, sharing a local tangent plane. According to an embodiment, said lateral sliding surface **40**, **140** forms at least one sharp edge **63** with the member on which it is built.

According to an embodiment, said lateral sliding surface **40**, **140** joins a surface of the member on which it is built with a continuity surface **64** on one side and on the other side forms one sharp edge **63** with the member on which it is built.

According to an embodiment, said sliding surface **40**, **80**, **140**, **180** is a joint sliding surface **80**, **180** that at least partially surrounds an axis of joint movement. According to an embodiment, said sliding surface **40**, **80**, **140**, **180** is a Joint sliding surface **80**, **180** that at least partially surrounds at least one of said pitch axis P-P and said yaw axis Y-Y, and in which said joint sliding surface **80**, **180** is oriented opposite with respect to at least one of said pitch axis P-P and said yaw axis Y-Y, such as to allow at least one intersection between the tendon path T-T of said tendon **90** and the tendon path T-T of said opposite tendon **190**. In other words, said joint sliding surface **80**, **180** is not suitable to face towards said joint axis of movement of the closest rotational joint **171**, when in operating conditions.

According to an embodiment, said joint sliding surface is convex and partially surrounds at least one of said pitch axis P-P or yaw axis Y-Y, such as to permit at least one intersection of two opposite tendons on itself.

According to an embodiment, the term "closest joint" refers to the rotational joint **141** which is closest in distance to the sliding surface **40**, **80**, **140**, **180** along the tendon path T-T.

According to an embodiment, on said joint sliding surface **80**, **180** the tendon path T-T of said tendon **90** and the tendon path T-T of said opposite tendon **190**, although they do not intersect, they at least partially overlap in a projection plane orthogonal to the direction of said axis of joint movement of the closest rotational joint **171**.

According to an embodiment, on said joint sliding surface **80**, **180** the tendon path T-T of said tendon **90** and the tendon path T-T of said opposite tendon **190** are distinct from each other and parallel on a projection plane parallel to the joint axis of movement of the closest rotational joint **171**.

According to an embodiment, the tendon path T-T of said tendon **90** overlaps with the tendon path T-T of said opposite tendon **190** at least on a projection plane orthogonal to the direction of said joint axis of movement of the closest joint. According to an embodiment, the tendon path T-T of said tendon **90** is substantially parallel to the tendon path T-T of said opposite tendon **190** on a projection plane parallel to said joint axis of movement of the closest rotational joint.

According to an embodiment, the tendon path T-T of each tendon **90** are substantially parallel to each other, on a projection plane parallel to the said joint axis of movement of the closest rotational joint **171**.

According to an embodiment, each tendon path T-T is substantially stationary over the joint member which it contacts. In other words, even when the tendon **90** is sliding, the overall tendon path T-T is substantially always in the same position with respect to the joint member of said medical instrument **60**, which it contacts.

Such a feature is uniquely realized by provisioning that said sliding surface **40**, **80**, **140**, **180** of said winding surfaces

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86 has a cooperative geometrical relationship with said tendon termination feature **82**, which is in turn fittingly positioned on a portion of said medical instrument.

According to an embodiment, said tendon path T-T remains substantially stationary over the joint member which it contacts for both tendon **90** and opposite tendon **190** that determines opposite joint movements.

According to an embodiment, the tendon path T-T of each tendon **90** is substantially stationary in its section over said frame **57**, except for said deflectable portion **93**. Said deflectable portion **93** is in fact suitable to be deflected by the pusher assembly **94**, not unlike a guitar string.

According to an embodiment, said at least one tendon **90**, **190**, when in operating conditions, follows a tendon path T-T that is entirely composed of successive straight in-flight sections **9**, which are not in contact with any sliding surface **40**, **80** or winding surfaces **86**, and curved sections which are in contact with sliding surfaces **40**, **80** or winding surfaces **86** of the joint members **71**, **72**, **73**, **74**, **75**, **77**, **78**.

According to an embodiment, said at least one tendon **90**, **190** describes a path around said first joint member **71**, such as to at least partially wind itself over said joint sliding surface **40**, **140** of said first joint member **71**.

According to an embodiment, said at least one tendon **90**, **190** describes a path around said distal second joint member **72**, such as to at least partially wind itself over said joint sliding surface **80**, **180** of said second joint member **72**.

According to an embodiment, said medical instrument **60** comprises a plurality of tendons.

According to an embodiment, the projections of said tendon path T-T of said tendon **90** and of said tendon path T-T of said opposite tendon **190** on a plane orthogonal to said joint axis of movement of the closest rotational joint **171** overlap at least at a point of intersection **16**.

According to an embodiment, a said in-flight segment **9** of said tendon path T-T of said tendon **90** is substantially parallel to at least one said in-flight segment **9** of said opposite tendon **190**.

According to an embodiment, the tendon paths T-T of each tendon **90** are substantially parallel to each other, on a projection plane parallel to the direction of said joint axis of joint movement of the closest rotational joint **171**.

According to an embodiment, each said tendon termination feature **82** is positioned such as to support each tendon **90**, **190** so as to keep its tendon path T-T substantially orthogonal to the joint axis of movement of the closest rotational joint **171**, such as to allow said tendon **90** to slide on said at least one sliding surface **40**, **80** following a tendon path T-T substantially parallel to the tendon path T-T of any other tendon.

According to an embodiment, each tendon termination feature **82** is positioned such as to support each tendon **90**, **190** such that its tendon path T-T is stationary with respect to the joint member closest to it.

According to an embodiment, said tendon termination feature **82** is positioned such as to maintain its tendon path T-T of each tendon **90** substantially always in contact with said winding surface **86**, when in operating conditions.

According to an embodiment, said tendon termination feature **82** is positioned such that the tendon path T-T of each tendon **90**, **190** does not enter in contact with the tendon path T-T of any other tendon **90**, **190**, when in operating conditions.

According to an embodiment, said tendon termination feature **82** is positioned such that each tendon **90**, when in operating conditions, slides on at least one sliding surface **40**, **80**, describing a curved section of the tendon path T-T

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substantially parallel to the curved section of the tendon path T-T described by any other tendon **90**, **190** when it slides on the same sliding surface **40**, **80**.

According to an embodiment, said medical instrument **60** is a surgical instrument, suitable to be applied in at least one of the following fields: microsurgery, minimally invasive surgery and laparoscopic surgery.

According to an embodiment, said medical instrument **60** is suitable for being used for a biopsy. According to an embodiment, said medical instrument **60** is suitable to be used for an endoscopic procedure.

According to an embodiment, said tendon **90**, **190** has a substantially circular cross section. According to an embodiment, the diameter of said tendon **90**, **190** is variable in different portions of said tendon **90**, **190**. According to an embodiment, the mechanical properties of said tendon **90**, **190** are variable in different portions of said tendon **90**, **190**. According to an embodiment, said tendon **90**, **190** is obtained by joining portions of tendons with different characteristics. According to an embodiment, the composition of said tendon **90**, **190** is variable in different portions of said tendon **90**, **190**.

According to an embodiment, said tendon path T-T in at least one portion of the tendon is substantially locally orthogonal to the generatrices of the sliding surface **40**, **80**, **140**, **180** on which the tendon slides, in every operating condition, that is for any rotational angle of the rotational Joints **171**. These characteristics contribute to avoiding that said tendon path T-T of each of said tendons is ever deflected, that is to say that it never bends in a direction parallel to the axis of joint movement of the closest rotational joint **171**.

According to an embodiment, said tendon path T-T is substantially locally orthogonal to the generatrices of the sliding surfaces **40**, **80**, **140**, **180** on which it slides.

According to an embodiment, said jointed device **70** is primarily fabricated from metallic materials.

According to an embodiment, said joint members are suitable to be polished with the aim of further reducing the friction generated by the sliding of said at least one tendon, when said tendon slides over it.

According to one aspect of the invention, a tendon drive system **50** for a medical instrument **60**, **160**, **260** comprises at least one pusher assembly **94**.

Said medical instrument **60**, **160**, **260** comprises a frame **57** and at least one tendon **90**, **190**, exclusively suitable to work under tensile loads applied at its endpoints, in which a tendon direction T-T is defined, or a tendon path T-T, substantially coinciding with the direction of longitudinal development of said tendon **90**, and in which said tendon **90** is fastened at its first endpoint **91**, or proximal tendon endpoint **91**, or first tendon termination **91**, to said frame **57**.

Said pusher assembly **94** is suitable to apply a force over at least one portion of said deflectable portion **93** of said tendon **90** along a pushing direction transversal to the tendon path T-T such as to deflect the tendon path T-T and induce an increased tensile load in said tendon **90**.

When said pusher assembly pushes in said pushing direction, transversal to the tendon path T-T, it tends to lengthen locally, only locally, said tendon path. Such a localized path lengthening, which create a larger, local tendon loop is directly related to the amount of advancement of the pusher assembly. The creation of such a larger local tendon loop results at the opposite end of the tendon, in a proportional moving back of the distal endpoint of the tendon **92** which

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is fastened to the tendon termination feature **82** on the joint member and hence results in a movement of the joint member.

According to an embodiment, said pusher assembly **94** acts as a unilateral constraint for said tendon **90**.

According to an embodiment, said pusher assembly **94** lengthens or shortens said tendon path T-T in at least one section of said tendon path T-T, which is substantially straight.

According to an embodiment, said pusher assembly **94** is suitable to retrieve a determined length of said tendon **90**. According to an embodiment, said pusher assembly **94** is suitable to release a determined length of said tendon **90**.

According to an embodiment, said pusher assembly **94** is suitable to retreat on said tendon deflectable portion **93** of said tendon **90**, in a direction transversal to the tendon path T-T such that the deflection of said tendon path T-T is decreased and the strain in said tendon **90** is decreased. In this way, a controlled movement of at least one portion of said jointed device **70** of said medical instrument **60** is allowed.

The terms “retreat” and “retrieve” mean that the pusher assembly, when pushing in said pushing direction, which is transversal to the tendon path T-T, locally and only locally, shortens the tendon path. Such a local shortening creates an increasingly smaller local loop, which is directly related to the amount of pulling back of the pusher assembly, and at the opposite end of the tendon, where it is fastened at its distal endpoint **92** to the joint member on which it acts, it allows a moving away of said distal endpoint, and hence enables the movement of said joint member.

According to an embodiment, said tendon **90** and opposite tendon **190** have lengths that result in said jointed device **70** of said medical instrument **60** being held in a reference position when said tendon **90** and said opposite tendon **90** are tensioned by the respective tensioning elements **99**, **199**.

According to an embodiment, said frame **57** comprises at least one shaft **65**, in which a longitudinal shaft direction X-X is defined, said direction coinciding or being parallel to the axis of longitudinal development of said shaft **65**.

According to an embodiment, said tendon **90** comprises at least one longitudinal tendon portion **19**, in which the tendon path T-T is substantially parallel to the longitudinal direction of the shaft X-X, determining a movement of at least said longitudinal portion of tendon **19** with respect to said shaft **65**, at least along the shaft direction X-X.

According to an embodiment, said pushing direction is parallel to the longitudinal direction of the shaft X-X.

According to an embodiment, said pushing direction is orthogonal to the longitudinal direction of the shaft X-X.

According to an embodiment, said tendon **90** is pretensioned. In this way, when said pusher assembly **90** stops exercising its pushing action on said tendon deflectable portion **93**, said tendon **90** remains substantially under tension. The provision of a pretensioned tendon allows a simple calibration of said tendon drive system **50**, making it possible to arbitrarily decide in which pose of the jointed device to position a zero pushing action pose.

According to an embodiment, said pusher assembly **94** always applies a minimum positive tension on tendon **90**. In this way, as said pusher assembly contacts said tendon deflectable portion **93**, said tendon **90** remain substantially always under tension. Provisioning a pretensioned tendon allows for the efficient control of the tendon path within the medical instrument **60**, under any operating conditions.

According to an embodiment, said tendon **90** also comprises a second tendon endpoint **92**, or distal tendon end-

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point **92**, suitable to pull a mobile element, which can be connected to said second distal tendon endpoint **92**.

According to an embodiment, following said tendon along its tendon path T-T one first encounters said first tendon endpoint **91**, then said at least tendon deflectable portion **93**, and then said second tendon endpoint **92**.

According to an embodiment, said mobile element is at least one portion of said medical instrument **60**, **160**, **260**, which is mobile with respect to said frame **57**.

According to an embodiment, when said tendon deflectable portion **93** is deflected by said pusher assembly **94**, said tendon **90** determines the movement of at least one portion of said jointed device **70** with respect to said frame **57**.

According to an embodiment, said pusher assembly **94** comprises at least one pushing element **95**, mobile with respect to said frame **57** and suitable to push a plunger **96**, such that said plunger **96** pushes on at least one tendon deflectable portion **93** of said tendon **90**.

According to an embodiment, at least one body is placed between said pushing element **95** and said plunger **96**.

According to an embodiment, said pushing element **95** is in contact with said plunger. In other words, said at least one pushing element **95** is suitable to push directly or indirectly on said plunger **96**.

According to an embodiment, said pushing element **95** is mobile with respect to said plunger **96** within a contacting position, in which said pushing element **95** is suitable to exercise a pushing action on said plunger **96**, and a non-contacting position, in which said pushing element **95** is disconnected from said plunger **96**, and it is not suitable to exercise any pushing action on said plunger **96**. According to an embodiment, in said contacting position said pushing element **95** is not necessarily in contact with said plunger **96**. In other words, according to an embodiment, said pushing element **95** exercises a pushing action via at least one intermediate body placed between said pushing element **95** and said plunger **96**.

According to an embodiment, said pusher assembly **94** also comprises at least one sterile barrier **87**, suitable to substantially impede mutual bacterial contamination of the two environments it separates.

According to an embodiment, said sterile barrier **87** is placed between said pushing element **95** and said plunger **96**.

According to an embodiment, said sterile barrier is of a form and material suitable to transmit the push of said pushing element **95** to said plunger **96**.

According to an embodiment, said pushing element **95** is mobile with respect to said frame **57** along a substantially linear trajectory.

According to an embodiment, said pushing element **95** is a piston.

According to an embodiment, said drive system **50** comprises at least two tendon guiding elements **97**, or guiding pulleys, positioned along said tendon direction T-T such that when said pusher assembly determines a deflection of said tendon path T-T, said at least said two tendon guiding elements **97** cooperate to confine the deflection of said tendon path T-T to the tendon path section between said two guiding elements **97**.

According to an embodiment, said plunger **96** comprises at least one plunger idle pulley **98**, suitable to push on said tendon deflectable portion **93**, and in which said plunger idle pulley **98** is suitable to freely turn around its axis, and in this way to reduce the sliding friction over said tendon deflectable portion **93** at least when pushed by said plunger **96**.

According to an embodiment, said plunger idle pulley **98** is a ball bearing.

According to an embodiment, said second tendon endpoint **92** is a boss or a loop or a knot.

According to an embodiment, said tendon **90** is suitable to be pretensioned.

According to an embodiment, said tendon drive system **50** comprises at least one pretensioning element **99**, suitable for maintaining said tendon **90** pretensioned.

According to an embodiment, said pretensioning element **99** is a spring, suitable to apply a force between the frame **57** and the plunger **96**, to impose a preload on said tendon **90** that is substantially proportional to the compression movement of said spring **99**.

According to an embodiment, said pusher assembly **94** comprises an electric motor, suitable to move said pushing element **95**.

According to an embodiment, said pusher assembly **94** comprises a lead screw and nut type actuator. According to an embodiment, said actuator comprises a ball screw.

According to an embodiment, said tendon **90** is at least partially made of a material that is softer than the materials of the surfaces over which it slides. In other words, said tendon **90** is at least partially made of material that is less hard than the surface over which it slides.

According to an embodiment, said tendon **90** is at least partially made of polymeric material. The provision of a tendon made at least partially of polymeric material allows a reduction in wear of the surfaces over which it slides, with respect to a tendon made of metal, for example.

According to one variant of an embodiment, said first tendon endpoint **91**, is fastened to said plunger **96**, instead than to said frame **57**.

According to an embodiment, said tendon drive system **50** comprises at least one further tendon **190**, or opposite tendon **190**, opposed to said tendon **90** and fastened or constrained in its first endpoint **91**, or proximal endpoint, to said frame **57**, said tendon **190** extending along the tendon direction T-T, or tendon path T-T.

According to an embodiment, said tendon drive system **50** comprises at least one further pusher assembly **94**, or opposite pusher assembly **194**, opposed to said pusher assembly **94** and suitable to push on at least one portion of tendon deflectable portion **93** of said opposite tendon **190**, along a transversal pushing direction of tendon T-T such as to deflect the tendon path T-T and to induce an increased tensile load in said opposite tendon **190** and said tendon **90**. In other words, said tendon **90** and said opposite tendon are suitable to work opposed to each other like antagonistic muscles of the human body that cooperate to determine the adduction and abduction movements of a joint.

According to an embodiment, said opposite pusher assembly **194** pushes on said tendon deflectable portion **93** of said opposite tendon **190** along a pushing direction transversal to said tendon path T-T, deflecting said tendon path T-T, inducing tensile load in said opposite tendon **190**, from its proximal portion **18** and inducing tensile load in said tendon **90**, from its distal portion **19**.

According to an embodiment, said tendon **90** and said tendon **190** are distally structurally connected by a junction between said tendon and said opposite tendon. According to an embodiment, said tendon and said opposite tendon are both distally structurally connected to a common junction element, such that the transmission of the force by said tendon to said opposite tendon is guaranteed.

According to an embodiment, said opposite tendon **190** comprises a second endpoint **92**, or distal endpoint **92**,

suitable to pull a mobile element associable to said second tendon endpoint **92** of said opposite tendon **190**.

According to an embodiment, said opposite tendon **190** comprises a second endpoint **92**, or distal endpoint suitable to pull a common, single mobile element, associable to both said second tendon endpoint **92** of said tendon **90** and said second tendon endpoint **92** of said opposite tendon **190**. According to an embodiment, said tendon **90** and said opposite tendon **190** have lengths such that said common, single mobile element is in a reference position when said tendon **90** and said opposite tendon **190** are pretensioned by their respective pretensioning elements.

According to an embodiment, said tendon **90** and said opposite tendon **190** are two portions of a single tendon **90**.

According to an embodiment, said second tendon endpoint **92** of said tendon **90** and said second endpoint **92** of said opposite tendon **190** coincide and are suitable for pulling a common mobile element, which can be associated both to said second endpoint **92** of said tendon **90** and to said second tendon endpoint **92** of said opposite tendon **190**.

According to an embodiment, said opposite tendon **190** comprises at least a longitudinal portion **19**, in which the tendon path T-T is substantially parallel to the longitudinal direction of the shaft X-X, such as to move at least said longitudinal portion **19** of said opposite tendon **190** with respect to said shaft **65**, at least along the longitudinal direction of the shaft X-X.

According to an embodiment, said tendon drive system **50** comprises at least a pair of tendons **90**, **190** for each degree of freedom, in which said tendon pair comprises a tendon **90** and an opposite tendon **190**.

According to an embodiment, said tendon **90** and said opposite tendon **190** are suitable to be pulled simultaneously, such that the force transmitted to the common mobile element by both said tendon **90** and said opposite tendon **190** is the sum of the force transmitted by said tendon **90** and said opposite tendon **190**.

According to an embodiment, said tendon **90** and said opposite tendon **190** are suitable to be simultaneously pulled with substantially the same amount of force.

According to an embodiment, said tendon **90** and said opposite tendon **190** are suitable to be pulled with a force on one of them higher than on the other.

According to an embodiment, said tendon **90** and said opposite tendon **190** are suitable to be simultaneously pulled, retrieving substantially the same tendon length from their proximal portion.

According to an embodiment, said tendon **90** is suitable to be pulled, retrieving a first tendon length from its proximal section and simultaneously said opposite tendon **190** is suitable to be released by its proximal portion, releasing a second tendon length from the opposite tendon, substantially equal to said first tendon length.

According to an embodiment, said tendon drive system **50** comprises an opposite pretensioning element **199**, suitable for maintaining said opposite tendon **190** pretensioned.

According to an embodiment, said opposite pretensioning element **199** is a spring **99**.

According to an embodiment, said pretensioning element **99** and said opposite pretensioning element **199** are suitable to cooperate to simultaneously maintain said tendon **90** and said opposite tendon **190** pretensioned, so that the pusher assembly **94** and said opposite pusher assembly **194** can work at the same time.

The provision that said pretensioning element **99** and said opposite pretensioning element **199** allows said tendon **90**

and said opposite tendon **190** to be kept in their pretensioned state, with a pretension value suitable to counterbalance the weight of said common mobile element attached to them. In this way, the gravitational force has no role in the drive system.

According to an embodiment, said tendon **90** and said opposite tendon **190** are suitable to connect their second endpoints **92** to their respective tendon fastening points **82**, or tendon termination features **82**, to one of: said second joint member **72** and said terminal member **77**, such as to move it in opposite directions. The cooperation between said characteristic and the provision of said pretensioning element **99** and said opposite pretensioning element **199** allows for all movements to be positively guided and controlled, avoiding any passive or free joint movements, such as from return springs.

According to an embodiment, said tendon drive system **50** comprises a plurality of tendons **90** and a plurality of opposite tendons **190**.

According to an embodiment, said tendon drive system **50** comprises a plurality of pusher assemblies **94** and a plurality of opposite pusher assemblies **194**.

According to an embodiment, said plurality of tendons **90** and said plurality of opposite tendons **190** are positioned on a portion of a drum **59**, or drum **59**, of said frame **57** such that the tendon path T-T of each tendon **90**, **190** runs separate with respect to the path of all other tendons **90**, **190**.

According to an embodiment, said plurality of tendons **90** and said plurality of tendons **190** are positioned substantially radially, or as rays, on said drum **59**. According to an embodiment, said plurality of tendons **90** and said plurality of opposite tendons **190** are configured one said drum **59** like a cylinder of a radial engine, and in which the paths of said tendon **90** and said opposite tendon **190** do not cross each other on said drum **59**.

According to an embodiment, each tendon **90** of said plurality of tendons **90** is suitable to be engaged by its respective pusher assembly **94** independently of other tendons **90**.

According to an embodiment, said tendon **90** of said plurality of tendons **90** is suitable to be engaged by its respective pusher assembly **94** independently of an associated opposite tendon **190**.

According to an embodiment, a drive system assembly for a medical instrument **60**, **160**, **260** comprises:

- at least one tendon drive system **50** according to one of any embodiments previously described,
- at least one medical instrument **60**, **160**, **260** comprising at least one jointed device **70**, **170**, **270** in which said jointed device **70**, **170**, **270** comprises at least one rotational joint.

According to an embodiment, said tendon **90**, **190** is fastened or constrained at its second endpoint **92** to at least a portion of said jointed device **70**, **170**, **270** mobile with respect to said frame **57**, such that said tendon **90**, **190** is suitable to pull on at least a portion of said jointed device **70**, **170**, **270**, moving it with respect to said frame **57**.

According to an embodiment, said tendon **90** and said opposite tendon **190** are both fastened to a same portion of said jointed device **70**, **170**, **270**, mobile with respect to said frame **57**, in their respective second endpoints **92**, such that said opposite tendon **190** is suitable to pull at least a portion of said jointed device **70**, **170**, **270**, moving it with respect to said frame **57** by a movement which is opposite to the movement determined by said tendon **90**.

According to an embodiment, said drive system assembly comprises a tendon pair **90**, **190**, and said tendon pair

comprises a tendon **90** and an opposite tendon **190**, for every degree of freedom of movement of said jointed device **70**, **170**, **270**.

According to an embodiment, when said tendon **90** and said opposite tendon **190** are pulled simultaneously and with substantially the same amount of force, the movement of at least a portion of said jointed device **70**, **170**, **270** of said medical instrument **60**, **160**, **260** is impeded.

According to an embodiment, when said tendon **90** and said opposite tendon **190** are simultaneously pulled with different amounts of force, where one amount of force is greater than the other, a controlled movement of at least a portion of said jointed device **70**, **170**, **270** of said medical instrument **60**, **160**, **260** results.

According to an embodiment, said medical instrument **60**, **160**, **260** is at least one of: a surgical instrument, a micro-surgical instrument, an instrument for laparoscopic surgery, an endoscopic instrument, an instrument for biopsies.

According to one aspect of the invention, a tendon **90**, **190** for a medical instrument **60**, said medical instrument **60** comprising at least one jointed device **70** and one frame **57**, is suitable to move at least a portion of said jointed device **70** with respect to said frame **57**.

Said jointed device **70** has at least one degree of freedom of movement with respect to said frame **57**.

Said tendon **90** is exclusively suitable for working under tensile load.

Said tendon **90** is fabricated in a material that is less hard than the material of said jointed device **70**.

The provision of this characteristic allows the fabrication of a medical instrument **60** comprising a jointed device **70** with greater resistance to wear, caused by the sliding of tendon **90** over at least a portion of said jointed device **70**. Furthermore, this characteristic avoids any wear and loss of material of the surface of the jointed device **70** over which the tendon slides. In other words, the provision of this characteristic avoids said jointed device **70** from becoming scratched due to the effects of the tendon **90** sliding over it, when in operating conditions.

According to an embodiment, said tendon **90** slides over at least one portion of said jointed device **70**, when in operating conditions.

According to an embodiment, said tendon **90** is made of a construction that is not suitable for transmitting pushing.

According to an embodiment, said tendon **90** is fabricated of a softer material than the material of said jointed device **70**.

According to an embodiment, said tendon **90** is fabricated in a polymeric material. The provision of a tendon that is at least partially fabricated in a polymeric material allows the wear of the surface over which it slides to be reduced, with respect to a tendon made of metal for example, and helps to preserve the geometric tolerances established during the design phase and subsequently prolongs the life of said tendon **90**, **190** as well as the life of said medical instrument **60**, **160**, **260**.

According to an embodiment, said tendon **90**, **190** is made of polyethylene. According to an embodiment, said tendon **90**, **190** is made of high molecular weight polyethylene, or UHMWPE. According to an embodiment, said tendon **90**, **190** is made of Kevlar. According to an embodiment, said tendon **90**, **190** is made of Vectran. According to an embodiment, said tendon **90**, **190** is made of Zylon, or PBO. According to an embodiment, said tendon **90**, **190** is made of a combination of the above materials.

According to an embodiment, said tendon **90**, **190** is made of polymer fibers.

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According to an embodiment, said jointed device **70** is made of a metallic material.

According to an embodiment, said jointed device **70** is made of at least one of: INOX steel or stainless steel; super-fast steel; widia; hardened steel; tempered steel; titanium.

According to an embodiment, said jointed device **70** is made of a ceramic conductive material.

According to an embodiment, said tendon **90** comprises at least one tendon endpoint **91**, suitable to be glued to said frame **57**.

According to an embodiment, said tendon **90** is unraveled into strands around its first tendon endpoint **91** such as to maximize the glued surface.

According to an embodiment, said tendon **90** comprises at least a second tendon endpoint **92**, suitable to connect to at least a portion of said jointed device **70**.

According to an embodiment, said second endpoint **92** is a boss. According to an embodiment, said second tendon endpoint is a loop. According to an embodiment, said second tendon endpoint **92** is a knot.

According to an embodiment, said second tendon endpoint **92** is glued to at least one portion of said jointed device **70**.

According to an embodiment, said first tendon endpoint **91** is terminated by wrapping said tendon around a portion of said medical instrument **60** multiple times. According to an embodiment, said second endpoint **92** is terminated by wrapping said tendon around a portion of said medical instrument **60** multiple times. According to an embodiment said tendon is wrapped around with a curvature radius that is substantially equal to its diameter.

According to an embodiment, said tendon **90, 190** has a diameter between 0.05 mm and 0.3 mm.

According to an embodiment, said tendon **90, 190** has an elastic module between 50 GPa and 100 GPa.

According to an embodiment, said tendon **90, 190** is fabricated such as to have a curvature radius inferior or substantially equal to one millimeter.

According to an embodiment, said tendon **90** is exclusively suitable to work under tensile load applied at the endpoints, avoiding said tendon to be pinched, to be laterally guided in a channel or to comprise a sheath.

According to an embodiment, said tendon **90, 190** is suitable to be pre-lengthened with a load cycle comprising at least two loads of an entity equal to at least half of the tensile breaking strength of said tendon **90, 190**.

According to an embodiment, said tendon **90** has a transverse dimension, that is a dimension that is substantially orthogonal with respect to said tendon path T-T, variable in different tendon portions.

According to an embodiment, said tendon **90, 190** has a substantially circular cross section.

According to an embodiment, the diameter of said tendon **90** is variable in different portions of said tendon **90**.

According to an embodiment, said tendon **90** is thinner at said second tendon endpoint **92**. According to an embodiment, said tendon **90** is thicker in said longitudinal portion **19**. This way, the tendon **90, 190** is suitable to be more flexible close to or at the tendon fastening point **82**, as well as being stiffer close to or on the inside of said shaft **65**.

According to an embodiment, the mechanical properties of said tendon **90** are variable in different portions of said tendon **90**.

According to an embodiment, said tendon **90, 190** is obtained by joining or juxtaposing tendon portions with different characteristics.

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According to an embodiment, the composition of said tendon **90, 190** is variable in different portions of said tendon **90, 190**.

According to an embodiment, said tendon **90, 190** has a diameter between 0.1 mm and 0.3 mm.

According to an embodiment, said tendon **90** is suitable to cooperate with an opposite tendon **190** to move at least a portion of said jointed device **70, 170, 270**.

According to an embodiment, when said tendon **90** and said opposite tendon **190** are suitable to be simultaneously pulled with one force being larger than the other, a controlled movement of at least a portion of said jointed device **70, 170, 270** or said medical instrument **60, 160, 260** results.

According to an embodiment, when said tendon **90** and said opposite tendon **190** are simultaneously pulled with the same force, the movement of at least a portion of said jointed device **70** of said medical instrument **60** is impeded.

According to an embodiment, a tendon pair **90, 190**, in which every pair comprises a tendon **90** and an opposite tendon **190** is foreseen for every degree of freedom of movement for said jointed device **70**.

In the following a driving method for a robotic assembly **100** is described.

A driving method of a surgical robotic assembly comprises the following phases:

provide a robotic assembly **100** according to one of any of the embodiments previously described.

employ at least a vision system associable to the robotic assembly **100** for the visualization of at least a portion of the patient **201**.

position said macro-positioning arm **30**, such that the work volume **7**, reached by at least a portion of said terminal portion **77** is within the field of view of said at least one vision system **103** associable to said robotic assembly **100**;

drive at least one micro-positioning device **41, 141, 241, 341**;

drive at least one jointed device **70, 170, 270** of a medical instrument **60, 160, 260, 360**.

According to one possible operating mode, a driving method of a surgical robotic assembly comprises at least one of the following further phases, listed in a preferred, but not necessary order:

release said macro-positioning arm **30** so as to be able to drag it.

position said macro-positioning arm **30**, so that the work volume **7** reached by said at least one terminal portion **77** is within the vision field of said at least one vision system **103**, associable to said robotic assembly **100**;

lock said macro-positioning arm **30**;

drive said at least one micro-positioning device **41, 141, 241** by means of said at least one control device **20**;

drive said at least one jointed device **70, 170, 270** of the medical instruments **60, 160, 260** by means of said control device **20**.

A control method for a control device for microsurgery for a microsurgical robotic assembly is described below.

A control method for a control device for microsurgery for a microsurgical robotic assembly comprises the following phase, listed in a preferred but not necessary order.

provide at least one microsurgical control device **20** according to one of any of the embodiments previously described;

manipulate said control instrument **21**;

move at least on portion of said control instrument **21** with respect to said detection device **22**.

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According to one possible operating mode, one method comprises at least one of the following further phases:

provide a microsurgical robotic assembly **100** according to one of the embodiments previously described;

move said surgical micro-instrument **60, 160, 260** by means of said control instrument **21**;

move said micro-positioning device **41, 141, 241** by means of said control instrument **21**;

use a microscope **103** associable to said robotic assembly **100** to visualize at least one portion of a patient **201**;

activate a teleoperation condition, or mode, according to which a movement of the control instrument **21** in a first direction, with respect to a coordinate system associated to at least one of said detection device **22** and said microscope **103**, corresponds to a movement of said surgical micro-instrument **60, 160, 260** in the same direction with respect to said coordinate system.

According to an embodiment, said portion of the patient **201** is comprised in said work volume **7**.

According to one possible operating mode, a method comprises the following further phases:

provide a further control device **20** such as to comprise a first control device **120** and a second control device **220**;

manipulate said first control device **120** with one first hand;

manipulate said second control device **220** with a second hand.

According to one possible operating mode, one method comprises the following further phases:

provide a further control instrument **21**, such as to comprise a first control instrument **121** and a second control instrument **221**;

manipulate said first control instrument **121** with one hand;

manipulate said second control instrument **221** with the other hand.

According to an embodiment, said joint members **71, 72, 73, 74** are obtained by wire electro-discharge machining.

According to an embodiment, said joint members **71, 72, 73, 74** are obtained by micro-injection moulding.

According to an embodiment, said joint members **71, 72, 73, 74** are obtained by 3D printing.

A method for the fabrication of said medical instrument **60, 160, 260** is described below.

According to one possible operating mode, a fabrication method for the medical instrument **60, 160, 260** comprises a phase of fabrication of a medical instrument **60, 160, 260** according to one of any embodiments previously described, by at least one additive manufacturing technique.

According to one possible operating mode, a fabrication method of a medical instrument **60, 160, 260** comprises a phase of fabrication a medical instrument by micro-injection molding. In other words, a fabrication method for the medical instrument **60, 160, 260** comprises a phase of fabrication of a medical instrument by means of micromolding.

A driving method of a tendon **90, 190** for a medical instrument **60, 160, 260** is described below.

A driving method of a tendon **90** for a medical instrument **60, 160, 260** comprises the following phases, listed in a preferred, but not necessary order of execution:

A') provide a tendon drive system **50** according to one of any of the previously described embodiments;

B') push at least on a portion of said tendon **90, 190** such as to deflect its tendon path T-T;

C') generate a tensile load in said tendon **90, 190**.

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According to one possible operating mode, a method comprises the further phase of providing a drive system assembly according to one of any of the embodiments previously described.

According to one possible operating mode, one method comprises at least one of the following further phases:

D') pretension said tendon **90** before phase B;

E') drive said pusher assembly **94** before phase B and after phase D;

F') after phase C, move at least one portion of said jointed device **70, 170, 270** of said medical instrument **60, 160, 260**;

G') after phase F'), drive said opposite pusher assembly **194**;

H') after phase G'), move said at least one portion of said jointed device **70, 170, 170** of said medical instrument **60, 160, 260** of phase F') in an opposite direction.

According to one possible operating mode, one method comprises the further phases of:

I') simultaneously drive said pusher assembly **94** and said opposite pusher assembly **194**.

J') pull said tendon **90** and said opposite tendon **190** with differing amount of forces, force on one being greater than on the other;

K') move at least one portion of said jointed device **70, 170, 270** of said medical instrument **60, 160, 260** by a controlled movement.

According to one possible operating mode, a method comprises the further phases of:

L') instead of phase J'), pull said tendon **90** and said tendon **190** with substantially the same amount of force;

M') instead of phase K'), impede the movement of at least a portion of said jointed device **70, 170, 270** of said medical instrument **60, 160, 260**.

According to one possible operating mode, one method comprises the following phases instead of the phases I'), J'), K'):

N') drive simultaneously said pusher assembly **94** and said opposite pusher assembly **194**;

O') simultaneously pull said tendon (**90**) to retrieve a first tendon length from its proximal portion and release said opposite tendon (**190**) by its proximal portion releasing a second length of the opposite tendon, substantially equal to the first tendon length,

P') move at least one portion of said jointed device **70, 170, 270** of said medical instrument (**60, 160, 260**) by a controlled movement in relationship to said tendon length and opposite tendon length.

According to one possible operating mode, one method comprises the further phases of:

drive said opposite tendon **190** by means of said opposite pusher assembly **194**;

move at least a portion of said medical instrument **60, 160, 260** by means of said pusher assembly **94**.

A method to replace a tendon **90, 190** for a medical instrument is described below.

According to one possible operating mode, a method for replacing a tendon **90, 190** comprises the following phases: provide a further tendon **90, 190** according to any of the embodiments previously described;

A") detach said tendon **90, 190** from said medical instrument **60**;

B") mount said further tendon **90, 190** on said medical instrument **60**.

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According to one possible operating mode, the tendon **90** is attached first at said second tendon endpoint **92** and then at said first tendon endpoint **91**.

According to one operating mode, a method comprises the following further phases:

C") before the phase A"), lock said plunger (**96**), in a position suitable to eliminate any pretension on the associated tendon **90**.

According to an embodiment, said plunger (**96**) is locked by the use of a pin inserted in the plunger locking hole **48**.

According to one possible operating mode, one method comprises the following further phases:

D") between the phase A") and the phase B"), clean said medical instrument **60**.

According to one possible operating mode, said phase D") comprises a further sub-phase, which entails the immersion of said medical instrument **60** in a bath of organic solvents.

According to one possible operating mode, said phase A") comprises a further sub-phase of dissolving said any remain of tendon **90**.

According to one possible operating mode, said phase A") comprises a further sub-phase of introducing said medical instrument **60** in an autoclave or other sterilization system.

According to one possible operating mode, said phase A") comprises a further sub-phase of introducing said medical instrument **60** in an oven at a temperature between 25° C. and 150° C.

According to one possible operating mode, said phase A") comprises a sub-phase of immersing of said medical instrument **60** in a chemical organic solvent bath.

According to one possible operating mode, said phase B") comprises the following sub-phases, preferably, but not necessarily, in the following order:

lock said jointed device **70** in a reference position and/or

lock said plunger **96** in its locked position;

connect said second endpoint **92** to said jointed device **70**;

thread said further tendon **90**, **190** inside said shaft **65**,

connect said first tendon endpoint **91** to said frame **57**.

According to one possible operating mode, one method comprises the following further phase:

E") after the phase B"), calibrate of said medical instrument **60**, **160**, **260** identifying a new zero position for the plungers.

A fabrication method of the jointed device **70**, **170**, **270** is described below.

According to one aspect of the invention, one fabrication method of a jointed device **70**, **170**, **270** comprises at least the following phases, in the preferred order indicated below:

(A") provide a machining fixture **112** on an EDM machine and arrange a plurality of workpieces **117** on said machining fixture **112**.

(B") cut the desired geometry on said workpieces **117** with cutting lines parallel to each other.

The provision of a single cutting step on said workpieces with cutting lines parallel to each other, allows the machining of surfaces that are parallel to each other on said workpieces, with an extreme precision of parallelism.

According to one possible operating mode, the machining method described above allows the machining of ruled surfaces characterized by parallel generatrices on said workpieces **117**.

According to one possible operating mode, one machining method as described above allows the cutting of workpieces of very small dimension, for example of millimetric or sub-millimetric dimensions.

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According to an embodiment, said machining method is suitable to fabricate at least one jointed device **70** that comprises a plurality of joint members **71, 72, 73, 74, 75, 76, 77, 78**.

According to one possible operating mode, said machining method is suitable to machine parallel cuts on said workpieces **117** such as to form joint members comprising surfaces parallel to each other.

According to one possible operating mode, said machining method is suitable for machining parallel cuts on said workpieces **117** such as to form joint members suitable to be assembled in a complementary fashion because they comprise surfaces that are parallel to each other.

According to one possible operating mode, said EDM machine is suitable to perform wire EDM and comprises a cutting wire **115**.

According to an embodiment, said cutting wire **115**, or EDM wire **115**, or electrical discharge machine wire **115** is of a diameter between 30 microns and 100 microns, and is preferably of 50 microns.

The provision of a machining method as described above allows exclusively thermal energy to be transferred to the piece being machined **117**, avoiding any mechanical energy to be transferred to the piece being machined **117**, for example inducing flexion, as it is the case when carrying out cuts with a milling machine.

According to an embodiment, said machining method is suitable to fabricate at least one jointed device for applications in the medical-surgical sector.

According to an embodiment, said machining method is suitable to fabricate at least one jointed device, suitable for applications in precision mechanics, for example suitable for use in watchmaking. According to an embodiment, said machining method is suitable to fabricate at least one jointed device, suitable for applications in the jewelry and/or fashion jewelry sector. According to an embodiment, said machining method is suitable for the fabrication of at least one jointed device, suitable for applications in the assembly of electromechanical products.

According to one possible operating mode, the phase (A") comprises the following sub-phases:

mount a plurality of workpieces on said machining fixture **112** in their respective member seats **116**.

According to one possible operating mode, a sub-phase is first carried out during said phase (A"):

(A1") provide a machining fixture **112** on an EDM machine; and then the sub-phase:

(A2") arrange a plurality of workpieces **117** on said machining fixture **112**.

According to a possible operating mode, one method comprises the following further phase between the sub-phase (A1") and the sub-phase (A2"):

(C") carry out a calibration.

According to one possible operating mode, one method comprises the following further phase between the phase (A") and the phase (B"):

(C") carry out a calibration.

According to one possible operating mode, one method comprises the following further phases after the phase (B").

(D") rotate said machining fixture **112**.

repeat said phase (B").

According to one possible operating mode, said phase of rotating said machining fixture **112** comprises a further phase of using a rotary table to rotate said machining fixture **112**, avoiding to dismount said machining fixture **112** from the cutting machine to carry out the following phases:

rotate said machining fixture **112**;

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carry out a second calibration, or cut calibration, exclusively on said reference rod **118**, repeat said phase (B').

According to one possible operating mode, said phase (C'''), carry out a calibration, comprises the following sub-phases:

switch on the EDM machine;

provide a reference rod **118** with its axis parallel to said member seats **116** of the workpieces **117**;

bring said cutting wire **115** in contact with a first portion **122** of said reference rod **118**, or portion facing towards the side of wire approach **122**;

measure, or register, the position of said wire;

and/or

measure, or register, the position of said cutting wire **115**, when it is in contact with a first portion of a first workpiece to be machined, or the portion facing the side of wire approach; execute the previous phase for each workpiece **117**;

and/or

bring the cutting wire **115** in contact with a second rod portion **123** of said reference rod **118**, or portion facing the side of wire departure **123**, opposite with respect to said first rod portion **122**;

measure, or register, the position of said cutting wire **115**; compute the position of the axis of said reference rod **118** as a midpoint between the position of said wire when in contact with said first rod portion and the position of said wire when in contact with said second rod portion.

and/or

measure, or register, the position of said cutting wire **115** when in contact with a second portion of said first workpiece, or the portion facing the side of wire departure;

compute the position of said first workpiece as a midpoint between the position of said wire when in contact with said first portion of the workpiece and the position of said wire when in contact with said second portion of the workpiece;

and/or

execute the previous phase for each workpiece **117**;

and/or

repeat the procedure for all cutting planes X-Y, Y-Z, X-Z.

According to an embodiment, said machining fixture **112** of a jointed device **70**, **170**, **270** is suitable to be mounted on a machine for EDM.

According to an embodiment, said machining fixture **112** is suitable to perform at least two cuts on different cutting planes on workpieces **117** by using a single cutting profile **110** per cutting plane.

According to one realization, said machining fixture **112** comprises a first pair of fixing surfaces **113**, **114**, which are rectified, opposite and substantially parallel to each other and substantially orthogonal to a first plane of cutting X-Y.

According to an embodiment, said machining fixture **112** comprises a second pair of fixing surfaces **134**, **135**, which are rectified, opposite and substantially parallel to each other and substantially orthogonal to a second plane of cutting Y-Z.

According to an embodiment, said first pair of fixing surfaces **113**, **114** and said second pair of fixing surfaces **134**, **135** are rectified.

According to an embodiment, each pair of locating surfaces comprises at least one base fixing surface **113**, **135** and at least one fixture fixing surface **114**, **134**.

According to an embodiment, said plurality of member seat **116** are sequentially arranged such that a translating

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straight line, substantially orthogonal to said first cutting plane X-Y, or substantially orthogonal to said second cutting plane Y-Z, would intersect at most only one of said workpieces **117** at a time, when said workpieces are mounted in respective member seats **116**.

According to an embodiment, said member seats **116** are substantially parallel to each other.

According to an embodiment, said machining fixture **112** also comprises a pair of locating surfaces, opposite and substantially parallel to each other and substantially orthogonal to a third cutting plane X-Z.

According to an embodiment, said third pair of locating surfaces comprises at least a guide hole **125**, and the EDM wire **115** of said EDM machine is inserted in at least one said guide hole **125**, to avoid the EDM wire coming into contact with at least one machining fixture **112**, during the cut.

According to an embodiment, said machining fixture **112** also comprises:

a plurality of member seats **116**, each suitable to receive at least one workpiece **117**, said workpiece **117** being suitable to realize at least one portion of said jointed device **70**, **170**, **270**.

According to an embodiment, said machining fixture **112** also comprises at least one reference rod **118**, suitable to allow for the cut calibration.

According to an embodiment, said machining fixture **112** comprises at least one fixing element, or fastening element, suitable to firmly connect said at least one workpiece **117** in its respective member seat **116**.

According to an embodiment, said at least one fastening element is conductive glue.

According to an embodiment, said at least one fastening element is a grub screw.

According to an embodiment, said grub screw is suitable to be mounted in a threaded hole supplied in said at least one fastening surface.

According to an embodiment, said fastening grub screw, is suitable to penetrate in said threaded hole of said fastening surface.

According to an embodiment, said machining fixture **112** comprises four member seats **116** and a reference rod **118**.

According to an embodiment, each member seat **116** is substantially positioned at the same distance from its respective fastening surface.

According to an embodiment, said fastening surfaces are positioned in a stepwise manner, such as to form a stair shape in profile. In other words, said fastening surfaces are positioned in a stepwise manner, such as to form a stair shape in profile with respect to at least one cutting plane X-Y, Y-Z, X-Z.

According to an embodiment, said machining fixture **112** has a surface facing towards any cutting plane X-Y, Y-Z, X-Z inferior to 10000 square millimeters.

According to an embodiment, said machining fixture **112** has a surface facing towards any cutting plane X-Y, Y-Z, X-Z inferior to 5000 square millimeters.

Known microsurgical procedures are carried out manually by the surgeon **200**, or micro-surgeon **200**, by the use of manual instruments, such as forceps, scissors and needle holders used to manipulate very fragile tissues and ducts with a diameter of 1 mm or less. The microsurgical procedure step most commonly performed is anastomosis, in which two small, severed vessels are sutured back together to reestablish blood flow. This procedure is carried out by holding the two adjacent vessel stubs with specific clamps and by using small caliber needles to perform the suture. The micro-surgeon **200** must hence perform very small move-

ments, trying to limit the natural tremor of the hand and to maintain a high level of both concentration and sensitivity in order to delicately manipulate the fragile tissues with which he/she interacts via the instruments. It is apparent that robotics can bring significant improvement to the performance of complex microsurgical procedures.

According to an embodiment, said robotic assembly **100** has the function of supporting the surgeon **200** in the execution of a microsurgical procedure by using jointed devices and robotic devices that guarantee extremely precise movements, that scale down the actual hand movement of the surgeon **200** eliminating any tremor while reproducing the kinematics of the human wrist on a small scale.

According to an embodiment, said surgical robotic assembly **100** comprises a support **104**, an articulated macro-positioning arm **30**, and a pair of micro-positioning devices **41**, **141**, **241**. A medical instrument **60**, **160**, **260**, which comprises a motor box **61** and a sterile jointed device **70**, **170**, **270** is attached to each micro-positioning device **41**, **141**, **241**.

Two control devices **20**, suitable for the robotic control of the two medical instruments **60**, **160**, **260** and of the micro-positioning devices **41**, **141**, **241**, are connected to the support **104** by communication cables **109**. All the electronic control circuit boards and the power sources of the robotic assembly **100** are integrated in the support **104**, while a control panel **108**, for switching on and off and the management of user messages from the robotic assembly **100** by an operator, is situated on its surface. A dedicated, external video-microscope entry allows the integration of any traditional external microscope **103** for microsurgery. A digital microscope **103** is integrated in the system to visualize the substantially overlapping work volume **7** of the two sterile jointed devices **70**, **170**, **270**.

According to an embodiment, a possible configuration of the surgical robotic assembly **100** is specifically dedicated to performing microsurgical procedures at the limb extremities or on free flaps. This is composed of an operating table **102** on which the limb to be operated on, or the free flap, is placed and comprises the use of a pair of jointed devices **70**, **170**, **270** connected to micro-positioning devices **41**, **141**, **241** and remotely controlled in real time by the microsurgeon **200** by their respective control devices **20**. Note that microscope **103** is not part of the surgical robotic assembly **100** but is an independent element, fundamental for the visualization of the work volume **7** during the performance of the procedure.

According to an embodiment, a possible configuration of the surgical robotic assembly **100**, particularly suitable for breast reconstruction procedures, but also suitable for carrying out microsurgeries on all other body parts, is composed of: a support **104** which allows for the support of the surgical robotic assembly **100** and for its transfer into the operating room to a position adjacent to the mobile operating table **102** on which the patient **201** is lying, one passive, articulated macro-positioning arm **30** that extends from the support **104** and allows the active part of the surgical robotic assembly **100** to reach the anatomical site involved in the procedure. A pair of precision micro-positioning devices **41**, **141**, **241**, or micro-positioning devices **41**, **141**, **241**, each with four degrees of freedom, to which the respective medical instruments **60**, **160**, **260** are attached, and which are used by the surgeon **200** to perform the microsurgical procedure by handling both the tissue and the small suture needles, are placed at the end of the surgical robotic assem-

bly **100**. The whole procedure is carried out under vision guidance provided by an external, traditional surgical microscope **103**.

According to an embodiment, the support **104** has both a structural and transport function for the surgical robotic assembly **100**, while the macro-positioning arm **30** connected to it allows the simultaneous positioning of a pair of micro-positioning devices **41**, **141**, **241** and the medical instruments **60**, **160**, **260** in proximity to the anatomical district which will be operated on. The micro-positioning devices **41**, **141**, **241** and the medical instruments **60**, **160**, **260** are actively moved and controlled in real time by the control devices **20**.

According to an embodiment, each control device **20** is equipped with a support clamp or bracket, which can be independently positioned, for example by connecting it to the operating table **102**. Said control devices **20** are connected to the surgical robotic assembly **100** by a power cable **107**, also suitable for the transmission of control data.

According to an embodiment, to simplify the transport of the surgical robotic assembly **100**, a retractable handle **106** and a foot platform **105** are positioned on a posterior side. The cart **104** has a control panel **108** on a posterior surface for the management of the parameters of the surgical robotic assembly **100** by the user and for the display of messages or warnings of the machine itself. On/Off switches (power buttons) and an emergency stop button are present on the same side. A power cable **107** supplies electrical current to the entire system, while the video data acquired by the digital microscope are passed to the surgical robotic assembly **100** via a communication cable **109**, such as to be able to integrate vision-derived information into the controls. According to an embodiment, said surgical robotic assembly **100** comprises a foot platform **105**, suitable to be used together or alternatively to a retractable handle **106** for the transport of the robotic assembly **100** during its positioning in the operating room, placed on the bottom of the posterior side of the cart.

Said foot platform **105** allows the foot of an operator responsible for the movement of said robotic assembly **100** to rest on it, such that the robotic assembly **100** can also be pushed from the base, eliminating the risk of its tipping over while it is moved.

According to an embodiment, the control device **20** has the function of controlling the robotic movement of the micro-positioning devices and of the medical instrument **60**, **160**, **260**. The control device **20** comprises a control instrument **21**, whose position in space is detected in real time by a magnetic tracking sensor. The magnetic tracking sensor is made of a magnetic field generator and of wired markers containing micro-bobbins, such as for example, but not limited to, the product "NDI AURORA V3 tracking system" comprising a "Planar field generator" and sensor "Mini 6DOF" by the company "NDI—Northern Digital Inc., 103 Randall Drive Waterloo, Ontario, Canada N2V1C5". The control instrument **21** integrates all the markers necessary for the detection of the six spatial coordinates of the control instrument **21** with respect to a base structure **67** and comprises an additional degree of freedom of gripping located in its tip portion **68**, whose angle of aperture is measured by a tip sensor **29**. Said tip sensor **29** is a position sensor or a proximity sensor. A connection tendon **23** connects the control instrument **21** to a base structure **67** that contains a magnetic field generator, suitable both for powering and data transmission between the control instrument **21** and said base structure **67**, particularly, but not necessarily when it comprises a detection device **22**. A power and

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communication tendon **24** connects the magnetic field generator to the external power source at the cart **104** of the robotic assembly, transferring the data relative to the position and orientation of the control instrument as well as the aperture angle of the forceps of the control instrument **21**. A further marker for the detection of the six spatial coordinates of the cart with respect to the base structure **67** is present on the support **104** and connected by the power and communication tendon **24** to the base structure **67**. Status signal lights **26** are integrated in the base structure **67** and communicate the activity of the control device to the user. A soft, dedicated, ergonomic operator support **27** is made to allow an ergonomic use of the control device **20**, while the control instrument **21** reproduces the geometry of traditional micro-instruments such as the forceps and needle holder to make their handling more intuitive and familiar to the surgeon.

According to an embodiment, the macro-positioning arm **30** allows the anatomical districts involved in the surgical procedure to be reached by the active parts of the robotic assembly **100**, such as, for example, the micro-positioning devices **41**, **141**, **241** and the medical instruments **60**, **160**, **260**. Said macro-positioning arm **30** is composed of four members **31**, **32**, **33**, **34** connected to each other in series by passive rotational joints each having vertical and parallel arm movement axes a-a, b-b, c-c. Inside each rotational joint, electromagnetic brakes allow the position of each single member to be locked in space. A dedicated brake release button **35**, positioned below on the bottom side of the fourth arm member **34** to facilitate its grasping and activation, allows all joint brakes to be simultaneously released and thus to reposition each arm member in space as required by the user. The new position can then be frozen by undepressing the release button **35**.

According to an embodiment, the first member **31** of the macro-positioning arm **30** is connected to a cart **104** by a rack and pinion mechanism that allows to manually control the movement of said macro-positioning arm **30** within a dedicated linear sliding guide **36** along a preferably vertical linear displacement axis, when a manual knob **37** is turned.

According to an embodiment, the fourth member **34** of the macro-positioning arm **30** has a rotational joint at its tip, which is manually activated by a dedicated rotational dial nut **43** that turns around a fourth axis of arm movement d-d, perpendicular to the third axis of arm movement c-c.

According to an embodiment, the macro-positioning arm **30** is connected to the support member **38** via the rotational joint, which is manually activated via the movement of said rotational dial nut **43**. A pair of micro-positioning devices **41**, **141**, **241** is connected to the two extremities of said support member **38** that also carries a video camera **45** in its middle section, which can display enlarged images of the work volume **7** in which the microsurgery is carried out. The medical instruments **60**, **160**, **260** are rigidly attached to a distal portion of the micro-positioning devices **41**, **141**, **241**.

According to an embodiment, the micro-positioning device **41**, **141**, **241** comprises three motorized slides **51**, **52**, **53**, orthogonally connected to each other and able to each move independently along respective three axes of linear displacement f-f, g-g, h-h, and a motorized rotary joint **46**.

According to an embodiment, said motorized slides **51**, **52**, **53** are motorized micro-slides. The medical instrument **60**, **160**, **260** is rigidly attached to the micro-positioning device **41**, **141**, **241** by a motorized rotary joint **46** that turns it around its longitudinal rotation axis r-r.

According to an embodiment, the medical instrument **60** has a motor box **61** that contains at least one tendon drive system **50** equipped to drive the jointed device **70** of said

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medical instrument **60** and its terminal device **77**. According to an embodiment, the transmission mechanism integrated inside the mechanical transmission box **62**, connected to the motor box **61**, transmits the motion to the medical instrument **60** via the shaft **65** to the jointed device **70** and to the terminal device **77**.

According to an embodiment the medical instrument **60** is made of a motor box **61** containing the actuators for driving the medical instrument **60**, the associated electronic control boards and motor driver boards. The mechanical transmission box **62**, which contains the mechanisms dedicated to transmit the motor motion via said shaft **65** along the longitudinal shaft direction X-X, to the jointed device and the terminal device **77**, is connected to said motor box **61**.

According to an embodiment, the motor box **61** contains six pushing elements **95** associated to three degrees of freedom of the medical instrument **60**. In particular, said pushing elements are moved by at least one pusher assembly **94**, which comprises electric micro-motors with linear transmission system lead screws. Actuation pistons **95** come out of the wall of motor box **61** facing the transmission box **62** and actuate the transmission mechanism integrated into the mechanical transmission box **62**.

According to an embodiment, the motor box **61** and the mechanical transmission box **62** are separated by a sterile barrier **87** and can be integrally connected with each other by connecting features, for example via a bayonet connection, as shown in FIG. **12**.

According to an embodiment, the shaft **65** is hollow, fabricated in metal, extends itself along the longitudinal shaft direction X-X and inserts itself into the mechanical transmission box **62**. The jointed device **70**, **170**, **270** with the terminal device **77** at the tip is inserted at the other shaft end or tip.

According to an embodiment, the six pushing elements **95**, implemented as actuation pistons connected to motors, couple with the respective plungers **96** of the mechanical transmission box **62** thus connecting the motor box **61** with the mechanical transmission box **62**.

According to an embodiment, said pushing element **95** and said plungers **96** are separated by a sterile barrier **87**.

According to an embodiment, the plungers **96** can move linearly along the piston movement axis and are maintained in a proper alignment by means of linear bushings not represented inserted in the first frame section **58**, or upper frame **58**, and by means of respective shoulder surfaces **88**.

According to an embodiment, the actuation of the jointed device **70** is assigned to six tendons **90**, or actuation cables **90**, which are independent and run from a tendon fastening surface **84** in the mechanical transmission box **62**, to the jointed device **70** of the medical instrument **60**, via the mechanical transmission box **62**, the tendon passage hole and the hollow shaft **65**.

According to an embodiment, in its section running inside of the mechanical transmission box **62**, each tendon **90** winds around each respective four guiding pulleys **97**, mounted on said lower frame **59**, such as to change its path direction until aligning with the instrument axis X-X. Such guiding pulleys **97** can be a fixed or idle pulleys and in a preferred configuration they are idle pulleys, with the exceptions of the first guiding pulley **197**, positioned closest to the first tendon endpoint **91**, which is a fixed guiding pulley **197**.

According to an embodiment, a further plunger idle pulley **98**, is positioned on each plunger **96** and moves integral with it along the linear piston pulley movement axis. Each actuation tendon **90** also partially winds around the respective plunger idle pulley **98**, fastened to the respective

plunger 96. Said plunger idle pulley 98 is located between said first guiding pulley 197 and a second guiding pulley 297.

According to an embodiment, the movement of the plunger 96 and hence of the plunger idle pulley 98 induced by the actuation piston 95, pushes the tendon 90 and hence varies its path length between said first guiding pulley 197 and said second guiding pulley 297. This change in length is transmitted by means of said transmission mechanism to the distal articulation of the medical instrument 60, 160, 260, resulting in its actuation.

According to an embodiment, a spring 99, suitable to work by compression, is inserted between the plunger idle pulley 98 and the upper frame 58 around the plunger 96.

According to an embodiment, said spring 99 generates a force directed along the plunger movement direction axis and establishes a variable preload on each plunger 96 sufficient to always keep the tendon 90 under a light tension and avoid its derailing from said guiding elements 97, 98, 197, 297 during changes in its tensile load.

According to an embodiment, a tendon guide element 89 maintains each tendon 90 in position and impedes its derailing, even in cases of anomalies such as a loss of tension in the tendons 90.

According to an embodiment, the jointed device 70 uses six low-friction, low minimum curvature radius and high stiffness polymeric tendons as movement transmission means for actuation of the three degrees of freedom of motion which the jointed device 70, 170, 270 is capable of. Each actuation cable, or tendon, 90, is glued with a low viscosity acrylic glue to tendon fastening surface 84 of the lower frame 59 and changes its direction by passing across four successive guide elements 97, 197, 297, integral to the lower frame 59 until it reaches the center of the transmission box 62 and enters through a central hole in the shaft 65 of the medical instrument 60, 160, 260 running in the direction of the instrument X-X, down to the jointed device 70, 170, 270.

As shown in FIG. 13, the first guiding pulley 197 of each actuation cables 90 is a fixed pulley 197 on which the tendon 90 winds. Successive guide elements are idle pulleys, around which tendon 90 is partly wound. Between said first guiding pulley 197 and said second guiding pulley 297 a space is provided allowing the linear motion of the plunger 96 actuated by the actuation piston 95.

According to an embodiment, at least one tendon 90 winds around at least four guiding pulleys 197, 297, 397, 497, thus defining a third guide element 397 and a fourth guide element 497. Between said third guide element 397 and said fourth guide element 497, a tendon guide element 89 keeps the tendon 90 in the correct position and avoids derailing of the tendon 90, even in cases such as an anomalous loss of tension.

According to an embodiment, the joint members that form the jointed device 70, 170, 270 and its terminal device 77, reproduce the kinematics of the human wrist adding a grasping degree of freedom of movement at the tip, for a total of three degrees of freedom of movement.

According to an embodiment, a first joint member 71 and a second joint member 72 are connected to each other by a rotational joint 171 around a first axis of rotation P-P, followed by a first portion of the terminal member 177 and a second portion of the terminal member 277, both connected to said second joint member 72, which freely rotate around a second axis of joint movement Y-Y, orthogonal to the first axis of joint movement P-P and providing a terminal device 77 at the tip.

According to an embodiment, the first member 71 locks on or is jointed in a concentric manner with the shaft 65 of the medical instrument 60 and is rigidly attached to it via fastening pins 76.

According to an embodiment, six actuation cables 90 run through the medical instrument shaft arranged respectively two planar groups of three symmetrically arranged with respect to a shaft section plane defined by the axis of the instrument X-X and by its first joint axis of joint P-P.

According to an embodiment, the tendon and opposite tendon 90, 190 associated to second joint member 72, providing for its clockwise and anticlockwise rotation around said first joint axis of movement P-P, are arranged opposite to each other with respect to said section plane, slide over two opposite lateral sliding surfaces 40 of the first member 71, then both cross said section plane before said first axis of joint movement P-P, then they wind around at least one joint sliding surface 80 of the second member 72 and finally they attach to said second member 72.

According to an embodiment, the tendon and opposite tendon 90 associated to the first portion of the terminal member 177, like the two tendons 90 associated to the second portion of the terminal member 277, both run on the same side of said shaft section plane, they both slide on the same lateral sliding surface 40, 140 of first member 71, then they both cross said section plane before the first axis of joint movement P-P, then they both wind around at least one same sliding surface 80 of the second member 72 and continue their path to end up winding in opposite directions on the winding surface 86 of the terminal member 77. When only the first portions of the terminal member 177 or only the second portion of the terminal member 277 are actuated, the tendons 90 associated to said first portion of terminal member 177 and associated to said second portion of terminal member 277 slide along the sliding surface 80 of the second member 72.

According to an embodiment, the movement of the jointed device 70 is realized by polymeric actuation cables 90, or polymeric tendons 90. These tendons 90 run through the mechanical transmission box 62, run along the whole hollow shaft 65 and arrive at jointed device 70 and terminal device 77.

According to an embodiment, the transmission of motion to the joints of the jointed device 70 is a function of the path of the tendons 90 in the jointed device.

Exploiting the low friction, the very small curvature radius of the tendons 90, the tendons slide across the joint members that make up the jointed device and they wind around the various joint axes of movement P-P, Y-Y.

According to an embodiment, the members that make up the jointed device 70 are in fact rotationally connected to each other by a axis support feature of the rotational joint 171. Each member has joint sliding surfaces 80, or joint winding surfaces 86 for the tendons 90, both around the joint axis of movement P-P, Y-Y and along its body.

According to an embodiment, a further elbow joint member 75, positioned before a wrist joint member 78, suitable to reproduce the kinematics of the human wrist, can be included by provisioning an elbow joint member 75 characterized by having two distinct parallel axes of joint movement P-P, P-P. According to an embodiment, the first member 71 is coupled to said elbow member 75 having two distinct and parallel axes of joint movement P-P, P-P, one more proximal and one more distal, that are the first joint and second joint respectively. Said elbow member 75 has two lateral sliding surfaces 40, 140 arranged laterally opposite to

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each other with respect to a second section plane, defined as the plane that contains the first axis P-P and the second joint movement axis Y-Y.

According to an embodiment, there are eight actuation cables **90**, **190**. Said eight actuation cables, or tendons, run on the lateral sliding surfaces **40**, **140** of the first member, arranged in one group of four opposite to another group of four with respect to said first section plane, and they cross said section plane before the first axis of joint movement P-P, hence they run on the first joint sliding surface **80** of the elbow member **75**.

According to an embodiment, two actuation cables **90**, **190**, dedicated to the motion of rotational joint **171** of the elbow member, are terminated on said elbow member **75**. The remaining six cables **90**, **190** continue along the lateral sliding surfaces **40**, **140** of the rotational joint **171** of the elbow, crossing a second section plane before said second joint axis. The following progression of the tendons around the second, third and fourth members **72**, **73**, **74**, to the first portion of terminal member **177** and the second portion of terminal member **277** is analogous to what has been previously described in the presentation of the wrist configuration.

According to an embodiment, all members that form the jointed device **70** and the terminal device **77** are fabricated by a wire EDM performed on two orthogonal work planes X-Y, Y-Z.

According to an embodiment, fabricating the first member **71** starting from a cylindrical piece to be machined **117**, said first member presents two circular surfaces that allow its concentric insertion into the shaft **65**.

According to an embodiment, said circular surfaces present mating features on a lower portion, such as through-holes, that permit the rigid attachment of said first member of shaft **65** by means of fastening pins **76**. Said first member **71** presents on a distal portion two features to support rotational joint **171**, each characterized by a cylindrical seat centered around said first axis of joint movement P-P and a lateral shoulder surface.

According to an embodiment, all holes, being machined by wire EDM such as the pin holes **79**, have extra machining grooves **49** resulting from the passage of the cutting wire **115**.

According to an embodiment, having defined said first section plane containing the axis of the instrument X-X and the first joint movement axis P-P, the first member **71** presents two opposite tendon sliding surfaces **40**, **140** each having rounded shapes that are symmetrically opposite i.e. mirrored with respect to said section plane.

According to an embodiment, being machined by wire EDM, each sliding surface **80**, **180**, **40**, **140** is resulting from the sweeping motion of parallel straight generatrices that move directly along a cutting profile **110**.

According to an embodiment, the actuation cables **90** slide in two groups of three, respectively along the two lateral sliding surfaces **40**, **140**, one opposite to the other on the first member **71** and they cross said section plane before the first axis of rotation to then continue onto second member **72**.

According to an embodiment, said second member **72** has a joint sliding surface **80** proximally, arranged around said first axis of joint movement P-P having a cylindrical portion.

According to an embodiment, said joint sliding surface **80** is formed by parallel straight generatrices following the wire EDM cutting profile.

According to an embodiment, a pin holding feature **76** and a lateral shoulder surface characterize the joint of the first

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member **71** around the first axis of joint movement P-P. Two tendon termination features **82** are laterally derived from the second member **72** allowing the fastening of second tendon endpoint **92** of the second member by knot or gluing. Distally, two support features for the third and fourth rotational joint are each characterized by a pin hole **79** around the second axis of joint movement Y-Y and a lateral shoulder surface.

According to an embodiment, the second axis of joint movement Y-Y is orthogonal to the first axis of joint movement P-P. Being machined by wire EDM, the pin hole **79** has machining grooves **49**, resulting from the cutting wire **115**.

According to an embodiment, the third member **73** is characterized by a pin hole **79** located around the second axis of joint movement Y-Y. The third member **73** is mated to the second member **72** by a seat for a joint pin and an associated lateral shoulder surface. A winding surface **86** of the actuation cables **90**, **190** allows the winding of the actuation cables **90**, **190** around that winding surface **86** that is concentric to the second axis of joint movement Y-Y.

Laterally to the third member **73** a tendon termination feature **82** and tendon fastening points **82** are derived. The tendon termination feature **82** allows the passage of the tendons **90**, and the tendon fastening point **82** holds the second tendon endpoint **92**, **192** of the third member **73**, defined by knots.

According to an embodiment, the first portion of the terminal member **177** and the second portion of the terminal member **277** are jointed to the second member **72**, sharing the same second axis of joint movement Y-Y.

According to an embodiment, the first portion of the terminal member **177** mirrors the shape of the second portion of the terminal member **277**.

According to an embodiment, the third member **73** can be individually mated to the second member **72** if the terminal device **77**, present on the third member **73**, is a medical instrument **60** of a surgical or microsurgical type, similar to, for example to a scalpel blade.

According to an embodiment, a terminal member **77**, can be individually jointed to said second member **72**, only if the terminal device **77** is itself a medical instrument **60** of a surgical or microsurgical type similar for example to a scalpel blade or to a fiber-optic tendon carrier for laser light treatments. In this case, the jointed device **70** will only comprise two degrees of freedom of movement, in particular of pitch and yaw, losing the degree of freedom for grasping.

According to an embodiment, the first portion of terminal member **177** and the second portion of terminal member **277** can mate with each defining different terminal devices **77**, such as a micro device for cutting, a terminal micro device providing a straight grasp, a micro device providing angled grasping, a needle holder and other traditional microsurgical instruments as illustrated in FIGS. **25-27**. The terminal devices reproduce the form, proportions and functionalities of traditional microsurgical instruments tips, in order to facilitate their recognition and use by the microsurgeon **200**.

According to an embodiment fastening pins **76** are inserted in the pin holes **79** of the members of the jointed device **70**. The fastening pins **76** are preferentially made of hard metal, rectified and polished to reduce sliding friction.

According to an embodiment the fastening pins **76** have interference in mating with the pin holes **79** located in correspondence to the axes of joint movement P-P, Y-Y, of rotational joint **171**.

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According to an embodiment, the fastening pins **76** have lee way, or clearance, in the pin holes **79** associated to the winding surfaces **86**.

According to an embodiment, the connection by fastening pins **76** between the first member **71** and the second member **72** forms a rotational joint, suitable to rotate around the second axis of joint movement P-P, with an associated actuation angle substantially comprised between $+90^\circ$ and -90° .

According to an embodiment, the connection by a single fastening pin between the second joint member **72**, the first portion of terminal member **177** and the second portion of terminal member **277** creates a rotational joint between said three members **72**, **177**, **277** with an associated actuation angular range substantially between $+90^\circ$ and -90° . Said joint defines two degrees of freedom, characterizing both the yaw and the grasp of the medical instrument **60**.

According to an embodiment, the polymeric tendons **90**, **190** can be terminated in several ways, provided that, as a result of strong fastening, they can be tensioned and such a tension is also transmitted to the joint member, or to the part to which they are connected, driving its motion.

According to an embodiment, the tendons **90** run through a tendon termination feature **82** and are locked by a knot formed by the tendon **90** itself, located at said tendon fastening point **82**.

According to an embodiment, a second method for fastening the tendon **90**, used for example for the actuation of the second member **72**, provisions the passage of a loop of the tendon **90** around a tendon fastening point **82** and the application of tension to both the extremities of the tendon **90**, such that the two sides of the tendon **90** act as a single tendon **90**, halving the loads to which it is subject.

According to an embodiment, a third fastening method of the tendons **90** provisions the insertion of tendon portions in tendon fastening points **82**, intended for this use, and the use of glues specific for the polymer of which the tendons **90** are made, such as those used for example for gluing the first endpoint **91** to the lower frame **59** of the mechanical transmission box **62** of the tendon drive system **50**.

According to an embodiment, the jointed device **70** is characterized by three degrees of freedom of movement, and in particular by one degree of freedom of pitch between the first member **71** and the second member **72**, one degree of freedom of yaw between the second member **72** and the third member **73**, one degree of freedom of gripping, or grasping, between the first portion of terminal member **177** and the second portion of terminal member **277**.

According to an embodiment, the second joint member **72**, the first portion of the terminal member **177** and the second portion of the terminal member **277** can move around respectively said first axis of joint movement P-P and said second axis of joint movement Y-Y independently. The movement of the medical instrument **60** is carried out by the actuation cables **90**, which run over the members jointed to each other by rotational joints.

According to an embodiment, a pair of tendons **90**, **190** comprises a tendon **90** and an opposite tendon **190**, which is suitable to work as a pair of agonistic and antagonistic tendons associated to a first portion of the terminal member **177** and a further pair of tendons **90**, **190**, comprising a tendon **90** and an opposite tendon **190**, suitable to work as a pair of agonistic and antagonistic tendons associated to the second portion of the terminal member **277**, and a further pair of tendons **90**, **190**, comprising a tendon **90** and an

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opposite tendon **190**, suitable to act as a pair of agonistic and antagonistic tendons associated to the second joint member **72**.

According to an embodiment, a pair of tendons, **90**, **190**, comprising a tendon **90** and an opposite tendon **190**, suitable to work a pair of agonistic and antagonistic tendons, transmit a rotational movement to the second portion of the terminal member **277**, around said second axis of joint movement Y-Y, running over the lateral sliding surface **40** of the first joint member **71**, crossing said section plane, running on the joint sliding surface **80** of the second joint member **72**, and then splitting to wind them respectively in opposite directions around the winding surface **86** of the second portion of terminal member **277** and terminating with a knot. When one of the two tendons **90**, **190** is tensioned or released, it slides on a sliding surface **40** of the first joint member **71** and over sliding surface **80** of the second joint member **72**, while it winds itself or unwinds over the winding surface **86** of the fourth joint member like over a fixed pulley.

According to an embodiment, a further tendon pair **90**, **190** consisting of a tendon **90** and an opposite tendon **190**, actuates the first portion of terminal member **177** in a similar fashion to the way the second portion of terminal member **277** is actuated.

According to an embodiment, a yet further tendon pair **90**, **190**, consisting of a tendon **90** and an opposite tendon **190**, suitable to work as a pair of agonistic and antagonistic tendons, move second joint member **72** around first axis of joint movement P-P, running over the lateral sliding surface **40**, **140** of the first member **71**, on one side with respect to said section plane of the medical instrument **60**, intersecting said section plane, winding themselves in the opposite directions on the joint sliding surface **80** of the second joint member **72**, and terminating at tendon fastening points **82**. In particular, each actuation tendon **90**, **190** of the second joint member **72** is formed in a loop that passes around respective tendon fastening point **82** and comes back doubled up, passing over the winding surface **86**, joint sliding surface **80** and lateral sliding surface **40** along a path analogous to that followed by the opposite tendon.

According to an embodiment, when moving the second joint member **72** around the first axis of joint movement P-P, in one rotation direction, both the end of the tendons **90**, **190** are subject to tension. Furthermore, differently from the two tendon pairs **90**, **190** that actuate the first portion of terminal member **177** and the second portion of terminal member **277** respectively, in the case of tendons **90** of the second joint member **72**, the tendons **90**, when moving, do not slide over the sliding surface of the second joint member **72**, but wrap or unwrap around said joint sliding surface **80**, as though it were a pulley.

According to an embodiment, six independent tendons **90** are used for the actuation of the three degrees of freedom of movement of the jointed device **70**, but eight cables intersect on said section plane between the lateral sliding surface **40** of the first joint member and the lateral sliding surface **80** of the second joint member **72**, because both loop ends of the actuation cables **90**, **190** of the second member **72** are tensioned during the movement in one direction around the first axis of joint movement P-P.

According to an embodiment, the sliding surfaces **80**, **180** between the actuation cables **90** and members of the jointed device **70** are reduced to a minimum surface area, such as to reduce friction. The tendons **90**, **190** are terminated at their second tendon end points **92** such a way that their tendon path T-T remains parallel to the instrument axis X-X as much as possible, avoiding transversal forces.

According to an embodiment, the intersection of the tendons **90** and their crossing of said section plane between the joint sliding surface **40** and the first axis of rotation P-P prevents the tendon **90** from leaving the joint sliding surface **80** during its movement and guarantees a constant length and angle of the tendons **90**, **190**.

A method for machining tridimensional, assemblable mechanical micro-components by EDM is described below. In particular it regards the fabrication of jointed devices **70** of a characteristic outer diameter inferior to 4 mm for application in micro-surgery. Furthermore, the main characteristics of a specific machining fixture **112**, which is a fundamental element for the set up of a production process in an economically sustainable fashion and which is capable of guaranteeing the required precision, are described below.

According to an embodiment, the need to produce micro-parts with many mechanical details and a high level of precision requires the use of hard metals as a structural material and requires wire EDM as the machining process for the parts. As is known, EDM is a subtractive fabrication process in which material is removed by a conductive piece with a series of current discharges between the piece itself and an electrode kept at an electrical voltage difference, separated by a dielectric liquid such as water or oil, until the desired shape is obtained. In particular, during wire EDM machining, the workpiece **117** is held fixed and is immersed in a bath of dielectric liquid while a metal cutting wire **115**, made of copper or brass for example, and of a diameter varying between 0.5 mm and 0.02 mm, continuously runs between two bobbins. The cutting wire **115** is sustained by an upper guide and a lower guide, which being driven by a computer numeric control system in the horizontal plane, carry out two-dimensional cutting profiles. The movement of the guides is very precise, and the overall machining resolution is close to 1 micron (μm), nevertheless, the planar cut substantially limits the fabrication of three-dimensional parts. Despite the fact that some advanced machines have an upper guide, which can move independently in the horizontal plane, the ability to produce complex 3D parts has not substantially increased.

The primary advantages of wire EDM comprise:

- the possibility of machining hard metals,
- absence of direct contact between the tool and the piece to be machined **117**
- delicate details can be machined without distortion,
- a good superficial finish can be obtained,
- complex shapes, otherwise difficult to produce with conventional cutting instruments can be produced, while maintaining very low tolerances.

The manual phases for the fastening each single, metallic workpiece to be machined **117** to the machine for each of the cutting planes and the following calibration of the machine itself, are very slow phases during the fabrication of the parts and are also the phases which result in the greatest geometric errors that hinder the perfect mating between micro-parts produced individually.

According to an embodiment, in order to substantially decrease the fabrication time and guarantee the precision required for the correct mating of the fabricated micro-parts, a machining fixture **112** is provided, which intended specifically for this use. It provides a mechanical support, which allows the simultaneous fastening and machining of all the workpieces **117**, simplifying assembly of at least a portion of a jointed device **70** on one or more difference planes, with a single cutting profile **110** and a single calibration step.

According to one possible operating mode, the frontal plane of the machining fixture **112** has member holes **116**, suitable to hold the workpieces **117** with very tight tolerance, that is to say at least H6h5.

According to one possible operating mode, the frontal plane of the machining fixture **112** has a "stepped" profile to allow threading short through holes on the stepped lateral planes.

According to one possible operating mode, grub screws **M2** fasten the workpieces **117** to the machining fixture **112** and guarantee a perfect electrical conductivity with said machining fixture **112**, which fundamental for a successful EDM process.

According to one possible operating mode, the grub screws disappear under the plane to which they are screwed, i.e. are headless, to avoid limiting securing the fixture along those planes, with a vise of an EDM machine.

According to one possible operating mode, an alternative to the grub screws and to the threaded holes associated to the grub screws, is the use of conductive glue, to fasten the workpieces **117** to the machining fixture **112** and guarantee a perfect electrical conductivity with said machining fixture **112**.

According to one possible operating mode, the arrangement of the workpieces **117** on the machining fixture **112** is such that they not overlap in the work planes, for example in the X-Y and Y-Z planes, such that different and independent details or profiles can be cut for each plane on each workpiece **117**, by providing a single and continuous cutting profile **110** for the wire.

According to one possible operating mode, the gap, or non-overlapping section, between two adjacent workpieces is minimized such as to keep the dimensions of the machining fixture **112** as compact as possible. In this way it is possible to minimize the distance between the upper and lower guides, improving the machining precision.

According to one possible operating mode, a metallic reference rod **118** is inserted in the machining fixture **112** and is used for calibration of the EDM machine once the machining fixture **112** and the workpieces **117** are mounted on the machine.

According to one possible operating mode, a first calibration is provisioned, which is carried out only once for a given machining fixture **112**, loaded with all the workpieces **117** and a given EDM machine being used for the machining. Said first calibration is capable to identify and compensate all errors related to the EDM machine and to the geometric errors of the machining fixture **112**, such as for example those related to the relative position between the reference rod **118** and the workpieces **117**.

According to one possible operating mode, once the positions of the workpieces **117** are defined with respect to the reference rod **118** in the various cutting planes, the cutting profiles **110** are generated, taking into account of any differences of the actual positions with the nominal ones.

According to one possible operating mode, said first calibration will be repeated only if the EDM machine is changed or a new machining fixture **112** is being used.

According to one possible operating mode, each time the machining fixture **112**, loaded with the workpieces **117**, is secured to the vise of the EDM machine before a cut, a second calibration procedure is foreseen, or a cut calibration, performed only on the calibration rod **118**. This cut calibration process eliminates geometric offset and errors related to the manual fastening of the fixture and identifies the origin of the machine reference system with respect to the axis of the reference rod.

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According to one possible operating mode, to allow the correct fastening of the machining fixture **112** to the vise of the EDM machine, said machining fixture **112** has at least a pair of fastening or fixing surfaces **113**, **114**, opposite and parallel to each other, and rectified, meant to be gripped by the jaws of the vise, and a flat posterior X-Z surface, rectified and orthogonal to the fixing surfaces **113**, **114**, meant to be flush with an reference surface of the machine, orthogonal to the vise's clamp.

According to one possible operating mode, by not using rotary table in the EDM machine, it is necessary that the machining fixture **112** have a pair of fixing surfaces **113**, **114** that are flat, parallel and rectified, opposite to each other for each cut plane provisioned for the fabrication of the micro-components.

According to one possible operating mode, other cutting planes can be produced by appropriately modifying the machining fixture **112**.

According to one possible operating mode, to machine in a third orthogonal plane, it is necessary to provision openings **125** in the machining fixture that allow the cutting wire **115** to be inserted on the inside of the machining fixture and hence avoid the cutting of portions of the machining fixture **112**, for example. Several independent cutting profiles must be used however without requiring further calibrations. Nevertheless, at the end of every cutting profile **110** in said plane, the cutting wire **115** must be cut and reinserted in the next opening **125**.

According to one possible operating mode, the fabrication process used for the fabrication of parts of a jointed device **70**, provisions the insertion of four workpieces **117** composed of metallic cylinders made of tool steel, into member holes **116** on the front side of said machining fixture **112** and then their fastening with grub screws of M2 size.

According to one possible operating mode, all three-dimensional micro-part that form the jointed device **70** for micro-medical applications, are machined from metallic workpieces **117**, in particular steel cylinders of 3 millimeter outer diameter and 12 millimeter length, that are machined by wire EDM on two planes, X-Y and Y-Z.

According to one possible operating mode, the machining fixture **112** loaded with the workpieces **117** is secured on the vise of the EDM machine by using the fixing surfaces **113**, **114** as reference planes for the fastening and then the calibration in the X-Y plane is performed using the axis of the reference rod **118**, rigidly attached to the machining fixture **112**, as a reference. The first cutting profile **110** is performed, machining all the workpieces **117** fastened to the machining fixture **112**, in the X-Y plane.

According to one possible operating mode, the machining fixture **112** is then removed from the machine and remounted, rotated by 90° to machine along said second plane Y-Z of the machining fixture **112**.

According to one possible operating mode, a second calibration for the second work plane Y-Z is performed and then the cut of the second cut profile **210** is carried out.

According to one possible operating mode, by equipping the EDM machine with a rotating or orientable table, it is possible to perform the cut calibration process just once and rotate the work plane as necessary between one cut profile and the next.

According to one possible operating mode, at the end of the second cut profile **210** the components produced are completely detached from the workpiece and can be collected in the EDM machine bath.

Due to the provision of a robotic assembly, according to one aspect of the invention, it is possible to control the

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positioning and motion of at least one jointed medical instrument within a work volume, in a reliable, precise and easily controllable manner.

Due to the provision of a robotic assembly, according to one aspect of the invention it is possible to control the positioning and simultaneous motion of at least two jointed medical instruments, each comprising one jointed device operative within a workspace, in a reliable, precise and easily controllable manner, potentially reaching every body part of the patient with the terminal portions of said medical instruments.

Due to the provision of a robotic assembly according to one aspect of the invention, comprising an image capturing system, but lacking an integrated microscope, it is possible to limit the cost as well as the physical volume of said assembly, resulting in a compact platform compatible with the installation of a pre-existing microscope, hence allowing retro-fitting operations.

Due to the provision of a robotic assembly according to one aspect of the invention, having as few moving parts as possible that require a large range of movement during the movement of the terminal portion of the medical instrument, it is possible to provide a microsurgical robotic assembly of low encumbrance, improving the comfort of the microsurgeon, who can, for example, tele-operate while being in the immediate vicinity of the operating table and hence can see and directly access the operating field, as well as improving the overall working conditions of the surgical team, by, for example, avoiding collisions with mobile parts of the robot while accessing the operating field, as well as simplifying the transport of the robotic assembly, or the flow of people or air around the robotic assembly. Equally, it becomes possible to use two or more robotic assemblies simultaneously on one patient.

Due to the provision of a control device according to one aspect of the invention, it is possible to simplify the tele-operation master interface and make it more intuitive and comfortable, without limiting its functionality. At the same time, the training time required by a surgeon, not necessarily specialized in microsurgical procedures, to achieve a sufficient level of mastery of the control device, is reduced.

Due to the provision of a microsurgical robotic assembly, according to one aspect of the invention, comprising a control instrument suitable to replicate the shape of a traditional surgical or microsurgical instrument, it is possible to provide a familiar master interface for teleoperation to the surgeon, without compromising the accuracy of the manipulation.

At the same time, according to one aspect of the invention, due to the provision of at least one sensor coupled to an electromagnetic 3 D tracking device, said control instrument is also suitable to replicate the functionality of traditional surgical or microsurgical instruments, while allowing a complete freedom of movement in the three dimensions of space and allowing easy repositioning of the control device, for example between the operating table and the microscope, still guaranteeing good performance of the robotic system in terms of response time.

At the same time, according to one aspect of the invention, due to the provision of a compact control device and at least one sensor, suitable for relating the robotic assembly and the detection device to a common reference system, it is possible to freely position said control device in a simple manner, for example said control device can be positioned next to the operating table, or on a support table close to the microscope, or in a position deemed ergonomic for the surgeon looking into the microscope.

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Due to the provision of a control instrument according to one aspect of the invention, which replicates the shape of a traditional microsurgical instrument having at least one joint at its tip, such as for example tweezers of forceps, equipped with at least one aperture sensor, it is possible to control the opening and closing, as well as grip movements of a jointed medical device in a familiar and precise manner.

The provision of a medical instrument comprising a jointed device moved by tendons according to one aspect of the invention, reduces the complexity of its machining, for example by eliminating the provision of channels or sheaths, allowing extreme miniaturization of the medical instrument, without reducing its reliability during use or assembly.

Due to the provision of a jointed device according to one aspect of the invention, comprising actuation cables, or tendons, made of non-metallic material, for example polymeric material, it is possible to reduce the curvature radius of said tendons, as well as the friction coefficient of said tendons and consequently miniaturize further the jointed device. Due to the provision of a jointed device according to one aspect of the invention, comprising ruled surfaces with all parallel generatrices for the sliding of said tendons as well as tendon termination features arranged in a specific geometrical relationship to said surfaces, it is possible to do without tendon guide channels or sheaths, still guaranteeing parallelism of the tendons and hence allowing an extreme miniaturization of the jointed device.

Due to the provision of a fabrication method according to one aspect of the invention, as well as a machining fixture, suitable to guarantee the simultaneous positioning of several workpieces in a manner that permits to their cutting lines to remain parallel to each other, it is possible to obtain a single cut path by a EDM cutting wire for each cutting plane, on a plurality of workpieces. In this way, it is possible to generate parallel surface on said pieces, with high tolerances, even in cases where very detailed, small shapes are machined.

Due to the provision of a fabrication method according to one aspect of the invention, it is possible to produce micro-mechanical parts guaranteeing a high degree of precision as well as surfaces suitable for medical and/or surgical applications.

Due to the provision of a fabrication method, according to one aspect of the invention, it is possible to produce a medical instrument more rapidly with respect to known solutions, and as a consequence, more cost-efficiently.

Due to the provision of a machining fixture, as well as a fabrication method, according to one aspect of the invention, it is possible to obtain a fast and efficient process, even for repeated positioning of the workpieces within the machine.

Due to the provision of an improved machining fixture for EDM according to one aspect of the invention, which accelerates the cutting process on a plurality of cut planes, it is possible to reduce the number and duration of the phases dedicated to calibrating the machine.

Due to the provision of a fabrication method for electro-erosion according to one aspect of the invention, which permits the machining of micromechanical parts comprising cavities and ridges, that, even when leaving a groove between two prongs **81** of material, are suitable to form pin holding features without having to machine holes, it is possible to significantly reduce the machining time.

Due to the provision of a tendon drive system according to one aspect of the invention, it is possible to guarantee the movement of said tendons exclusively by a pusher assembly, suitable to push the tendons and produce tensile load on at least a portion of said tendon. In this way the drive system

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avoids pulling on the tendons, for example by clinging to a portion of the tendon or by wrapping a portion of the tendon around a winch.

Due to the provision of a tendon drive system according to one aspect of the invention, the number and complexity of the components of said drive are reduced and any backlash of the parts when they are not loaded can be avoided, making the system suitable for extreme miniaturization, without diminishing its reliability or its precision.

The provision of a tendon according to one aspect of the invention, allows the reduction of an outer diameter dimension of said tendon and as a consequence, of the medical instrument, without reducing its performance in terms of durability or reliability.

Due to the provision of a tendon according to one aspect of the invention, it is possible to guarantee improved performance in term of sliding friction of said tendon on at least a portion of said medical instrument, with respect to known solutions.

Due to the provision of a tendon, as well as a tendon replacement method, according to one aspect of the invention, it is possible to increase the working lifespan of said instrument with respect to known solutions.

Due to the provision of a tendon according to one aspect of the invention, produced of non-metallic material, for example polymeric material, it is possible to reduce the curvature radius of said tendon, as well as the friction coefficient of said tendon, and consequently increase the miniaturization of the medical instrument that comprises said tendon.

Due to the provision of a tendon according to one aspect of the invention, it is possible to do without the provision of tendon guide canals or sheaths in the medical instrument, still guaranteeing the parallelism between a plurality of tendons and hence allowing an extreme miniaturization of the medical instrument.

Due to the provision of a tendon **90** comprising a second tendon endpoint **92** as described above, it is possible to obtain a jointed device **70** in which its members do not require tendon guides or channels to facilitate the tendon **90** routing, without said tendons **90** interfering with each other. In fact, the geometric location of said tendon endpoints **92** is chosen in a way that said tendons **90** run substantially parallel to each other and parallel to said sliding surface **40**, **80**.

Due to the provision of a sliding surface, for example lateral sliding surfaces **40** and joint sliding surfaces **80**, as previously described, it is possible to for said tendons to slide over the jointed device with low friction.

Due to the cooperation between said sliding surfaces **40**, **80** and the geometric location of said first tendon endpoints **91** and said second tendon endpoints **92** it is possible to guarantee that the friction forces between the tendon and the sliding surface, as well as the fastening reactions at the first and second tendon endpoints **91** and **92** are substantially parallel to each other and along a same axis.

Due to cooperation between said sliding surfaces **40**, **80** and the geometric location of said first tendon endpoints **91** and said second tendon endpoints **92**, it is possible to obtain an extreme miniaturization of said medical instrument **60**. For example, in this way it is possible to do without pulleys and/or other tendon guides, which are not suitable to be miniaturized beyond a certain threshold. For example, according to an embodiment, the shaft **65** of said medical instrument can measure 3 millimeter in outer diameter.

Due to the provision of tendons **90** sustaining a curvature radius smaller than or substantially equal to 1 millimeters, it

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is possible to design a tendon path T-T, that at least partially wraps around said members **71, 72, 73, 74, 75, 77, 78, 177, 277** of said jointed device **70**, such as to avoid the formation of loops, when for example at last a portion of said jointed device **70** moves with respect to an axis of movement P-P, Y-Y.

Due to the provision of said tendon drive system **50**, as well as a tendon **90, 190** having said first tendon endpoint **91** and said second tendon endpoint **92**, being a boss, and/or a knot, and/or glued as previously described, it is possible to mount as well as easily replace a tendon **90, 190** with high precision, prolonging the working lifespan of said medical instrument **60**. Furthermore, due to the provision of tendons made of polymeric material, the members of said jointed device **70** are not damaged during working conditions.

Due to the provision of a tendon drive system **50** comprising at least a pusher assembly **94** suitable to push, while resting on a tendon deflectable portion **93** of a tendon **90**, it is possible to actuate said tendons without squeezing them or wrapping them around a capstan.

In this way it is possible to avoid damaging them when in working conditions, and hence increase the lifespan of said tendons, as well as of said medical instrument **60**, diminishing maintenance costs.

Due to the provision of a tendon drive system **50** as previously described, it is possible to reduce to a minimum the backlash within the tendon drive system **50**, always providing a defined preload.

Due to the provision of a substantially linear pusher assembly, it is possible to integrate micrometric actuation systems, such as slides and piezoelectric actuators, to control the tensile load of the tendons, as well as to release and pull exact lengths of tendon, allowing to move at least a portion of said medical instrument by a desired amount, for example around a movement axis.

The provision of a tendon drive system suitable for cooperating with a jointed device across a sterile barrier allows the production of a medical instrument, which is highly reliable and sterile.

Due to the provision of a fabrication method based on EDM as previously described, it is possible to fabricate an entire jointed device with only one placement step in a machine, decreasing the fabrication time and cost, without decreasing the reliability or precision of machining.

Due to the provision of a fabrication method according to one aspect of the invention, it is possible to produce joint members of a jointed device having ruled surfaces with parallel generatrices, such as to allow a tendon sliding over them maintain a stationary path with respect to said joint member. This allows the friction between the tendon and the sliding surface of the joint member to be reduced to a minimum, facilitating the miniaturization of the jointed device.

Due to the provision of a fabrication method based on EDM as previously described, suitable to transfer only thermal stimulation to the workpieces, it is possible to obtain parts of submillimeter dimensions, allowing an extreme miniaturization of said medical instrument **60**, still maintaining a satisfying cut precision due to the provision of cutting on a plurality of workpieces in a single passing.

Due to the provision of a tool, as well as a method of EDM according to one aspect of the invention, suitable for performing, with a single wire passing, the cut of parts in a plurality of workpieces which will be assembled together after machining, it is possible to obtain matings with millimetric precision, particularly suitable for building rotational joints features such as prongs, pivot holes, profiles of joint

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members, allowing hence to reliably mount pieces by snap-fit, or with controlled backlash between the same parts.

Due to the provision of a robotic assembly **100**, comprising at least one control instrument that replicates a traditional surgical instrument as well as a control device comprising an ergonomic support element for the operator, it is possible to improve the familiarity and ergonomics of the surgeon, improving the outcome of the surgical operation and patient comfort as a consequence.

Due to the provision of a robotic assembly according to one aspect of the invention, comprising a macro-positioning arm having a mechanical structure of arm members, as well as highly rigid joints, it is possible to avoid structural mechanical vibrations at the terminal portion of the instrument, and hence facilitate the surgeon's work.

Although some combinations of embodiments described above can be seen in the attached figures, an expert of the field will also be able to configure combinations not shown in the figures, without departing from the scope of the following claims.

To satisfy specific and temporary needs, a person skilled in the art can carry out a number of modifications, adaptations and substitutions of elements with other functionally equivalent elements, without departing from the scope of the following claims.

REFERENCE LIST

- 7** work volume, or common workspace volume
- 9** tendon
- 16** point of intersection
- 18** proximal tendon portion
- 19** distal tendon portion
- 20** control device
- 21** control instrument
- 22** detection device
- 23** connection cable
- 24** communication and power cable
- 25** operator support surface
- 26** status signal light
- 27** operator support element
- 28** position sensor
- 29** tip sensor
- 30** macro-positioning arm
- 31** first arm member
- 32** second arm member
- 33** third arm member
- 34** fourth arm member
- 35** release button, or brake release button
- 36** linear sliding guide
- 37** manual knob
- 38** support member
- 39** attachment feature
- 40** sliding surface
- 41** micro-positioning device
- 43** rotation dial nut
- 45** video camera
- 46** motorized rotary joint
- 47** base portion
- 48** plunger locking hole
- 49** machining groove
- 50** tendon drive system
- 51** first motorized slide, or first motorized micro-slide
- 52** second motorized slide, or second motorized micro-slide
- 53** third motorized slide, or third motorized micro-slide
- 54** first slide rail

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55 second slide rail
 56 third slide rail
 57 frame
 58 first frame portion, or upper frame
 59 second frame portion, drum, or lower frame 5
 60 medical instrument or micro-instrument or surgical micro-instrument
 61 motor box
 62 mechanical transmission box
 63 sharp edge of lateral sliding surface 10
 64 continuity surface of lateral gliding surface
 65 shaft, or hollow shaft
 67 control device base structure
 68 tip portion of control device
 69 forceps articulation of control device 15
 70 jointed or articulated device
 71 first member or first joint member, or first link
 72 second member or second joint member, or second link
 73 third member or third joint member, or third link
 74 fourth member or fourth joint member, or fourth link 20
 75 elbow member, or elbow link
 76 fastening pin
 77 terminal device, or terminal member, or terminal portion
 78 wrist member or wrist joint member 25
 79 pin hole
 80 sliding surface or joint sliding surface
 81 prong
 82 tendon termination feature, or tendon fastening point.
 83 surface 30
 84 tendon fastening surface
 86 winding surface, or ruled winding surface
 87 sterile barrier
 88 shoulder surface
 89 tendon guide element 35
 90 tendon, or actuation cable, or tendon of a first pair of tendons
 91 first endpoint or first tendon endpoint, or proximal tendon endpoint, or first tendon termination
 92 second endpoint or second tendon endpoint, or distal tendon endpoint, or second tendon termination 40
 93 tendon deflectable portion or deflectable portion
 94 pusher assembly or pushing means
 95 pushing element, piston, actuation piston or linear actuation piston. 45
 96 plunger or sliding shaft
 97 guiding elements, or tendon guiding elements, or guiding pulleys
 98 plunger idle pulley
 99 tensioning element, or pretensioning element, or spring 50
 100 robotic assembly, or robotic surgical assembly, or surgical robotic assembly, robotic assembly for micro-surgery or microsurgical robotic assembly
 102 operating table
 103 vision system, microscope, or surgical microscope 55
 104 support or cart
 105 foot platform
 106 retractable handle
 107 power cable
 108 control panel 60
 109 communication cable
 110 cutting profile, or cutting line
 111 display
 112 machining fixture
 113 first fixing surface of the first pair of fixing surfaces 65
 114 second fixing surface of the first pair of fixing surfaces

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115 cutting wire, or EDM wire, or electrical discharge machine wire
 116 member holes or member seats
 117 workpieces or pieces to be machined
 118 reference rod
 120 first control device
 122 first rod portion
 123 second rod portion
 125 guide hole or opening
 134 first fixing surface of the second pair of fixing surfaces
 135 second fixing surface of the second pair of fixing surfaces
 141 first micro-positioning device
 145 first portion of plunger
 146 second portion of plunger
 147 pushing surface
 148 reciprocal pushing surface
 150 sensor
 151 force sensor
 152 pressure sensor
 153 proximity sensor
 160 first medical instrument
 170 first jointed device
 171 rotational joint
 172 jointing portion
 173 spherical joint
 177 first portion of terminal member
 190 opposite tendon, or opposite tendon of a first pair of tendons
 191 tendon of a second pair of tendons
 192 opposite tendon of a second pair of tendons
 194 opposite pusher assembly or opposite pushing means
 197 first guiding element, or first guiding pulleys
 199 opposite tensioning element, opposite pretensioning element, or opposite spring
 210 second cut profile
 220 second control device
 221 second control instrument
 241 second micro-positioning device
 260 second medical instrument
 270 second jointed device
 277 second portion of terminal member
 297 second tendon guiding element, or second tendon guiding pulley
 397 third tendon guiding element, or third tendon guiding pulley.
 497 fourth tendon guiding element, or fourth tendon guiding pulley.
 200 surgeon, or microsurgeon
 201 patient
 202 surgical needle
 341 third micro-positioning device
 360 third medical instrument
 T-T tendon direction or tendon path
 X-X longitudinal shaft direction, or instrument axis
 P-P pitch axis, or first axis of joint movement
 Y-Y yaw axis, or second axis of joint movement
 a-a first axis of arm movement
 b-b second axis of arm movement
 c-c third axis of arm movement
 d-d fourth axis of arm movement
 e-e longitudinal axis of base portion of macropositioning arm
 f-f first slide direction
 g-g second slide direction
 h-h third slide direction

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r-r longitudinal axis of rotation
 X-Y first cutting plane
 Y-Z second cutting plane
 X-Z third cutting plane
 θ shaft angle

The invention claimed is:

1. A medical instrument comprising:

a frame; and

a jointed device, said jointed device comprising:

a first link;

a second link;

a terminal member; and

a first rotational joint defining a first axis of rotation,
 wherein said first link is connected to said second
 link via said first rotational joint;

wherein said second link is connected to said terminal
 member via a second rotational joint defining a
 second axis of rotation;

at least one tendon for moving said terminal member with
 respect to said second link by pulling;

wherein at least one of said first link and said second
 link comprises in a single piece at least one sliding
 surface, on which said at least one tendon slides for
 moving said terminal member with respect to said
 second link;

wherein each of said at least one sliding surface is a
 convex, ruled surface formed by a plurality of gener-
 ator lines, said generator lines are all portions of
 straight lines that are all parallel to each other;

wherein said terminal member comprises at least one
 winding surface onto which said at least one tendon
 wraps without sliding, and wherein said at least one
 winding surface is a convex, ruled surface formed by
 a plurality of portions of generator lines, said gener-
 ator lines are straight lines and all parallel to each
 other.

2. The medical instrument of claim 1, wherein the gener-
 ator lines of said at least one winding surface of the
 terminal member are all parallel to said second axis of
 rotation.

3. The medical instrument of claim 1, wherein the straight
 generator lines of said at least one winding surface of the
 terminal member are either parallel or orthogonal to said
 second axis of rotation.

4. The medical instrument of claim 1, wherein the straight
 generator lines of said at least one winding surface of the
 terminal member are either parallel to said first axis of
 rotation or parallel to said second axis of rotation.

5. The medical instrument of claim 1, wherein said at least
 one winding surface is free of any grooves or channels to
 guide said at least one tendon.

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6. The medical instrument of claim 1, wherein the straight
 generator lines of a sliding surface of said at least one sliding
 surface are parallel to said first axis of rotation.

7. The medical instrument of claim 1, wherein the straight
 generator lines of said at least one sliding surface are either
 parallel to said first axis of rotation or parallel to said second
 axis of rotation.

8. The medical instrument of claim 1, wherein said first
 axis of rotation is orthogonal to said second axis of rotation,
 and wherein said first axis of said first rotational joint is a
 pitch axis and said second axis of said second rotational joint
 is a yaw axis.

9. The medical instrument of claim 1, wherein said
 terminal member consists of a single piece forming a
 terminal link.

10. The medical instrument of claim 1, wherein said
 terminal member comprises two links defining a grip degree
 of freedom.

11. The medical instrument of claim 10, wherein said grip
 degree of freedom comprises relative rotation of the two
 links of said terminal member about said second axis of
 rotation.

12. The medical instrument of claim 1, wherein each of
 said first link and said second link comprises a sliding
 surface of said at least one sliding surface.

13. The medical instrument of claim 12, wherein the
 sliding surface of said first link is parallel to the sliding
 surface of said second link.

14. The medical instrument of claim 12, wherein the
 sliding surface of said first link faces opposite to the sliding
 surface of said second link.

15. The medical instrument of claim 1, wherein said at
 least one sliding surface is free of any groove or channel to
 guide said at least one tendon.

16. The medical instrument of claim 1, wherein the path
 of said tendon is locally orthogonal to the straight generator
 lines of the convex, ruled surface of said sliding surface for
 any angle defined by any rotational joints when in operative
 conditions.

17. The medical instrument of claim 1, wherein said at
 least one tendon comprise a pair of tendons working in
 opposition.

18. The medical instrument of claim 1, wherein mechanical
 properties and/or composition of said tendon vary in
 different portions of said at least one tendon.

19. The medical instrument of claim 1, wherein said at
 least one tendon comprises polymer fibers.

20. A robotic surgery assembly comprising at least one
 medical instrument according to claim 1.

21. The robotic surgery assembly of claim 20, wherein the
 at least one medical instrument is detachably mounted to the
 robotic surgery assembly.

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